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WHAT IS GROWTH NEED STRENGTH?

Jane Sutton

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For the Degree of Doctor of Philosophy

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What is Growth Need Strength?

This study investigated the nature of the individual difference factor Growth Need Strength (GNS), found by Hackman and Oldham (1976), to moderate the relationship between job characteristics and job satisfaction. Studies 1 and 1a report the results of a partial test and extension of the Job Characteristics Model using data from 692 employees working on 71 jobs in 4 organisations. The model was extended to investigate the moderating effects of personality, general values and work values, and the psychometric properties of the measures for GNS were assessed. The results found support for the moderating effect of GNS on the relationship specified above, employing moderated multiple regression analysis using multiplicative interactions terms, but no moderating effect was found for other variables tested. A relationship was found between GNS, Openness to Experience three of the general values and three work values domains. but GNS was found to be psychometrically distinct from them. In Study 2, the nature of GNS was examined using the Interpretative Phenomenological Approach (IPA), for data collected from experts in human resources. Using results from the two studies, a model was proposed in Study 3, for how and under what conditions GNS predicts those who take up of training opportunities. This model was tested using data from 64 employees in one organisation and the results show that, paradoxically, high GNS predicted low intention to train and low training activity. It was also predicted that employees' general and work values would mediate the influence of their GNS on the taking up of training opportunities. No support was found for this. Overall the results suggest that GNS may be a higher-order factor of personality and this has numerous implications for research and practice in occupational psychology, especially in the field of employee training and development.

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WHAT IS GROWTH NEED STRENGTH?

CONTENTS		Page
Chapter		
1.	Introduction and Overview	1
	The Job Characteristics Model	7
	The Research Question	10
	Summary of Aims	29
2.	Introduction	31
	Study 1 – Method	35
	Pilot Work	41
	Main Study	44
	Measures and Procedure	45
	Ethics	46
	Results	46
	Discussion	58
3.	Study 1a – Introduction	66
	Method	67
	Results	68
	Discussion	77
4.	Introduction and Overview	84
	Conceptual Framework	87
	Method	89
	Pilot Study	93
	Main Study 2	95
	Data Analysis	96
	Findings of Study 2	99
	Discussion	120
5.	Introduction	124
	Method	133

	Results	137
	Discussion	144
6.	What is Growth Need Strength?	148
	What Does Growth Need Strength Predict?	154
	Future Research	160
	Summary of Findings	161

REFERENCES

APPENDICES

A.	Table of Selected Studies
B.	Questionnaire for Studies 1 and 1a
C.	Letter to Participants
D.	Study 1- Correlations Among Questionnaire Scales Scores
E.	Questionnaire Assessing Job Characteristics
F.	Study 1a - Correlations Among Questionnaire Scales Scores
G.	Conceptual Framework
H.	Interview Schedule
I.	Explanation for Participants in Pilot Study
J.	Explanation for Participants in Interviews
K.	Demographic Details of Participants for Study 2
L.	Initial Coding
M.	Theming the Analysis for the Interview Data
N.	Theming the Analysis for the Interview Data (132 Themes)
O.	Final Themes
P.	Example of Summary Tables of Themes
Q.	Study 3 - Questionnaire 1
R.	Study 3 - Questionnaire 2
S.	Study 3 - Correlations Among Questionnaire Scales Scores
T.	Results of the Test of the Moderating Effect of Barriers to Attending Training on the Intention to Train-Training Activity Relationship
U.	Studies 1 and 1a – Scree Plots and Rotated Components Matrices

CHAPTER 1

Introduction and Overview

The Job Characteristics Model (JCM) proposed by Hackman and Oldham (1976) sets out the conditions under which individuals will become motivated to perform effectively in their jobs. The model suggests that differences among people moderate how they react to work and one such moderator is called Growth Need Strength (GNS) which Hackman and Oldham (1975) describe as '*a malleable individual difference characteristic which is predicted to influence how positively an employee will respond to a job with objectively high motivating potential*' (p.163). Following the publication of the JCM a number of studies examined the moderating effect of GNS on various relationships within the work setting but a review of these studies found that the nature of the construct had not been questioned. The definition put forward by Hackman and Oldham is felt to be inadequate because it does not reveal anything about the origins of GNS and this thesis attempts to address this inadequacy by comparing GNS to other related constructs in the hope that this may shed some light on its true nature. However, before exploring the issue of definition further, it is helpful to look back at some of the developments in the study of work behaviour which lead eventually to the design of the JCM.

Organisational Psychology, a branch of applied psychology, is concerned with the study of work related behaviour. The term is now widely used to encompass occupational, organisational, industrial, business and vocational behaviour and has most recently become known as 'work-psychology'. The discipline covers a wide range of organisational behaviour including selection and assessment, training, performance, organisational change, well being in work and work design. Being 'in employment' is something, which concerns almost everyone at some time during his or her lives. Employment enables people to earn money to live and to enjoy a life outside working hours, but working life in the twenty first century is significantly different to the working life experienced by people say, following the second world war. Organisational behaviour at that time was typified by two things – "fitting the person to the job" and "fitting the job to the person" (Furnham, 1997 p. 1). Work was carried out between specified hours, there was a clear delineation between working time and leisure time and the majority of work was carried out by men. In 2002 the concept of work has

changed for many groups of employees. For example, some organisations operate 24 hours a day, seven days a week and for many, the bulk of the work is carried out using computers and telephones. In order to work in such organisations individuals must conform to those work patterns specified by their employers. Sometimes those patterns are difficult for people to fit into, leaving them dissatisfied and unfulfilled in terms of their working lives and causing disruption to their lives outside work. In many cases organisations expect individuals to 'fit the job' without any regard for the idea that the job may not 'fit the person', and in doing so they are denying themselves the opportunity to reap the benefits that a good person – job fit can bring. Such benefits can include increased job satisfaction, improved performance and the reduction of absenteeism and turnover. However, before moving to a detailed discussion of the specific subject that forms the basis of the current research, it is helpful to look back to the origins of the study of work behaviour and some of the figures in history that have played a role in influencing organisational behaviour today.

The Study of Behaviour at Work

Although men and women have carried out 'work' for thousands of years the major influences on the organisation of work in the West are more recent. Until the Protestant Reform Movement during the sixteenth century, the system of feudalism dominated the way in which people's lives were organised. The religious reformers, the most famous of whom were Martin Luther in Germany and John Calvin in Switzerland, gained influence due to the fact that by the end of the Middle Ages the Church was regarded as a corrupt institution. Statt (1994) suggests that the Protestant reformers '*revived the idea of the ennobling power of work and of earning one's living while saving one's soul*' (p.7). The following centuries saw massive changes in the development of trade and industry that brought with them significant changes in the organisation of labour. These changes formed the Industrial Revolution which is widely accepted as beginning in the late seventeen hundreds, with the advent of steam power.

Scientific Management

The factories which were created at the end of the eighteenth century provided an enormous challenge to the management of labour. The emphasis was on production and central to this was the employment of large numbers of people to operate the machinery,

which would maximise that production. Individual differences amongst workers were ignored as people came to be seen as endless pairs of hands, no more important than the machines they were operating – and in some cases less so. Thus came the division of labour where rather than an entire job being done by one individual, each job was broken down into component parts, each part being carried out by individuals specifically trained for that one task. In 1835 Charles Babbage related human labour to the most effective use of machines and it was not until the late nineteenth century that further changes in the division of labour were made.

F.W. Taylor was born in 1856 and worked for an American steel company. He made a meteoric rise from labourer to chief engineer by the age of 31 years and was the first person to carry out systematic studies of work. Taylor's 'Scientific Management' studies analysed human behaviour at work with the overall aim of removing human variability. Each task was broken down into its smallest unit and an engineer worked out how each unit of work could be carried out. Following this, individuals would be taught how to perform that unit of work using only the 'motions' which were essential to the task. Taylor was, basically, restricting behavioural alternatives for the individual and looked at issues such as environment, capacity, speed and cost. His model had a profound effect on productivity, which rose dramatically as a result - as did the emergence of departments dealing with personnel and quality control.

At a similar time, Henri Fayol (1841 –1925), a French engineer, was also writing about the organisation of work. Whilst both Fayol's and Taylor's work focussed on the task, their approaches were quite different, with Fayol being more concerned with authority and its implementation. Fayol identified 14 principles that he saw as common to all organisations – specialisation/division of labour; authority with responsibility; discipline; unity of command; unity of direction; subordination of individual interests; remuneration; centralisation; chain/line of authority; order; equity; lifetime jobs (for good workers); initiative and esprit de corps. This approach was ahead of its time in that it made managers question their function.

Classical Theory

In discussing what he calls the 'great thinkers' in organisational behaviour, Furnham (1997) suggests that Max Weber was probably the most influential because of his ideas on the work ethic and bureaucracy and also because his work both came from and influenced many disciplines, as well as psychology (p.58). Weber's interest was in the nature of power and authority within organisations and he suggested that a bureaucracy was a special type of organisation which displayed certain characteristics. These were a division or specialisation of labour; a well-developed hierarchy; a system of procedures and rules by which the rights and duties of employees are defined; interpersonal relations based on position rather than personality and promotion and selection based on technical competence. However, Weber also noted that bureaucracies were to some extent dysfunctional and he argued that they had led to the depersonalisation of the modern world.

Classical theorists of management, such as Fayol and Weber were concerned with the structure of the organisation and how this could bring about greater efficiency and productivity. Up to this point in time, emphasis was on work and its management in order to maximise productivity - the human beings who were to operate the machinery and create the wealth for the large organisations were hardly given a mention. It was as a result of this clinical approach and the 'coldness' of scientific management and bureaucracy that the Social Behavioural School gained momentum. With the advent of the Human Relations movement, which as part of the Social Behavioural School, was concerned with worker well being, there was a shift of emphasis from focus on the task to focus on the individual as being important to production.

Human Relations

In the early 1920's Elton Mayo (1880 – 1949) conducted a series of experiments called the Hawthorne studies, named after the location of the Western Electric Company in Chicago, where the first set of these studies was carried out. In one study a group of 6 women were selected, working in the Relay Test Assembly Room, making telephone equipment. The study focussed on how their productivity was affected by a number of environmental variables such as lighting, hours of work and rest periods. Each change brought an increase in productivity, even when one change resulted in a reversion to

previous working practice. This indicated to the researchers that what was actually making the difference to productivity was the fact that workers were being taken notice of and were trying hard to perform well, whatever the conditions. This change in performance when attention was paid to people has become known as the Hawthorne Effect.

The value of the Hawthorne studies was in their discoveries concerning individuals, individuals in groups and organisational design. These studies marked the end of an era in the study of organisational behaviour, as since the beginning of the twentieth century what were known as 'industrial psychologists' saw their role as improving efficiency. Furnham (1997) suggests that such psychologists were '*not interested in worker-fatigue out of compassion for the workers, but rather how to ensure the most cost-efficient production from the workers*' (p.82). The Hawthorne studies marked a significant change in the way psychologists viewed worker behaviour, in that they began to realise that psychological factors, as well as physical working conditions, could affect human performance in work. The threat of and the eventual reality of a second war in Europe meant psychologists needed to find new ways to assess people for recruitment, to find men and women who could take leadership roles and to help rebuild those nations affected by war, once it was over.

Motivation

This second phase in the approach to organisational behaviour placed more emphasis on human relations and in particular the relationship between motivation and satisfaction. At the beginning of the 1950s Trist et al (1951) proposed their Socio-technical Systems Theory, which suggested that motivation was brought about by the interaction between technical aspects of work and the social situation in which work was carried out. However, the theory did not actually specify how the two aspects of work affect each other. One of the greatest influences on the design of work was proposed by Herzberg (1966) who suggested that for a job to be 'motivating' to the individual, both motivators e.g. satisfaction and achievement, and hygiene factors e.g. pay, must be built into jobs.

During this period psychologists were also beginning to realise that human beings had certain needs and if some of these could be satisfied then performance and thus

productivity would increase. One method of approaching the satisfaction of human needs was through the design of work and this chapter continues with a detailed examination of the study of higher order needs and how they may be satisfied.

Higher Order Needs

The suggestion that the satisfaction of higher order needs such as needs for personal growth, autonomy and self esteem comes from within the individual, is derived in part from Maslow's (1943) Theory of Human Motivation and from Alderfer's (1969) Theory of Human Needs. Maslow (1943) described the arrangement of human needs as follows: *'Human needs arrange themselves in hierarchies of pre-potency. That is to say, the appearance of one need usually rests on the prior satisfaction of another more pre-potent need. Man is a perpetually wanting animal'* (p.370). Maslow suggests that as soon as one need is satisfied other higher needs emerge which *'dominate the organism'* (p. 373) and when these are satisfied more, still higher needs, emerge. Maslow's hierarchy of needs begins with basic physiological needs such as the need to satisfy hunger and ends with the need for self-actualisation which he describes as *'the desire to become more and more what one is, to become everything one is capable of becoming'* p.382. Maslow does not present any empirical evidence for his theory claiming however that it derives most directly from clinical experience. Alderfer (1969) on the other hand, suggests that the satisfaction of higher order needs such as growth needs concerns *'Engaging problems which call upon him to utilise his own capacities fully and to include requiring him to develop additional capacities. A person experiences a greater sense of wholeness and fullness as a human being by satisfying growth needs'* (p.147). Alderfer believed that needs could be divided into three sets - existence needs, relatedness needs and growth needs and that growth needs were the least 'concrete' of these because they were dependent on the uniqueness of individuals. Alderfer describes growth needs as a person *'Seeking opportunities for his unique personal development and growth'* (Ibid p.145).

The early 1970s saw the start of research interest in the area of job design. Lawler & Hall (1970) conducted a study to attempt to differentiate between three factors which they suggested might be linked to work outcomes. Their theory was that the way a job is designed affects the attitudes, beliefs and feelings of the jobholder and they examined

individuals' attitudes towards (a) job involvement (b) need satisfaction and (c) intrinsic motivation. Lawler (1969) suggests that intrinsic motivation should exist when job holders feel that their job provides them with feedback about their performance, tests abilities that they personally value and provides them with the opportunity to set their own goals. These activities, he suggests, will result in the jobholder believing that good performance will lead to feelings of higher order need satisfaction. Higher order need satisfactions are defined by Lawler & Hall (1970 p.306) as being satisfactions which relate to the work itself, such as feelings of autonomy and growth. They concluded that "higher order need satisfaction" was probably an individual difference factor which people carry with them from job to job.

Hackman & Lawler (1971) question what it is about people that moderates the way that they react to their jobs and suggest that lower level needs are relatively easily satisfied, which is not the case with higher order needs such as personal growth and development. They suggest that the individual difference factor which moderates whether a person derives satisfaction of higher order needs is a high need for personal growth, feelings of worthwhile accomplishment, autonomy and self esteem, which come from within the individual. At the end of their study they suggest: *'All in all the data make a strong case for the moderating effect of individual higher order need strength in determining the effects of job characteristics on employee behaviour and attitudes at work'* Hackman & Lawler (1971 p. 280). With regards to the origins of higher order need satisfaction Hackman & Lawler say *'..either individuals with a particular pattern of needs tend to redefine the tasks they perform to be more consistent with those needs than is actually the caseor apparently individuals who are desirous of higher order need satisfaction will, when working on a job high on the core dimensions, gradually develop and/or verify the hypothesis that personally valued rewards can be obtained by working hard and effectively on the job'* (ibid., pp. 282 & 283).

The Job Characteristics Model

In 1976 Hackman & Oldham published their paper 'Motivation Through Design of Work: Test of a Theory' which proposed a theory of work motivation that would enable job design/redesign programmes to be devised and their effectiveness measured.

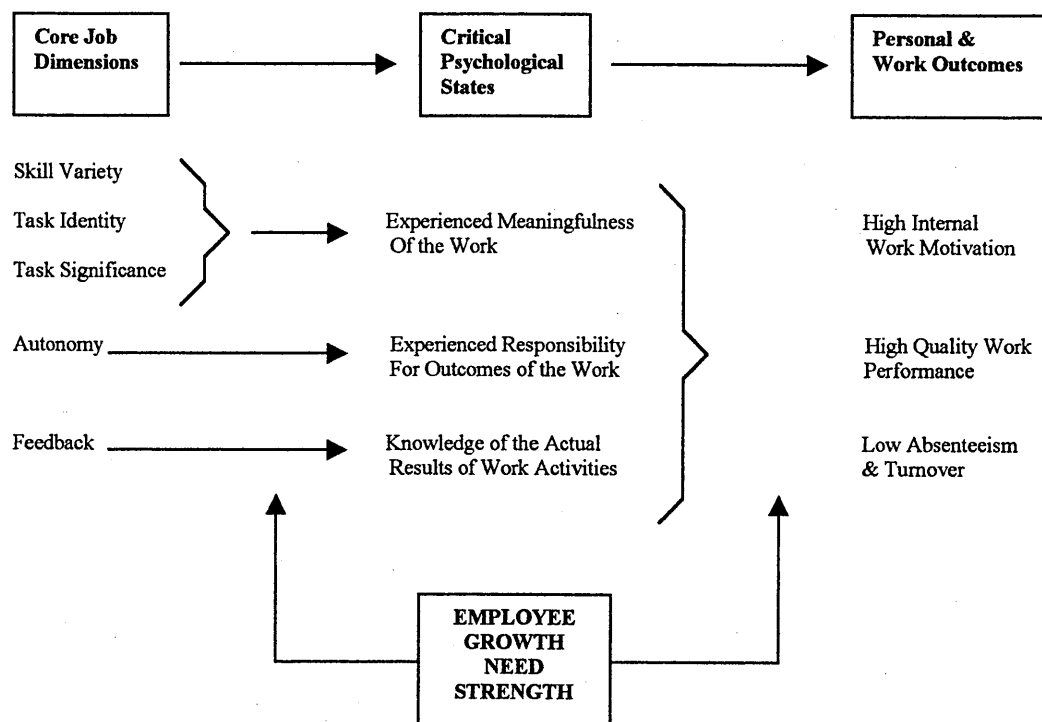
Hackman and Oldham's specific interest was in the effects of job enrichment on worker

behaviour. The model proposed by Hackman & Oldham - the Job Characteristics Model (see overleaf), suggests that if jobs are high in certain characteristics, these will bring about critical psychological states in individuals. As a result of individuals experiencing positive feelings about their jobs, personal and work outcomes will improve. Hackman and Oldham suggest that the relationship between job characteristics and critical psychological states and between critical psychological states and outcomes, is moderated by an individual difference factor which they call 'Growth Need Strength' (GNS). The rationale behind the Job Characteristics Model (JCM) is that '*An individual experiences positive affect to the extent that he **learns** (knowledge of results) that he **personally** (experienced responsibility) has performed well on a task he **cares about** (experienced meaningfulness)*' (Hackman & Oldham 1976, p.255). The model takes account of individual differences by introducing them as moderators which affect how people react to their work. These are individual growth need strength, context satisfactions, knowledge and skill. Although context satisfactions and knowledge & skill are mentioned in the theory, they are somewhat overlooked; with growth need strength playing the key role in the moderation process. Hackman and Oldham (1976) suggest that growth need strength moderates the relationship between job characteristics and work outcomes such as job satisfaction, performance, turnover and absenteeism.

The Job Characteristics Model sets out the conditions under which people will become '*internally motivated*' to perform effectively in their jobs (Hackman & Oldham, 1976). They state that internal motivation is brought about by three classes of variables '*(1) Psychological states of employees which must be present for internally motivated work behaviour to develop, (2) The characteristics of jobs which can create these psychological states and (3) The attributes of individuals that determine how positively they will respond to a complex and challenging job*' (p. 250). The purpose of the theory was to develop a model which would form the basis of a strategy for work redesign, thus improving productivity and quality of the work experience. Hackman & Oldham originally set out to discover why 'enriched' work sometimes resulted in positive outcomes and sometimes not.

Figure 1. The Job Characteristics Model of Work Motivation.

Hackman and Oldham, 1976.



The relationships shown in the model are tested by means of the Job Diagnostic Survey, a questionnaire specifically designed for the purpose. By using the questionnaire each job is given a Motivating Potential Score which Hackman and Oldham (1974a) describe as *'a score reflecting the potential of a job for eliciting positive internal work motivation on the part of employees (especially those with high desire for growth need satisfaction).'*' (p.73). This formula used by Hackman and Oldham is given below:

Motivating Potential

$$\text{Score (MPS)} = \frac{[\text{Skill Variety} + \text{Task Identity} + \text{Task Significance}]}{3} \times [\text{Autonomy}] \times [\text{Feedback}]$$

3

Hackman and Oldham suggest that the overall potential of a job to promote internal motivation should be highest when three conditions are met. These are a) the job is high on at least one of the three dimensions which lead to experienced meaningfulness, b) the job is high on autonomy and c) the job is high on feedback. The above authors suggest that the Motivating Potential Score (MPS) is a measure of the degree to which the above conditions are met and it is derived from the propositions of the Model. Hackman and

Oldham (1974a) suggest that MPS scores can range from 1 to 343 with the average being about 125 (p.39). Methods of combining the job characteristics to produce different motivating potential scores are provided on page 37.

According to Hackman & Oldham (1976) GNS is a way to '*conceptualise and measure*' (p.258) individual differences in how people react to their work. They say that people with high GNS will respond more positively to a job with a high motivating potential score than people with low GNS. According to the model, moderation takes place in two places the first being between the core characteristics and the critical psychological states, where the model specifies that people with high GNS will experience the psychological states more strongly when the job is high in motivating potential, than people with low GNS. The second operates between the critical psychological states and the personal and work outcomes, where high GNS individuals will respond more positively to the psychological states, and work outcomes will be more positive as a result

The publication of the JCM was followed by a flurry of research into the subject of job characteristics and their effects on the person-work relationship. Of the some 76 studies relating to the job characteristics-work outcomes relationship reviewed for the current research, the dimensionality of job characteristics (Dunham, 1976; Dunham et al, 1977; Pokorney et al, 1980; Stone and Guetal, 1985; Birnbaum et al, 1986; Fried and Ferris, 1987; Khandewal and Aleem, 1990; Kahn and Robertson, 1992) and the moderating effect of growth need strength (Oldham et al, 1976; Evans et al, 1979; Orpen, 1979; Abdel-Halim, 1979; Arnold and House, 1980; Champoux, 1980; Pokorney et al, 1980; Aldag et al, 1981; Griffin, 1982; Kemp and Cook, 1983; Loher et al, 1985; Spector, 1985; Graen et al, 1986; Fried and Ferris, 1987; Das, 1991; Tiegs et al, 1992; Johns et al, 1992; Kelly, 1992; Lee-Ross, 1998; de Jong et al, 2001) were the most frequently studied.

The Research Question

The question which arose from the literature review was 'What is Growth Need Strength?' Whilst the moderating effects of this individual difference factor have been studied at length only one study discusses what 'growth need strength' actually is.

Evans et al (1979) compared the moderating effects of GNS with the moderator 'need for achievement' as measured by the Thematic Apperception Test (Murray, 1938) using McClelland's (1973) label nAch for achievement. They suggest that need for achievement '*appears to be a more stable personality trait than growth need strength which appears to be more developmental.*' (p.377)

The central aim of this thesis was to gain an understanding of the origins of Growth Need Strength and to compare it with other individual difference factors in order to attempt to define it. This thesis continues with a discussion of the research which followed the Job Characteristics Model, examines the research into the individual difference factor Growth Need Strength and the support for GNS as a moderator of the Job Characteristics/Personal and Work Outcomes relationship. The chapter then provides information on previous research into individual and situational differences which might affect the way people respond to their jobs. These include work values, general values and personality

Growth Need Strength

Alderfer (1969) implied that growth needs exist in all people at some level regardless of education, intelligence, social and cultural background and Hackman and Oldham's (1976) study demonstrated that all of the participants they tested had a positive level of Growth Need Strength but that some were higher in this factor than others. Alderfer (1969) suggested that crucial to his theory is that if a person sees him/herself as learning from any experience and as a result becomes a '*fuller, more differentiated, more competent human being*' (p.152), then the need for growth will be enhanced.

The causal core of the JCM is made up of the three critical psychological states experienced meaningfulness of the work, experienced responsibility for the outcomes of the work and knowledge of the results of the work activities. As stated earlier, Hackman and Oldham (1976) suggest that an individual experiences positive affect depending on the extent to which the three critical psychological states are fostered within the individual by the presence of certain job characteristics. They suggest that this positive affect provides an incentive to the individual to continue to perform well in the future. The rationale behind this is that if a person does not perform well he/she will not

experience positive inner feelings and may try harder in the future to regain these inner feelings which are derived from good performance. Hackman & Oldham (1976) suggest however, that the individual will only continue to strive for inner feelings of reinforcement for as long as he/she values the internal rewards which have been previously experienced by good performance. This perpetuation of positive inner feelings is described by Hackman & Oldham (1976) as '*self-generated motivation*' (256), and is at its highest when all three psychological states are present.

Hackman & Oldham (1976) suggest that variations in the outcomes which result from the design of jobs, occur because of differences in individuals and say that there are three human characteristics which help to understand who will and will not respond positively to jobs with a high motivating potential score. The three characteristics are psychological needs, satisfaction with the work context and knowledge and skill and these are identified as 'moderators' in the model. Hackman & Oldham (1976) describe psychological needs as 'Growth Need Strength' and suggest that people who have a strong need for personal growth and self-direction at work are those who are more likely to respond positively to opportunities for personal accomplishment which can accompany the more complex jobs. Hackman & Oldham (1975) describe GNS as '*A malleable individual difference characteristic which is predicted to influence how positively an employee will respond to a job with objectively high motivating potential.*' (p.163).

Baron and Kenny (1986) suggest that whereas mediating variables explain *how or why* external physical events take on internal psychological significance, moderating variables specify *when* certain outcomes will occur (p. 1176). They suggest that a moderator is a quantitative or qualitative variable that affects the direction and/or strength of an independent and a dependent variable. Baron and Kenny (1986) summarise the essential properties of a moderator variable by using the model which is reproduced on the following page. Applying the model shown in Figure 2 to the JCM, there are three causal paths that feed into the outcome variables. Path 'a' is the impact of the independent variable Motivating Potential Score on the dependent variable work outcomes, Path 'b' is the impact of GNS on work outcomes and Path 'c' is the

interaction between the two other causal paths. Support for the moderator hypothesis is found if the interaction is significant.

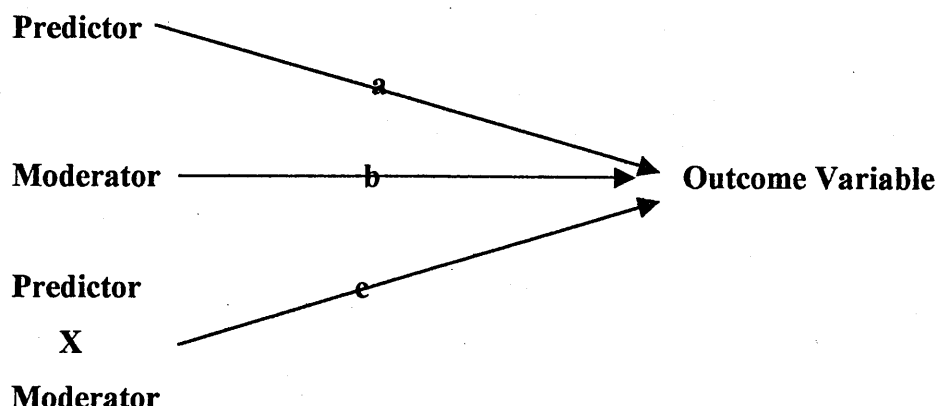


Figure 2. Moderator Model. Baron and Kenny, 1986 p. 1174.

The study Hackman and Oldham conducted in 1975 reported on the development of the Job Diagnostic Survey and the results were obtained from 658 employees working on 62 jobs in 7 organisations. The jobs were highly heterogeneous and were from both industrial and service organisations. The organisations were based in different parts of the United States both in urban and rural settings. Two measures of growth need strength were used, the first asking employees what attributes (from a list) they would like in their jobs ('would like' format). The second asked employees to choose between two jobs with different attributes ('job choice' format) and the authors report that the scale for GNS was constructed to be a highly reliable predictor of individual needs. The reported reliability for the 'would like' format was .88 and for the 'job choice' format .71. Hackman & Oldham say that their results suggest that internal consistency reliability of the scales and the discriminant validity of the items were satisfactory. They report support for their hypothesis that the relationships between job dimensions and the work outcomes would be stronger for individuals with high GNS than for individuals who do not strongly desire growth satisfactions.

Hackman & Oldham explored these relationships further in their 1976 study and reported that there was evidence from previous research that individual differences do moderate how people react to their work. Further, studies using direct measures of individual needs gave stronger support than measures of aspects such as sub cultural

background or generalised work values. (See 'Individual and Situational Differences' p.16). The results of the study showed that high GNS people were more likely to experience the critical psychological states when the objective job was high in motivating potential and that behavioural outcomes were higher. However, although all the differences between correlations for high versus low GNS individuals were in the predicted direction statistical significance was only achieved for the dependent measure of internal motivation.

In their study, Hackman & Oldham also examined the relationships between a summary measure of GNS and demographic and sub cultural variables which had been reported in previous research as moderating the job characteristic-work outcomes relationship. The results of this analysis showed that a '*typical*' high GNS person was '*A young and well-educated male who works or lives in a suburban or rural setting..... the individual's **present** place of work and residence relate most substantially to measured need for growth, whereas the locality of socialisation is rather weakly associated*' Hackman & Oldham (1976 p. 275). They suggest that an individual's current experiences are more responsible for determining desire for growth satisfaction than are aspects of personal history. However, they also say that individuals may change or adjust to meet the demands of their current situation, becoming more 'growth oriented' when the situation requires more independent thought and action in work.

Spector (1985) conducted a meta-analysis to test the moderating effect of higher-order need strength on the relationship between job scope and employee outcomes. He suggests that although higher-order need strength is an important component of the job characteristics model the concept of this individual difference factor and its measurement are somewhat weak. He discusses the nature of higher-order need strength and concludes that there are some variations in its definition but that generally the term is used to refer to the top two or three categories in Maslow's hierarchy. Spector points out that whilst Hackman & Oldham adopted the term 'growth need strength' to encompass these categories of need they failed to elaborate on its nature. In his work, Spector grouped together all higher order needs including higher order and growth need strength, achievement, autonomy and self-actualisation needs. His results showed that employees high on higher-order need strength demonstrated stronger correlations with

all employee outcomes than employees low in need strength. Spector's point about the lack of definition of Growth Need Strength is an important one in view of the fact that the job characteristics model generated a considerable amount of research. Some 75 studies have been made since the inception of the JCM in 1976 and although some of these include detailed examination of GNS as a moderator of the job characteristics-job satisfaction relationship, only Evans et al, (1979) discuss what this individual difference factor actually is and then only fleetingly. These authors base their suggestion that GNS is 'developmental' (p. 377), partly on Hackman and Oldham's original findings that GNS is related to education, age and sex and that typically high GNS individuals were young, white males with a relatively high level of education. This, they say, makes it more likely that this group of people has been '*socialised and exposed to growth need oriented environments*' (Evans et al, 1979 p.378).

Salancik and Pfeffer (1977) examined need-satisfaction models of job attitudes and concluded that such models should be re-examined with particular reference to origins of human needs. They suggest that the stability and usefulness of needs as concepts for explaining behaviour in a predictive way are important issues (p. 453). Salancik and Pfeffer also point out that consistency and priming effects may play a large role in the outcome of job attitude studies, based on the premise that individuals' attitudes are derived from the information available to them at the time they are asked about the attitude (Ibid, p. 450).

Growth Need Strength as a Moderator of the Job Characteristics/Personal & Work Outcomes Relationship

Of all of the studies relating to the Job Characteristics Model reviewed, the dimensionality of job characteristics and the moderating effects of Growth Need Strength were the most frequently studied. The review was refined to concentrate on 31 studies which specifically examined the moderating effects of Growth Need Strength on various relationships within the work setting and which were undertaken during the period 1976 to 2001 (See Appendix A).

Hackman & Oldham's measure of job characteristics - the Job Diagnostic Survey (JDS) was used in most of the 31 cases but some employed other measures of the job scope

dimension. For example Sims & Szilagyi (1976) used a measure taken from the Job Characteristics Inventory (JCI) which includes two items – ‘dealing with others’ and ‘friendship opportunities’ not found in the JDS as did Orpen (1979), Griffin (1982) and Birnbaum et al (1986). Outcome measures in the foregoing studies included job satisfaction, internal motivation, productivity, absenteeism, and turnover. In most cases the job scope - outcomes relationship included **mediation** by the three critical psychological states identified by Hackman & Oldham and in most, proposed **moderators** (though not always GNS) of the job characteristics-work outcomes relationship were tested. For example, Kahn & Robertson (1992) used training & experience as an additional moderator of the relationship.

In 16 studies (Oldham, 1976; Oldham et al, 1976; Hackman et al, 1978; Abdel-Halim, 1979; Evans et al, 1979; Arnold & House, 1980; Champoux, 1980 (study 3); Griffin, 1982; Lee et al, 1983; Loher et al, 1985; Graen et al, 1986; Mecof, 1991; Das, 1991; Johns et al, 1992; Lee-Ross, 1998; de Jong et al, 2001) support or partial support was found for the moderating role of growth need strength. However, in some cases the ‘job choice’ format for GNS was used, in others the ‘would like’ format and in others still, the format was not specified. In only 13 of the 31 cases was the subject base over 200, the remainder used relatively small samples of employees; indeed Wall et al (1978) used a sample of only 47 factory workers and found no support for the moderating role of GNS. In fact, Wall and his colleagues state ‘*The Job Characteristics Model as currently formulated is not fully consistent with either the findings upon which it was developed or those of the present study. It cannot therefore be regarded as valid.*’ (p. 194). In spite of this they do state that the JCM still ‘seems useful for job redesign purposes’ and their use of path analysis on the whole of the JCM shows some relationships which were not brought to light by the original study. Their analysis showed that some of the core job characteristics related directly to the outcomes and to the critical psychological states, other relationships specified by the original model were found not to exist at all.

Tiegs et al (1992) raised the issue of sample size when they examined the moderating effects of GNS and suggested that the results obtained in the studies that used smaller samples were subject to sampling error. In association with this they also found that a

number of studies (e.g. Abdel-Halim, 1979; Arnold & House, 1980; Bottger & Chew, 1986; Oldham, 1976; Oldham et al, 1976; Orpen, 1979; Pokorney et al, 1980) used data from incumbents of a restricted range of jobs which they say calls into question the reliability and dimensionality of the more common self-report job characteristics measures, which this use of homogeneous samples brings.

Fried & Ferris (1987) conducted an extensive review and meta-analysis of the validity of the JCM and concluded that one of the most important results of their study was that it indicated the importance of the effects of individual or situational differences on the relationship between job characteristics and work performance. They also found some evidence that GNS moderates the relationship between job characteristics and employee performance.

Stone, Mowday and Porter (1977) suggest that GNS is related to need for achievement and need for autonomy which they describe as two personality traits. They found that need for achievement moderated the perceived job scope-job satisfaction relationship but that need for autonomy did not. Job scope is defined by these authors as a combination of job variety, autonomy, task identity and feedback scores on a 13-item instrument developed by Stone (1976). Stone et al (1977) suggest that in looking at the relationship between job characteristics and job satisfaction the most consistent variable is higher-order need strength (this includes the strength of an individual's need for personal growth, accomplishment and achievement), but they report that there is no empirical evidence for a relationship between need strength as measured by such instruments as the JDS and such needs assessed by personality measures e.g. Jackson's (1967) Personality Research Form. Roberts & Glick (1981) included the above study in their critical review of the job characteristics approach to task design and suggest that future research should compare not only the psychometric properties of sets of needs but those of values and personality traits as alternative moderator variables.

Individual and Situational Differences

It is individuals' perceptions of what attributes or characteristics a job holds for them which form the basis of the JCM. Aldag, Barr & Brief (1981) state that an individual's perceptions of job characteristics or task dimensions have been found to be associated

with motivation, satisfaction, performance, absenteeism, expectancy perceptions, job involvement and alienation - to name but a few. In 1974 Wanous examined individual differences as moderators of the relationship between the presence of certain job characteristics and individual reactions to those differences. The three approaches he used were a) urban versus rural background b) strong versus weak belief in the protestant Work Ethic and c) high versus low strength of higher order needs. His results, which were based on job satisfaction outcomes, showed that higher order need strength was the most useful way to measure these individual differences and suggested that such differences arose as a result of a developmental sequence. Wanous (1974) suggests that early socialisation leads to a *value* orientation towards work and says that this '*emphasises the importance of work in one's total self-esteem and reinforces the belief that work can hold intrinsic satisfaction*' (p. 621). Wanous found that higher-order need strength was the 'best' moderator, followed by Protestant Work Ethic and the urban/rural difference. Sims & Szilagyi (1976) compared higher order needs with locus of control as moderators of the relationship under discussion and also added occupational levels to the equation. They used 'self-actualisation need strength' as an individual moderator, which they said, refers to the '*extent to which the individual values the importance of higher-level work outcomes*' (p. 218). This was found to be a statistically significant moderator of many of the relationships they tested.

In a much earlier paper by Lawler & Hall (1970) it is suggested that people do differ in the extent they are likely to become involved in their jobs because of their background and personal situation. However, they further suggest that there is no reason to suppose that just because a person's job is important to him/her that she/he will be motivated to perform well. This they say is because people *value* different aspects of their jobs more than others and thus pay, status or personal relationships may be, to some, the more important parts of the jobs and are not necessarily related to performance.

Two distinct areas are clear from the research into individual and situational moderators of the job scope-employee response relationship beginning with Hulin & Blood's (1968) study into alienation of the worker from middle class work-related values and norms and the protestant Work Ethic (discussed in the next section) as moderators of the job size-job satisfaction relationship. Secondly, the research into individual higher-

order needs which has been discussed at length thus far. All in all, however, research into individual difference moderators of work effects (Robey, 1974, Work Values; Wanous, 1974, Higher Order Needs Satisfaction, Protestant Work Ethic, Sub cultural Background; Hackman & Oldham, 1975, Growth Need Strength; Sims & Szilagyi, 1976, Growth Need Strength, Locus of Control) indicates that there is evidence in support of individual higher order need satisfaction as a more effective moderator of the relationships between jobs & outcomes, than work values or sub cultural background.

Work Values

Blood (1969) suggests that the way a person evaluates work in general should be related to his/her specific job. In order to find support for this hypothesis he devised an 8 item scale intended to measure the individual's amount of agreement with the protestant Ethic, which is described in Blood's study as a person feeling that '*Personal worth results only from self-sacrificing work or occupational achievement..*' (p. 456). Blood proposed that the amount of agreement with the Protestant Ethic would moderate the relationship between a person's job and the level of satisfaction they experienced with both the job in general and with life in general. His study of 448 airmen and non-commissioned officers from the United States Air Force found support for his hypothesis and concluded with his assumption that work values both precede and influence job satisfaction.

Wanous (1974) found support for the Protestant Ethic as a moderator of the job characteristic/job satisfaction relationship but concluded that it was not 'as good as' higher order need strength in that role. Stone (1976) tested the moderating effect of work-related values as measured by a Protestant Ethic scale on the job scope-job satisfaction relationship and found no support for the Protestant Ethic as a moderator and concluded that the Protestant Ethic was probably not an important individual difference factor to consider in the relationship between satisfaction with work and job scope. In a much later study Orpen and Fishendon (1993) tested the relationship between job enrichment and worker attitudes using work values and enrichment needs as moderators. They too used the Protestant Ethic scale and work values were found not to moderate the relationship. Robey (1974) tested the relationship between task design and worker response with work values as a moderator of this, in a laboratory experiment

where 60 participants carried out two routine decision tasks. He used Friedlander's (1965) measure of work values which included items concerned with specific aspects of the job such as 'receiving recognition for a job well done' and 'opportunities for promotion'. Robey (1974) found support for his hypothesis that the interaction between job content and work values affected job satisfaction. More recent studies into work values and job satisfaction (Ushasree & Chandariah, 1990; Drummond & Stoddard, 1991) have used Super's (1970) Work Values Inventory, neither finding strong support for work values as a moderator of the job content-job satisfaction relationship using this measure. Cook et al (1981) in their compendium and review of measures and their use suggest that although Super's measure is relatively easy to use there is little evidence of positive findings using full time employees, as in the majority of cases student samples were used.

Knoop (1994b) studied the relationship between importance and achievement of work values and job satisfaction using Elizur's (1984) measure of work values. Knoop found that both important and achieved values as measured by this scale related significantly to satisfaction. The purpose of Elizur's and Elizur et al's (1984; 1991) work was to define the structure of work values using a facet-theory approach and was based on data collected from 2280 respondents in 8 countries. Their 24 item Work Values Questionnaire (WVQ) was designed to represent various aspects of work values which were divided into what they call 'modalities of outcome' of which there are three - **instrumental**, which includes areas such as pay and working hours; **affective**, which includes areas such as esteem and co-workers and **cognitive**, which includes areas such as advancement and use of abilities. In describing their questionnaire Elizur et al (1991) suggest that it represents the major perspectives outlined by the basic theories of motivation including need theories (Maslow, 1954; Alderfer, 1972; McClelland, 1961; Herzberg, 1974 and Hackman & Oldham, 1976) and the items do appear to do so satisfactorily. Their results showed a similar structure in the samples from each country except that there were cultural differences in the reported relative importance of some of the work values. For example, the China sample rated contribution to society as very important but the more material outcomes of work such as pay, less so. Knoop (1994) suggests that work values '*denote the degree of worth, importance and desirability of what happens at work*' (p. 683). He further suggests that work values represent learned

qualities which people either want or desire in a job. Throughout the literature reviewed reporting studies into work values, whilst there appears to be little argument over the basic definition of work values, the items representing alleged desired or valued work outcomes vary considerably. The protestant Work Ethic largely represented the cultural attitudes towards work of the time, suggesting that if a person worked hard she/he could not fail to find satisfaction, both with the job and with life in general. Later theories of work values such as that of Elizur et al above, reflect the needs of individuals within the world of work, within organisational climates, which have a greater tendency towards change and development.

Dose (1997) proposes a comprehensive definition of work values and a structural framework, which she says *'reflects the central elements of the construct and reduces confusion over its conceptual boundaries'* (p.219). Dose's definition is that *'work values are evaluative standards relating to work or the work environment by which individual's discern what is 'right' or assess the importance of preferences.'* (pp.227-228). Dose's structural framework is in the form of a two-by-two dimensional structure with one continuum ranging from 'moral' to 'preference' work values and the other from 'personal' to 'social consensus' work values. Each of the major theories of work values and business ethics sits within this framework, which draws them together acknowledging relationships between them. However, the measurement of work values within such an integrated structure is an issue not dealt with by Dose.

General Values

Rokeach (1973) defined a value as *'an enduring belief that a specific mode of conduct or end state of existence is personally or socially preferable to an opposite or converse mode of conduct or end-state of existence'* (p.5). Rokeach devised lists of values which were either related to behaviour (instrumental values) or desirable goals (terminal values) which he called The Value Survey. He then related values to other variables - race, nationality and age. Following Rokeach, theorists have related values to personality (Furnham, 1984; Luk & Bond, 1993; Bilsky and Schwarz, 1994, Roccas et al, 2002), culture and gender (Bond, 1988). Furnham's (1984) results indicated differences in the value systems of people who differed in levels of neuroticism and extraversion as measured by the Eysenck Personality Questionnaire (Eysenck and

Eysenck, 1975) and concluded with the suggestion that '*people value what they most want but least have*' (p.485). Luk and Bond (1993) suggest that people have different needs according to their personality, which define what is desirable and should be valued, and found that with the exception of neuroticism, personality and values endorsement are linked through peoples' needs. Using a version of the Rokeach Values Survey (RVS) adapted for the Chinese culture, Bond (1988) perhaps understandably, found cultural differences in values which were not evident in the RVS. Roccas et al (2002) discuss the links between values and the five-factor model of personality but draw distinctions between the two. Values, they say, refer to what people consider to be important and the goals they wish to pursue, whereas traits are enduring dispositions. People believe their values are desirable, whereas traits can be either positive or negative (ibid., p. 790). Roccas et al (2002) found that the value self-direction was related to the personality domain Openness to Experience and in particular to the sub-scale of that domain representing the facet 'Ideas'.

Some of the most comprehensive studies into the nature of values have been carried out by Schwartz and colleagues over the last decade (Schwartz and Bilsky, 1987; Schwartz, 1990; 1992; 1994a; Schwartz and Sagiv, 1995; Schwartz, Verkasalo, Antonovsky and Sagiv, 1997). Schwartz and Bilsky (1987) use a 'mapping sentence' to formally define values which specifies three facets - goal type, interests served, motivational domain and is given below

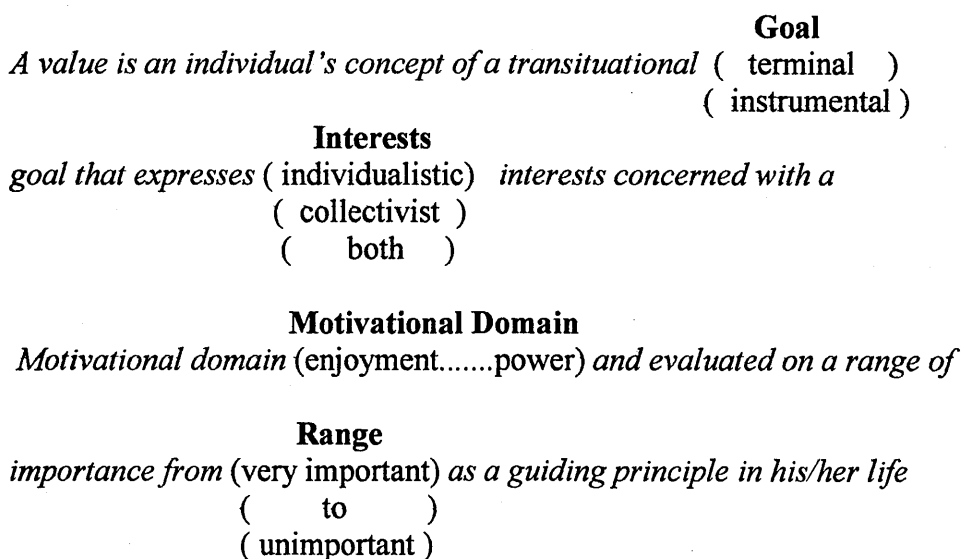


Figure 2. Schwartz and Bilsky (1987, p. 553)

Schwartz's studies over a period of 10 years have resulted in the identification of 10 motivational types of values (originally called motivational domains) - power, achievement, hedonism, stimulation, self-direction, universalism, benevolence, tradition, security and conformity. Recent research (Menezes and Campos, 1997) using a Portuguese sample of 600 subjects continues to find support for Schwartz's theory. Schwartz's extensive research indicates that both the meaning of the individual values and the presence of the motivational types are evident across several cultures however in his 1992 paper Schwartz puts forward two different theories about the formation of values. He first suggests that '*value structures probably evolve over time as social conditions are transformed*' (p. 47) and later suggests that individuals with particular value structures will be attracted to value priorities depending on their needs and goals, implying that values are both situationally and individually constructed.

Stern, Dietz and Guagnano (1998) presented a brief inventory of values based on Schwartz's instrument. This instrument consists of what these authors describe as 'the major clusters' in Schwartz's theory called self-transcendence, self-enhancement, openness to change and conservation (or traditional) values and was used to study peoples attitudes and actions towards the environment. Stern et al found the Schwartz measure to be robust but suggest that their shorter version is more appropriate for carrying out research using larger populations where the use of the full 56-item measure might be impractical.

Dollinger et al (1996) examined personality traits and values and suggest that both of these are '*broad categories of individual differences important to the study of persons that are, by definition assumed to be cross-situationally and cross-temporally consistent*' (p. 24). Their evidence for this statement comes in part from Rokeach (1973) who defines the nature of values as being 'enduring' beliefs, which determine action and because of this Dollinger et al suggest that values can be expected to converge with personality traits.

Personality

Atkinson et al (1990) describe personality as '*The characteristic patterns of thought, emotion and behaviour that define an individual's personal style and influence his or*

her interactions with the environment’ and suggest that there is an *‘interplay of forces that shape an individual’s personality through the course of life; promote, disrupt or transform the continuity of personality across time; and enable the individual’s personality, in turn, to affect the life course itself.’* (p.475). The study of personality is thought to be the oldest in the field of psychology. Early work proposed that individuals could be categorised into distinct types that are qualitatively different from each other (Atkinson et al, 1990). The earliest such work was by Hippocrates in 400BC, who suggested that there were four basic personality types associated with biological functions. In this theory, an excess of black bile produces the melancholic (depressed) type; an excess of yellow bile produces the choleric (irritable) type; blood produces the sanguine (optimistic) type and phlegm produces the phlegmatic (calm, solid) type.

Later a successor of Aristotle, Theophrates, proposed a set of 30 personality types. Each type was described by its dominant features and a set of behaviours expected to be typical of that type. Atkinson et al (1990) suggest that throughout history people have related body type to personality with popular stereotypes such as *‘fat people are jolly’* (p. 504). Sheldon (1940) reported a relationship between physique and temperament and proposed three “types” - endomorphic, being ‘soft and round’ in physique and having a relaxed, sociable temperament; mesomorphic, being ‘muscular and athletic’ in physique and having an energetic, assertive and courageous temperament; and ectomorphic, being ‘tall and thin’ in physique and having a retrained, fearful, introverted and artistic temperament.

In the late 18th and early 19th centuries Freud proposed his psychoanalytic theory of personality. Here personality is composed of three interrelated systems - id, ego and superego, and develops as a sequence of three stages (the oral, the anal and the phallic) which are completed by approximately 5 years of age. This marked a change in the conceptualisation of personality itself as Freud suggested that it was possessed of constituent parts, which had a subconscious effect upon each other.

Allport (1937) is widely credited as being the first person to propose a grand theory of personality and his work firmly placed personality psychology in the field of social sciences research. Allport’s theory was based on the premise that personality

psychology was the study of the individual person and his definition of personality, given below, is still used in contemporary psychological research '*The dynamic organisation within the individual of those psychopathological systems that determine his (sic) unique adjustments to his environment*'. Allport (1937). In 1961 Allport revised this definition replacing '*unique adjustments to his environment*' with '*characteristic behaviour and thought*'. Allport's theory is a single-trait theory which is one which is concerned with '*the role played by one particular part of the personality structure in the determination of behaviour*' Hampson (1988 p. 16). By the beginning of the 1950s personality psychology had evolved into the multi-trait or factor theories which form the basis of current research. Of these, Cattell (1950) defined personality as '*that which permits a prediction of what a person will do in a given situation*' (p.2). This meant the collection and measurement of a number of variables which could be grouped into 'traits' which Cattell saw as mental structures which may be inferred from observable behaviour.

Whereas Cattell proposed a minimum of sixteen traits which were necessary for describing the structure of personality, Eysenck (1953) initially proposed two factors which he felt were adequate, introversion-extraversion and neuroticism-stability. Later, Eysenck and Eysenck (1985) proposed the addition of a third factor, psychoticism-normality. Eysenck's theory varied from that of Cattell in that the former saw personality as being made up of factors which were present in individuals along dimensions, thus a person could be located at any point along each of the dimensions.

Cattell and Eysenck, as 'state-trait' theorists share the view that '*the interactive influence of traits and situations produces transient internal conditions known as "states"*'. Eysenck and Eysenck (1980, p. 191). However, Mischel (1968) argued that personality might be a function of the situation in which the behaviour occurs. Ickes et al (1997) suggest that common to the assumptions of Mischel and other researchers who wrote in support of personality as a state (for example Block, 1968; Magnusson, 1990), is the notion that individuals can choose to be in certain situations and choose to avoid others. They elaborate on this by suggesting that once in their chosen situations, individuals' words and actions genuinely reflect their personalities. They say '*..the fact*

that they display these behaviours in settings they have specifically chosen ensures a substantial degree of consistency in their behaviour.' Ickes et al (1997, p.166).

Some current thinking is that personality is divided into traits and such research is now concerned with the examination of the precise nature and number of traits which form the structure of human personality. The most influential work in this area is the 'Big Five' trait taxonomy which is based on studies by Fiske (1949), Norman (1963) and Tupes & Christal (1961) and there is substantial evidence (Digman, 1990; Goldberg, 1981; John, 1990; McCrae & John, 1992; McCrae & Costa, 1987; Ostendorf & Angleiter, 1994) that personality trait dimensions can indeed be reduced to five basic bipolar categories. McCrae & Costa (1987) identify the five factors as 1) extraversion-introversion (E) 2) neuroticism (N) 3) openness to experience (O) 4) agreeableness-antagonism (A) and 5) conscientiousness-undirectedness (C).

John (1990) suggests that during the 1980s researchers from many different traditions concluded that the above factors were fundamental to the structure of personality and have been found using various types of measurement, in natural languages and across the spectrum of age, gender and culture. McCrae and Costa (1985 p. 710) continued the premise made by earlier writers (Allport & Odbert, 1936; Cattell, 1946; Tupes & Christal, 1961; Norman, 1963) that natural languages such as English would have 'evolved terms' for all fundamental individual differences. The theory behind this is that a thorough analysis of language would provide a model of personality traits and there appears to be a general consensus, using this lexical approach, regarding the nature of the first four factors - extroversion, agreeableness, conscientiousness and neuroticism. The fifth factor, according to Barrick & Mount (1991) has been the most difficult to identify and has been variously interpreted, they say, as Intellect or Intellectence (Digman & Takemoto-Chock, 1981), Culture (Norman, 1963) and Openness to Experience (McCrae & Costa, 1985). Digman (1997) compared 14 studies supporting the five factor model and suggest that two higher-order factors emerged which reflect the 'hierarchical ordering of personality constructs' (p. 1248). The second of these factors is of most interest to the current research as it represents what Digman (1997) describes as a very broad concept: Personal growth versus personal constriction. Digman suggests that evidence for the existence of this factor can be seen in the

Extraversion and Intellect factors of the Big Five and that such evidence is characteristic of the work of personal growth theorists such as Rogers and Maslow, who emphasised the individual's tendency to strive to experience all that he or she can. McCrae & Costa (1987) however, draw the distinction between intelligence and Openness to Experience *'Intelligence may in some degree predispose the individuals to openness or openness may help to develop intelligence, but the two seem best construed as separate dimensions of individual differences.'* (1987, p.88).

Openness to Experience

The earliest reference to Openness to Experience is made by Fitzgerald (1966) who used a modification of Schachtel's (1959) concept 'openness to experience' in the development of a scale known as the Experience Enquiry. Fitzgerald used this scale in a study of regression in the service of the ego and describes Schachtel's concept as meaning *'a loosening of fixed anticipations and sets so that one approaches the objects of his experience in different ways and from different angles'* Fitzgerald (1966 p. 656). Fitzgerald goes on to suggest *'Originality, the ability to shift from more to less regulated thinking with facility, and the disposition to greet novel and unusual experiences without undue anxiety and without repression and with strength of ego are, indeed, characteristics of the person who is truly open to experience.'* (ibid., p. 656). Coan (1972) devised a questionnaire to measure Openness to Experience called the Experience Inventory and suggests that people vary considerably in the range and types of experience to which they are open (p.346). Coan believed that people could be very open to one kind of experience but very closed to another and so measured openness in a variety of different areas such as memory, ideas, impulses and feeling states. As a result of his study of 383 college students, Coan identified seven factors - a) aesthetic sensitivity versus insensitivity b) unusual perceptions and associations c) openness to theoretical or hypothetical ideas d) constrictive utilisation of fantasy and dreams e) openness to unconventional views of reality versus adherence to mundane, material reality f) indulgence in fantasy versus avoidance of fantasy and g) deliberate and systematic thought.

McCrae and Costa (1985) used Coan's methods to conduct their research into the identification of Openness to Experience. McCrae (1993-94) gives a definition of this

factor which comes from his work with Costa published in the handbook of Personality Psychology (1997). *'Openness is seen in the breadth, depth and permeability of consciousness and in the recurrent need to enlarge and examine experience'* (p.826). McCrae (1994) discusses this fifth factor of personality and concludes that whilst lexical studies interpret this factor as intellect, his own studies (and his work with Costa, 1985; 1985a; 1985b) which were carried out using questionnaires, define the construct as Openness to Experience. This, he says, is a broader construct encompassing more of the characteristics which appear to be associated with the fifth factor. McCrae & Costa (1987) suggest that Openness to Experience is best characterised (according to their adjective-factor results) by 'original', 'imaginative', 'broad interests' and 'daring' but that questionnaires may be better than adjectives as a basis for interpretation as they use phrases rather than single words.

McCrae (1990) continues this line of thinking by suggesting that his conception of an open individual as being *'interested in experience for its own sake, eager for variety, tolerant of uncertainty, leading a richer, more complex, less conventional life'* (p.123) is not easily described by single adjectives as there is a paucity of such words in the natural languages. McCrae & Costa's measure of Openness to Experience in the revised version of their 'Neuroticism, Extraversion, Openness - Personality Inventory' (NEO-PI (R)) (1992a) consists of forty eight items in the form of sentences relating to six areas to which individuals may be 'open' or 'closed'. These areas are fantasy, aesthetics, feelings, actions, ideas and values.

Although Eysenck (1992) argues that openness is not a unique dimension of personality and that McCrae & Costa's psychometric evidence is not enough to prove that it is, other researchers (Barrick & Mount, 1991; Bergemen et al, 1993; Dollinger & Orf, 1991; Saucier, 1992; Trapnell, 1994) have found substantial support for the five factor theory of personality with Openness to Experience as Factor Five. Barrick & Mount conclude their 1991 research on performance criteria by saying *'We believe that the robustness of the five factor model provides a meaningful framework for formulating and testing hypotheses relating individual differences in personality to a wide range of criteria in personnel psychology...'* (p.23). These authors examined the relationship between training proficiency and openness to experience and suggested that openness to

experience was expected to correlate with this criterion because '*Openness to experience appears to assess individuals' readiness to participate in learning experiences*'. (ibid. p.7). Their results supported this prediction.

de Jong et al (2001) examined the moderating effects of both GNS and Openness to Experience on the job characteristics-job satisfaction relationship and found a substantial relationship between GNS and Openness. These authors suggest that the relationship between GNS and job satisfaction is actually accounted for by the 'broad personality factor Openness to experience' (p.355). The current research will investigate the possibility that the origins of growth need strength either lie in or are closely linked to the personality domain 'Openness to Experience' and because of this it may too, predict individuals within the world of work who are ready to take up learning opportunities.

Summary of Aims

The research conducted for this thesis has a single overriding aim. This is to attempt to define the individual difference factor Growth Need Strength

The literature review so far indicates that the nature of GNS has been poorly defined, with the majority of researchers accepting GNS 'at face value' without examining its precise properties. Whilst this does not diminish the validity of past research, it is suggested that an in depth examination of the nature of GNS may make it possible to isolate the specific individual difference which is at the root of this need for personal growth. This in turn may make it possible to identify those individuals who are likely to respond more positively to the factors in the Job Characteristics Model in terms of what it is about people which prompts certain work behaviour, thus expanding the Model and increasing its generalisability.

The next four chapters of this thesis describe the studies which were carried out to investigate the nature of Growth Need Strength. Chapters 2 and 3 report on the first two studies undertaken to partially replicate the work of Hackman and Oldham (1976) and examine the relationships between GNS, Job Characteristics, Job Satisfaction, Personality, Values and Work Values. Study 2, reported on in Chapter 4, explores GNS

further through methods of data collection and analysis. Chapter 5, reports the final study, which proposes and tests a model for the relationship between GNS and Training Activity. The final chapter discusses the findings of all of the studies and their implications for future research.

CHAPTER 2

Introduction

Hackman and Oldham (1975) describe GNS as '*a malleable individual difference characteristics which is predicted to influence how positively an employee will respond to a job with objectively high motivating potential*' (p. 163) and as the literature review progressed a pattern began to emerge of GNS being related to values, beliefs, personality and need for achievement. The literature review was thus extended to include theories of personality, general values and work values. Other individual difference factors, which had influenced the work of Hackman and Oldham, such as Higher Order Need Strength and Self Actualisation Need Strength, were also examined. The results of the review indicated that elements from a person's cultural/social background, personality, education and intelligence might combine to form that person's values. The values held may be a reflection of an individual's level of growth need strength and may in turn, determine the level of personal and work outcomes (internal work motivation, quality of work performance, satisfaction with the work, absenteeism and turnover). It was therefore decided to test the relationship between job characteristics and job satisfaction which had been tested in the study by Hackman and Oldham (1976), using their measure of GNS 'job choice' from The Job Diagnostic Survey (JDS) as a moderator of this relationship. Roberts and Glick (1981) suggested that future research into the job characteristic approach to task design should compare both the psychometric properties of sets of needs and of values and personality traits as alternative moderator variables. Thus, in an attempt to answer the question 'What is Growth Need Strength?' this chapter examines the psychometric properties of GNS and other related concepts, specifically personality, general values, work values, higher order need strength and self-actualisation need strength, in an attempt to identify any relationships which might be present amongst them and to provide some evidence for the discriminant validation of GNS. Measures for all of these constructs were included in the questionnaire (see Appendix B).

The logic of the study is that, if Hackman and Oldham (1976) are correct, there will be a positive correlation between job characteristics and job satisfaction which is moderated by GNS (*Hyp. 1*). However this factor has thus far been poorly defined and other factors have been found to moderate the same relationship. An examination of the correlates of

growth need strength and of the moderating effect of other variables which might be related to it, may shed some light on what it actually is or, equally important, what it is not. Thus it would be expected that if these other variables perform in the same way as GNS, there will be a positive correlations between job characteristics and job satisfaction which is moderated by the personality factor Openness to Experience (*Hyp. 2*); the values domain Achievement (*Hyp. 3*); the values domain Self-direction (*Hyp. 4*); and the values domain Stimulation (*Hyp. 5*). The main measures selected for the questionnaire were as follows:

Growth Need Strength, Higher Order Need Strength and Self Actualisation Need Strength

Hackman and Oldham (1974a) suggest a measure for the individual difference factor that they term Growth Need Strength. The measure is in two parts with wording of the items in the first part closely resembling those of the measures for Higher Order Need Strength and Self Actualisation Need Strength. The second part, which was used in their examination of the moderating effect of GNS, has a different format. It was decided to use all three measures in the current study in order to assess their similarities and differences. Each of these measures, along with those for personality, general values and work values, is described in detail later in this chapter.

Personality

Costa and McCrae (1992a) suggest a five-factor model of personality which consists of the domains neuroticism, extraversion, openness to experience, agreeableness and conscientiousness. Whilst considerable debate has taken place about the nature of the factor 'Openness to Experience' it was felt that this particular factor might relate strongly to GNS. Questionnaire items which tapped individual desires for the opportunity to learn new things and to develop their own special abilities, and a dislike for work which was boring and monotonous seemed to relate to the terms used in the literature to describe people who are 'open to experience'. Barrick and Mount (1991) hypothesised that Openness to Experience would be a valid predictor of the performance criterion 'training proficiency' because it '*assesses personal characteristics such as curious, broad-minded, cultured and intelligent, which are attributes associated with positive attitudes towards learning experiences*' (p. 6). They

predicted that such individuals would be more likely to be motivated to learn on a training programme and were therefore more likely to benefit from the training. The hypothesis was supported by their data and as a result Barrick and Mount suggest that measures of Openness to Experience may help to identify those individuals who are '*most willing to engage in learning experiences*' (p. 19). Thus, the current research will attempt to relate GNS to the personality trait Openness to Experience with the specific prediction that individuals who show a high level of GNS will also score highly on Openness to Experience. This research will also ask whether Growth Need Strength is the same as Openness to Experience and thus whether Growth Need Strength is in fact a personality trait of the 'Big Five'. If a relationship is found between GNS and Openness to Experience then it is intended in a later study, to assess whether GNS may also help identify individuals who take up learning opportunities in the workplace more readily than others. In this case there would be a positive correlation between Growth Need Strength and Openness to Experience (*Hyp. 6*).

General Values

The work of Shalom Schwartz (1992) suggests a structure of human values which divides them into 10 domains - universalism, benevolence, conformity, tradition, security, power, achievement, hedonism, stimulation and self-direction. Of these 10 domains, achievement, stimulation and self-direction might be expected to have a relationship to GNS. Some of the values included in the achievement domain are ambitious, successful, capable and intelligent, those in the stimulation domain - exciting life, varied life and daring and those in the self-direction domain choosing own goals, independent, curious and freedom. Rim (1970) is credited with stimulating research interest into the link between individual differences and values and found differences in the ranking of values between participants high or low in dogmatism, authoritarianism, Machiavellianism and intolerance of ambiguity. Later work found links between extraversion and neuroticism (Furnham, 1984); agreeableness (Luk and Bond, 1992); openness to experience (Yik and Tang, 1995); personality and the variance in work performance (Furnham, Forde and Ferrari, 1998); extraversion, agreeableness and neuroticism (Roberts and Robins, 2000) and all five of the Big Five personality factors (Roccas, Sagiv, Schwartz and Knafo, 2002). In the latter research Extraversion was found to correlate positively with achievement (.31) and Openness to Experience was

found to correlate positively with self-direction (.48) and stimulation (.33). Thus, the current research will attempt to relate the individual difference factor Growth Need Strength to the value domains achievement, stimulation and self-direction with the specific prediction that individuals who show a high level of GNS will score highly on the values domains achievement, stimulation and self-direction. Here the question is whether value preferences, which have also been found to correlate positively with personality traits, reflect individual GNS. If so it would be expected that there would be positive correlations between Growth Need Strength and the general values Achievement, Self-direction and Stimulation (*Hyp. 7*).

Work Values

The concept of work values has been poorly defined. However, Elizur et al (1991) have devised a Work Values Questionnaire and hypotheses regarding the structure of relationships among components of work values. The questionnaire contains 24 items which are divided into 3 work value domains - instrumental, which includes aspects of work such as pay and hours of work; affective, which includes recognition and esteem and cognitive work values which include meaningful work, personal growth, advancement and achievement. The review of the research on Work Values shows that past studies into the moderating effect of the Protestant Work Ethic (for example Blood, 1968; Wanous, 1974; Stone, 1976; Orpen and Fishendon, 1993) had found that work values measured in this way influence job satisfaction. Other studies (for example Ushasree and Chandariah, 1990; Drummond and Stottard, 1991) had used Super's (1970) Work Values Inventory and did not find strong support for work values as a moderator of the job content/job satisfaction relationship using that measure. Thus it was decided to use Elizur et al's measure above as the items included appeared to represent more effectively contemporary values of individuals within the world of work.

No specific prediction is made about the relationship between work values and GNS but it might be expected that individuals high in GNS might also score highly in the cognitive domain of work values. The reason that no specific prediction is being made here, is that whilst individuals may have strong needs for personal growth it may not always be possible to satisfy these needs as the need to, say, earn a good rate of pay and to have convenient working hours may overshadow need for growth. On the other hand,

the latter ‘needs’ could be regarded as functional rather than as ‘values’. Thus, as Maslow (1943) suggested, they would be somewhat necessary in order to realise values rather than values themselves. It would be expected, then, that there will be a relationship between Growth Need Strength and the three Work Values domains (*Hyp. 8*).

Study 1

Research Design

Study 1 aimed to see whether a measure of Growth Need Strength behaved in a way consistent with theoretical prediction and whether it was related to theoretically similar constructs. This was done in two ways, firstly by replication of the original research by Hackman and Oldham (1976), to ascertain whether support could be found for the moderating effect of GNS. The original study had tested the moderating effect of GNS on the relationship between job characteristics and work outcomes in two places. One was between the job characteristics and critical psychological states and the other was between the critical psychological states and work outcomes. Hackman and Oldham (1976) also reported on the correlations computed directly between the overall motivating potential of the job and the outcome variables, which they say was ‘*in effect bridging the mediating function of the critical psychological states*’ (p.271). The current research proposed that if GNS was a distinct individual difference factor it should moderate the relationship between job characteristics and job satisfaction regardless of how they were measured i.e. it should be possible to take GNS out of the Job Characteristics Model and set it in a different measurement context. Secondly, an attempt was made to define GNS by comparing its correlates with those of the personality, values and work values measures described in the previous section. If GNS was the ‘same as’ any of the domains measured by the latter instruments then factor analysis ought to reveal this.

Method

The Questionnaire (see Appendix B)

A questionnaire was designed to measure the following:

1. Job Characteristics
2. The Importance of Job Characteristics
3. Job Satisfaction

4. Growth Need Strength
5. Higher Order Need Strength
6. Self Actualisation Need Strength
7. Personality
8. General Values
9. Work Values
10. Age, Educational Attainment and Gender
11. Whether respondents had ever undertaken a course of study outside their working hours and whether this was mainly for career purposes or for pleasure

A description of each measurement tool is given below.

Job Characteristics

Section 5 of the questionnaire asks respondents to indicate how much of each of the five job characteristics – skill variety, task identity, task significance, autonomy and feedback from the job itself, they thought was present in their jobs. The items were modified from those in the Job Diagnostic Survey to Anglicise the American the wording and to reduce it to the minimum that was felt necessary to enable respondents to answer effectively. Warr, Cook and Wall (1979) conducted a study using the Job Characteristics Model and one of their conclusions was that there was a need for short, robust scales which could be easily completed by respondents of a variety of educational attainments. In view of the comments of the above authors it was felt that shortened versions of questions regarding job characteristics would be adequate. The questionnaires would be completed by individuals working in a variety of jobs, in different organisations within organisational climates which had changed significantly from the era of the original study. The shorter versions were thus devised to enable all participants to complete the questionnaire as quickly and as accurately as possible in circumstances where they may be reluctant to have further burdens placed upon them. Responses were obtained on a seven-point scale running from ‘very little’ (of the characteristic present in the job) to ‘very much’.

As stated in the previous chapter Hackman and Oldham (1975) developed a formula for combining the core job dimensions. This formula is given on the following page:

Motivating Potential Score (MPS) =

$$\left[\frac{\text{Skill Variety} + \text{Task Identity} + \text{Task Significance}}{3} \right] \times (\text{Autonomy}) \times (\text{Feedback}).$$

According to the above formula, autonomy and feedback are the most important factors in the MPS because if either is zero then the MPS is zero. However, skill variety, task identity and/or task significance can still be low or zero and the MPS can be relatively high, if autonomy and feedback are high. Although Hackman and Oldham (1976) used the above formulation as it derived directly from the propositions of the Job Characteristics Model (see page 9), they nonetheless highlight the importance of comparing the empirical performance of the MPS with simpler alternative models. Their particular concern was the presence of the two multiplicative terms in the formula which can compound the effects of measurement unreliability. They developed five different models for combining the job dimensions and found that the results did not vary significantly. In the current research only three of these models were used – the MPS as above, a simple additive version and a full multiplicative version.

Importance of Job Characteristics

Research into job design/redesign was based on not only how much of each characteristic employees perceived was present in their jobs, but also on what employees attitudes were to their jobs. Measures such as those for Higher Order Need Strength and Self-actualisation Need Strength (see below) ask individuals how important different aspects of their work are to them and so the importance of the individual job characteristics was included as a measure in this questionnaire. Importance, however, could be described as a meta-attitude which is defined by Bassili, 1996 as an impression of 'properties of one's attitudes' (p. 637) and as such may not possess the predictive validity of measures which call for a direct assessment of the properties of a job, by the job holder. Nonetheless, it was decided to include 'importance' in the questionnaire to assess it's predictive validity and thus after each question about the amount of each characteristic respondents felt was present in their jobs, they were asked how important this was to them. It is acknowledged that there may be some ambiguity regarding this measure but it was hoped that by asking about

importance immediately after respondents had made an assessment of each characteristic, they would be responding to the importance of the *level* of each characteristic to them. Responses were obtained on a seven-point scale running from 'not at all important' to 'extremely important'. A mean score for each individual was then calculated to provide a scale score.

Growth Need Strength

The two measures of growth need strength were taken from the Job Diagnostic Survey (Hackman and Oldham, 1976) and are reproduced in Sections 1 and 2 of the questionnaire. The two scales are stated by Hackman and Oldham (1974a) to tap an individual difference among workers which they describe as follows '*The strength of the respondent's desire to obtain 'growth satisfactions' from his/her work*' (p. 6).

1) 'Would Like' format -

This asks respondents to indicate on a seven-point scale the degree to which they would like to have each attribute present in their jobs. Six items (2, 3, 6, 8, 10 & 11) relate to Growth Need Strength, the other six relate to alternative job attributes which might be desirable to employees.

b) 'Job Choice' format

This scale gives examples of two types of jobs per question and asks respondents to indicate on a five-point scale whether they prefer JOB A or JOB B. In each item a job with characteristics relating to growth need satisfaction is paired with a job that has the potential for satisfying one of a variety of other needs. In half of the items the choice is between jobs which both have positive characteristics. In half the choice is between jobs which both have predominantly negative features (e.g. a job where there is a real chance of being laid off versus a job with little chance to do challenging work). The growth relevant job is presented in half the items as JOB A and half as JOB B. Hackman and Oldham (1974b) report internal consistency reliabilities of .88 for the 'would like format' and .71 for the 'job choice' format. They also discuss the empirical validity of the Job Characteristics Model and state that, in general, the results provide strong support for this. However, the internal validity of the GNS scales is not reported. The following two measures were included in the questionnaire to compare with the measures of Growth Need Strength because Hackman and Oldham (1975) reported that

their instrument was based on earlier work on higher order needs as determining the effects of job characteristics on employee behaviour (see page 6).

Higher Order Need Strength

This instrument was designed by Warr, Cook and Wall (1979) and is a modification of the measure designed by Hackman & Lawler (1971). Warr et al (1979) describe 'Higher Order Need Strength' as 'a dispositional characteristic extending across jobs' (p. 131). They suggest that at the time of designing their measure, attention was being focussed on employees' needs for satisfaction and achievement through work and used the term Higher Order Need Strength recognising its derivation from Maslow's (1970) hierarchical theory. The six-item scale was developed following research involving two investigations of 200 and 390 workers and Alpha reliabilities of .91 and .82 respectively, are reported. It is reproduced in Section 7 of the questionnaire and responses were obtained on a seven-point scale running from 'not at all important' to 'extremely important' and summed to provide a scale score.

Self Actualisation Need Strength

This measure is taken from the work of Sims and Szilagyi (1976) and is reported in their examination of individual and structural moderators of the job characteristics relationships. They defined self-actualisation need strength as the '*Extent to which the individual values the importance of higher level work outcomes*' (p. 218) and report the Alpha reliability of the scale as .83. This scale is reproduced in Section 3 of the questionnaire and responses were obtained on a seven-point scale running from 'not at all important' to 'extremely important' and summed to provide a scale score.

Personality

Personality is measured in Section 4 of the questionnaire using the NEO Five Factor Inventory (Costa and McCrae, 1985). This version is scored for the five domains of personality only and was used as it is a shorter form of the NEO Personality Inventory, which is described by Costa and McCrae (1992) as being appropriate where global information on personality is considered sufficient. They report Alpha reliabilities of - neuroticism .90; extraversion .78; openness .76; agreeableness .86 and conscientiousness .90. This section of the questionnaire contains 60 statements and

respondents were asked to indicate on a five-point scale running from 'strongly disagree' to 'strongly agree', how much they agreed with each statement. Responses were summed to provide a scale score for each of the domains specified in the model. Permission to reproduce these items was sought and received from the Publisher, Psychological Assessment Resources, Inc., on payment of a fee.

General Values

General values are measured in Section 6 of the questionnaire using Schwartz's (1992) 48-item measure. No Alpha reliabilities are reported, by Schwartz (1992) for this instrument. Responses were obtained on a nine-point scale running from -1, 'opposed to the principles that guide you' to 7, 'of supreme importance as a guiding principle in your life' and were then recoded from 1-9 and summed to provide a scale score for each of the 10 domains specified in the model. The recoding was carried out to assess whether the negative value on the scale had an overall effect on each scale score.

Work Values

Work values are measured in Section 8 of the questionnaire using Elizur's (1984) 24-item measure. No Alpha reliabilities are reported for their scale. Responses were obtained on a five-point scale running from 'not at all important' to 'extremely important' and a mean score for each individual was calculated for each of the three work values domains to provide a scale score.

Job Satisfaction

Job satisfaction is measured in Section 9 of the questionnaire using Price and Mueller's (1981) seven items scale. The Alpha reliability co-efficient for the original scale was .87. Responses were obtained on a five-point scale running from 4, 'strongly agree' to 0, 'strongly disagree'. Four of the items were negatively worded (2,3,5 & 6) and recoded 0-4. All statements were then recoded 1-5 and the means calculated a scale score. The items were recoded so that '0' could be reserved for missing values in all scales should this be necessary.

Age, Education and Gender

Demographic details of respondents are obtained in Section 10 of the questionnaire.

Hackman and Oldham (1976) reported that their analyses showed that the typical high GNS person was 'a well educated male'. It would be interesting to assess whether this was so for the contemporary sample almost 30 years on as the results of Hackman and Oldham (1976) suggested that education and gender were associated with people who were high in GNS. If the high GNS participants in the current sample were also found to be those who had attained a higher educational status then this might indicate that GNS is a personality trait which is in part acquired through the process of education.

Courses of Study Outside Work

Respondents were asked to indicate at the end of the questionnaire whether they had ever undertaken a course of study outside working hours and whether this was mainly for career or pleasure purposes. The reason respondents were asked about courses of study outside work rather than training courses within the work situation was because such courses are more likely to be undertaken voluntarily. Barrick and Mount (1991) had suggested that openness to experience seemed to assess peoples' readiness to participate in learning experiences. Respondents were therefore asked whether they had undertaken courses of study outside working hours so that their readiness to participate in learning opportunities might be examined in conjunction with levels of GNS. This might show if GNS predicts peoples' readiness to participate in learning opportunities which would, in turn, help in the design of the final study.

Pilot Work

A small group of 5 full time employees from a local government organisation completed an initial draft of the questionnaire and they were asked to make comments about any difficulties they encountered in completing it. This pilot work was carried out to ensure that no major changes were required prior to the first pilot study and as a result of comments made by the five respondents, some changes were made to the layout of the questionnaire. It had been anticipated that the questionnaire would take approximately 40 minutes to complete and all respondents reported taking between 30 and 40 minutes.

Pilot Study 1

Aims

The aims of carrying out the pilot study were twofold. Firstly, to ensure that the questionnaire was as simple as possible to administer, did not take too long to complete and that respondents understood the instructions and carried them out. Secondly, to ensure that the items all had reasonable variances and thirdly, the responses to the questionnaires obtained from the study would provide a small data set with which to plan methods of empirical analysis to be used in Study 1.

Method

Hackman and Oldham (1976) had tested the job characteristics model using data obtained from 658 employees. The jobs they were employed in were reported as being highly heterogeneous, so it was important that the first main study followed the same pattern. The current researcher had worked as a consultant for a Local Government Organisation and, with the permission of the Chief Officer was able to ask for a small number of volunteers to take part. There were 6 departments within the organisation and the researcher asked for 5 volunteers from each to complete the questionnaire. In all 15 males and 15 females volunteered from a range of jobs from junior clerical to senior manager. The questionnaires were distributed by mail with each respondent being given the same specific instructions by letter, on how to complete the questionnaire. Ethical clearance for this research was not required under the departmental guidelines at the time.

Results

27 completed questionnaires were received with only one containing a small amount of missing data. The data were screened for outliers and none were found.

Table 1 shows that although both means and variances for the job dimension for the two samples are quite similar, the means for the current sample are slightly higher. Hackman and Oldham (May 1974) reported that the mean of the GNS 'job choice' format would be close to the mid-point (3.0) which is the case for the current sample.

Table 1 Means and Standard Deviations of the JDS, HONS, SANS, Job Satisfaction and Importance of Job Characteristics

	Current Study		Hackman & Oldham (1975)	
	Mean	SD	Mean	SD
1. Job Dimensions				
Skill Variety	5.19	1.44	4.69	1.67
Task Identity	5.41	1.72	4.87	1.43
Task Significance	4.81	1.82	5.49	1.29
Autonomy	5.26	1.56	4.80	1.43
Feedback from the job	5.07	1.38	4.98	1.41
GNS 'would like'	5.52	0.89	5.62	1.28
GNS 'job choice'	2.94	0.61	-	-
MPS	147.14	82.84	128.31	72.73
2. HONS	5.51	0.80	-	-
SANS	5.99	0.80	-	-
Job Satisfaction	19.67	1.41	-	-
Importance of JCs	5.24	0.96	-	-

Table 2 Means and Standard Deviations of the NEO-FFI from Two Samples

	Current Sample		Costa & McCrae (1991)	
	Mean	SD	Mean	SD
Neuroticism	24.22	8.74	17.76	7.29
Extraversion	36.00	11.23	27.96	6.81
Openness	26.07	6.43	26.22	5.58
Agreeableness	29.86	5.39	33.01	6.90
Conscientiousness	37.63	5.55	36.16	7.06

The means for the current sample are somewhat higher the Costa and McCrae sample for Neuroticism, Extraversion and Agreeableness and variance is similar except for Extraversion where it is substantially greater for the current sample. This was probably due to the way the sample was collected i.e. participants were well known to the current researcher and volunteered to complete the questionnaire.

Table 3 Means and Standard Deviations for the Values and Work Values Scales

	Mean	SD
General Values:		
Achievement	6.31	1.15
Benevolence	6.78	0.93
Conformity	6.70	1.07
Hedonism	6.96	1.30
Power	4.83	1.64
Security	6.27	1.31
Self-Direction	6.61	1.06
Stimulation	6.28	1.22
Tradition	5.13	1.27
Universalism	6.75	0.88
Work Values:		
Affective	4.14	0.60
Cognitive	3.77	0.50
Instrumental	3.85	0.81

For the Work Values each scale is measured using five points so it would be expected that the means would be close to the mid-point. As can be seen, for Affective Work Values the mean is somewhat higher. The variance in the scores is not as high as would be desirable but the sample was small and again the method of data collection may have contributed to this. The internal consistency reliabilities for all scales reached acceptable levels, including those for the Importance of Job Characteristics.

Conclusions

Only one questionnaire contained missing data indicating that in general there were no problems with its design and there was no negative feedback from the participants. The means and variances for the scales were acceptable given the size of the sample.

Although the pilot study data set was too small to be able to make any significant predictions about the relationships measured in the questionnaire, some interesting issues arose from the analysis which could be pursued in the main study. The mean motivating Potential Score was higher (147) than that reported by Hackman and Oldham (1974a), probably due to the way the sample was selected. It was intended that the main study participants would be more randomly selected and from a wider variety of jobs. The measure of the importance of the job characteristics appears to be reliable which meant that it was reasonable to include it in the main study.

Main Study

Method

Participants

There were certain criteria that it was felt needed to be met by the respondents to the questionnaire, in order to replicate as far as possible the original study by Hackman and Oldham (1976). Their model was tested for 658 employees, working on 62 different jobs in 7 organisations, so organisations were selected to provide data for the current research in which the jobs were highly heterogeneous. Firstly, attempts were made to find a large sample of people in work who were employed in a variety of jobs.

Secondly, the respondents were required to be in full time employment. Finding representative samples within organisations is however often difficult, as employers are not always willing to allow their employees to take part. In the course of finding suitable organisations for this research the researcher contacted nearly 17 organisations

either by telephone or letter asking them if they would participate. In the majority of cases organisations did not reply and some of those who did, declined to take part because one of the issues being researched was job satisfaction. One of the arguments against taking part was that if employees were asked (amongst others things) how satisfied they were with their jobs and if they were dissatisfied, employee expectations would be raised and employers would have to act on the findings of the research. Eventually three organisations agreed to take part in the study. A hospital trust, a local government organisation and a chocolate factory. The largest organisation was the National Health Service hospital trust, which included employees from all categories of staff - medical, nursing, & midwifery, administrative & clerical, professions allied to medicine (physiotherapists, radiographers etc.), professional & technical (biomedical scientists, pharmacy, estates etc.), directors and managers, maintenance and ancillary staff. This broad range of jobs within one organisation made the Hospital trust especially useful for the purposes of this research. The employees in the local government organisation ranged from environmental health professionals to clerical workers. It had been hoped that it would be possible to include staff from the assembly line in the chocolate factory but permission for this was refused because of the risks of disruption to production. Permission was however granted to approach the administrative staff within the company.

Measures and Procedure

The primary data collection instrument was the questionnaire, items and content of which were given in the previous section. No changes were made to the questionnaire as a result of the pilot study. All data were collected by mail and procedural steps were as follows.

1. The nature of the research was explained to a senior manager in each organisation and permission to administer the questionnaire was secured.
2. 1500 questionnaires in total, were sent out by internal post to the employees of each organisation.
3. A letter explaining the purpose of the research and asking each employee to respond by completing the questionnaire and returning it direct to the researcher in a pre-paid envelope provided accompanied each questionnaire.

It was emphasised that all information obtained would be held in confidence and that no one in the organisations would have access to the individual responses. (see Appendix C)

4. As an incentive to take part in the research participants were offered the opportunity to enter a prize draw and if they wished to do so, were asked to put their name and department on a tear off strip at the end of their questionnaire. These were removed so that names could be entered in the draw and destroyed once the winners had been notified, to maintain confidentiality.

Ethics

This research was carried out with due regard for the British Psychological Society's 'Code of Conduct: Ethical Principles and Guidelines, January 2000. All prospective participants were informed in writing of all aspects of the research, which might reasonably influence their willingness to participate, and it was made clear that any participation was entirely voluntary. Participants were informed that the researcher would be available in person at certain times and by telephone, to answer any queries they may have and to discuss any aspect of the research which may cause them concern.

Results

Screening of Data

Prior to analysis all data were examined for accuracy of data entry, missing values and distributions. 515 questionnaires were returned and of these 29 had either large amounts of missing or illegible responses. These were excluded from the analyses. The majority of the remaining missing data was small and randomly distributed and so these were left as missing and dealt with on an analysis-by-analysis basis. However, there were 8 cases where the all responses to the 'would like' format of the GNS scale were missing.

Following the advice of Tabachnick and Fidell (1996) means were calculated from the available data and used to replace the missing values prior to analysis. Although the above authors suggest that '*variance of a variable is reduced because the mean is probably closer to itself than to the missing value it replaces*' (p. 63), it was judged that the comparatively small amount of missing data involved would make a negligible difference to the analysis. Three missing values for the motivating potential score were

replaced with the mean in all cases. There were three extreme values found during the screening process which on rechecking the responses from the questionnaire were found to be input error and were thus easily remedied. To detect univariate outliers among continuous variables z scores were calculated and the number of extreme standardised scores was less than 0.001%. This was as expected for the large sample size (Tabachnick and Fidell, 1996 p. 67). Scores among the variables to be tested were all reasonably distributed although the scores for GNS 'would like' format and Self Actualisation Need Strength were somewhat negatively skewed (-.976 and -.954 respectively). The 29 cases containing large amounts of missing data were excluded from the analyses, leaving 486 cases.

The hypotheses were tested using data from 486 employees who worked on 70 different jobs in 3 organisations, a response rate of 32.4%. The following analyses were carried out:

1. Reliability and validity analysis of all scales.
2. Bivariate correlation of the relationships of the job characteristics with the outcome variable job satisfaction (*Hyp. 1*).
3. Simple analysis (by bivariate correlation) of the relationships of growth need strength with the five personality domains, the ten general values domains and the three work values domains (*Hyp. 6, 7 & 8*).
4. Test of the degree to which the relationship between job characteristics and job satisfaction is moderated by individual growth need strength as specified by the Job Characteristics Model (*Hyp. 1*).
5. Test of the degree to which the relationship between job characteristics and job satisfaction is moderated by the personality factor Openness to Experience (*Hyp. 2*).
6. Test of the degree to which the relationship between job characteristics and job satisfaction is moderated by the general values domains self-direction, achievement and stimulation (*Hyp. 3, 4 & 5*).
7. Test of the degree to which the relationship between job characteristics and job satisfaction is moderated by instrumental, cognitive and affective work values.

Discussions regarding the outcomes of all of the analyses carried out are given at the end of this chapter.

The Validity and Reliability of the Scales

Growth Need Strength

Principle components analysis with direct oblimin rotation of all GNS items revealed 3 factors.

Factor I All items from the 'would like' format accounting for 24.9% of the total variance. The Alpha reliability for these items was .85.

Factor II Items 2, 5, 6, 7, 9, 11 & 12 from the 'job choice' format. The Alpha reliability for these items was .69

Factor III Items 1, 4, 8 & 10 from the 'job choice' format. The Alpha reliability for these items was .58.

Item 3 from the 'job choice' format did not load onto any factor. The items used in the GNS sections of the questionnaire are proposed by Hackman and Oldham (1975) to all be measuring one construct but the findings of the above analysis indicate that this may not be the case. Examination of these items failed to find any clear links between the items designed to measure GNS which might account for this, but the response alternatives in the 'job choice' format on factor 1, apart from item 9, were all concerned with the area of human relationships. For example, item 2 asks respondents whether they would prefer a job where there are many pleasant people to work with and item 7 asks whether respondents would prefer a job with a supervisor who treats them fairly. The response alternatives in the second factor were concerned with areas unconnected with human relationships such as pay and job security. Thus, the items from the 'would like' format are measuring one clear construct but the 'job choice' format is complicated, possibly due to the format of the GNS questions and their response alternatives. However, internal consistency reliability for Growth Need Strength in the 'would like' format was .85 and for the 'job choice' format .69, both of which were consistent with the findings of Hackman and Oldham (1975).

Growth Need Strength, Self-actualisation Need Strength (SANS) and Higher Order Need Strength (HONS)

Principle components analysis with direct oblimin rotation of GNS items from the 'would like' format, SANS and HONS revealed 3 factors with Eigenvalues greater than 1. When items from the 'job choice' format were added the number of factors increased

to five. The GNS items from the two formats accounted for 43% of the total variance, in the factor said to be GNS, explained by those items. When items measuring HONS and SANS were added, the total variance explained by all the items increased to 61%. The purpose of analysing the three measures together was to see whether they were similar psychometrically (see page 31 and the discussion on the background to the GNS measures on page 13). Reliability analysis of items from the two GNS scales, HONS and SANS are shown in Table 4

Table 4 Reliabilities of GNS, HONS & SANS Scales

Scale	n ^a	N	Internal Consistency Reliability (Current Study)	Internal Consistency Reliability (Original Study)
Growth Need Strength 'Would like' format	6	486	.85	.88
Growth Need Strength 'Job choice' format	12	481	.69	.71
Higher Order Need Strength	6	485	.88	Sample 1 .91 Sample 2 .82
Self-actualisation Need Strength	6	485	.85	.83
GNS 'would like' + HONS	12	475	.89	N/A
GNS 'would like' +SANS	12	475	.87	N/A
GNS 'would like' +HONS+SANS	18	474	.90	N/A

n^a Number of items composing each scale.

The internal consistency reliabilities of all the scales measured by the questionnaire are summarised in Tables 5 and 6. As can be seen from Table 5 the variance in the scores for the items taken from the Job Diagnostic Survey is similar to that found by Hackman and Oldham (1975), though the variance for Skill Variety and the MPS is less for the current sample. The variance in the scores for the GNS 'job choice' is not reported by Hackman and Oldham (1975). However Hackman, Oldham and Stepina (1979) reported normative data for the Job Diagnostic Survey obtained from 6,930 employees working on 876 jobs in 56 organisations and report the mean for GNS 'job choice' across respondents as 4.23 and the standard deviation as .81.

Table 5 Psychometric Properties of the JDS, HONS, SANS, Job Satisfaction and Importance of Job Characteristics

	Current Study			Hackman & Oldham (1975)		
	Alpha	Mean	SD	Alpha	Mean	SD
1. Job Dimensions						
Skill Variety	-	5.06	1.44	0.71	4.69	1.67
Task Identity	-	5.01	1.61	0.59	4.87	1.43
Task Significance	-	5.95	1.37	0.66	5.49	1.29
Autonomy	-	5.16	1.41	0.66	4.80	1.43
Feedback from the job	-	4.30	1.51	0.71	4.98	1.41
GNS 'would like'	0.85	5.63	1.01	0.88	5.62	1.28
GNS 'job choice'	0.69	3.04	0.48	0.71	-	-
MPS	-	126.43	71.34	-	128.31	72.73
2. HONS	0.88	5.55	0.80	0.88	-	-
SANS	0.85	6.05	0.74	0.85	-	-
Job satisfaction	0.89	27.51	4.78	0.89	-	-
Importance of JCs	0.71	5.62	0.80	0.71	-	-

Personality, Values and Work Values

Correlations were computed for two of the personality domains both prior to and following recoding. The correlations were identical in each case indicating that recoding of the items made no difference to the overall results. Factor analysis revealed one factor each for instrumental and affective work values but 3 factors for cognitive work values. The cognitive work values scale is the largest of the three consisting of 14 items concerning areas such as job status, contributions to society, opportunity for personal growth and job interest. The items in the three factors revealed from the cognitive work values appear to be divided into three distinct groups – the first concerns the relationship between the individual and the organisation the second, the relationship between the individual, the organisation and society and the third, the individuals' personal feelings about work. Further discussion about the implications of these results is contained in the final section of this chapter.

The Table 6 shows that the reliabilities for the personality scale for the current analyses were somewhat lower than those reported for the original study, particularly for the scale Openness to Experience. As suggested in Chapter 1, there has been much discussion about the nature of this factor and the current results, perhaps, indicate that further examination of the scale is warranted. Nonetheless, the Alpha reliability of .66 was regarded as adequate for the analyses which follow.

Table 6. Psychometric properties of the personality traits, values and work values domains

	Alpha Coefficient	Mean	SD	No. of Items
Personality Traits:				
Neuroticism	.76 (.93)	31.73 (17.76)	7.73 (7.29)	12
Extraversion	.76 (.90)	42.03 (27.96)	5.89 (6.81)	12
Openness	.66 (.89)	39.27 (26.22)	5.62 (5.58)	12
Agreeableness	.70 (.95)	45.04 (33.01)	5.20 (6.90)	12
Conscientiousness	.80 (.92)	46.75 (36.16)	5.95 (7.06)	12
Value domains:				
Achievement	.80	6.58	1.12	4
Benevolence	.66	6.98	0.90	7
Conformity	.62	6.83	1.07	4
Hedonism	.60	7.00	1.13	2
Power	.66	4.91	1.31	3
Security	.65	6.62	1.06	5
Self-Direction	.72	6.76	0.94	5
Stimulation	.72	6.18	1.23	3
Tradition	.72	5.37	1.18	6
Universalism	.77	6.79	0.97	9
Work value domains:				
Affective	.71	4.19	0.52	5
Cognitive	.87	3.92	0.50	14
Instrumental	.78	3.94	0.66	5

The figures in brackets show the results for Costa & McCrae (1991).

Comparative figures are not available for values and work values.

The current research predicts that people high in GNS will also be high in the values domains achievement, self-direction and stimulation and the coefficients for these had acceptable consistency. There is concern though, as some domains in the scale are measured by using as few as two items so the reliability of the measures should be treated with caution.

In general, the results suggest that the internal consistency reliability of the scales is satisfactory. Although the 'would like' format for measuring GNS has a greater reliability than the 'job choice' format it is very similar to the scales used to measure higher order need strength and self-actualisation need strength. The 'job choice' format was the one used by Hackman and Oldham in their analyses and was therefore used in all of the current analyses.

Relationship Among the Measures

Correlations among the questionnaire scales scores are presented in Appendix D. The job characteristics themselves are moderately positively intercorrelated as has been

found previously by Hackman & Lawler (1971) and Hackman & Oldham (1974a). Hackman & Oldham (1974) suggest that this is to be expected because if a job is regarded by its incumbent as “good” then it is good in a number of ways and jobs which are regarded as “bad” are frequently generally bad. The correlations of the job characteristics with job satisfaction are shown below. The results are consistent with expectations, correlations are similar to those reported by Hackman and Oldham (1976) also shown in the table.

The Relationship of Job Characteristics with Job Satisfaction

The summary motivating potential score (MPS) relates more strongly to job satisfaction than do any of its component job characteristics except skill variety which is $r = .51$ for the current study. The simple additive motivating potential score (MPSA) produces a higher correlation than either the Motivating Potential Score or the Full Multiplicative version of this score (see Tables 7 & 8).

Table 7 Correlations of Job Characteristics with Job Satisfaction

	Job Satisfaction Current Study	General Satisfaction Hackman & Oldham 1976 (n = 658)
Skill Variety	.51**	.32**
Task Identity	.36**	.22**
Task Significance	.25**	.21**
Autonomy	.30**	.38**
Feedback	.37**	.38**

n = 485 * $p < .05$ ** $p < .01$

Table 8 Comparisons of Three Methods for Combining the Job Characteristics

	Job Satisfaction Current Study	General Satisfaction Hackman & Oldham 1976 (n = 658)
MPS $\frac{(SV+TI+TS) \times A \times F}{3}$	.47**	.49**
Full multiplicative (MPSM)	.47**	.45**
Simple Additive (MPSA)	.57**	.52**

n = 486 ** Correlation is significant at the 0.01 level (2-tailed)

The Importance of the Job Characteristics

The table below shows that the correlations between job satisfaction and the individual job characteristics were higher for those individuals who reported that each characteristic was important to them. (High importance was taken to be the top quartile of scores for each characteristic).

Table 9

Job Characteristic	Job Satisfaction Current Study N=485	Job Satisfaction for respondents who reported high importance for each job characteristic	z (for difference between rs)
Skill Variety	.51**	.61** (n=292)	1.97 (sig.)
Task Identity	.36**	.42** (n=272)	.93 (ns)
Task Significance	.25**	.33** (n=203)	1.74 (ns)
Autonomy	.30**	.35** (n=133)	.57 (ns)
Feedback	.37**	.43** (n=251)	.91 (ns)

* $p < .05$ ** $p < .01$

Although these results indicated that importance of job characteristics might moderate the relationship between job characteristics and job satisfaction there was a high correlation between 'Importance' and MPS ($r = .52$) which raises doubts about what the importance variable is actually measuring.

Relationship of GNS and the Five Personality Domains

Table 10 Intercorrelations Among GNS & Personality Domain Scale Scores

	GNS 'job choice'	Neuro.	Extra.	Openness	Agree.	Conscien.
Neuro.	-.15**					
Extra.	.12*	-.36**	-			
Openness	.29**	.04	.11*	-		
Agree.	-.18**	-.13**	.16**	-.05	-	
Conscien.	.04	-.30**	.31**	-.04	.19**	-

$n = 486$ * Correlation is significant at the 0.05 level (2-tailed) ** Correlation is significant at the 0.01 level (2-tailed).

As predicted the highest correlation between GNS and the personality factors was with Openness to Experience though it is modest and shows that GNS is not identical to this factor.

Relationship of GNS and the Ten General Values Domains

The table below shows the intercorrelations between GNS and the Values scales scores.

As predicted the GNS correlated most highly with the values domains achievement, self-direction and stimulation although these correlations are again modest. Furthermore there was some intercorrelation between the scales.

Table 11

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.
1. GNS 'job choice')	-										
2. Achievement	.27**										
3. Benevolence	.03	.49**	-								
4. Conformity	-.12**	.48**	.62**	-							
5. Power	.11*	.50**	.09*	.24**	-						
6. Self Direction	.28**	.64**	.50**	.38**	.34**	-					
7. Security	-.07	.44**	.51**	.60**	.33**	.51**	-				
8. Stimulation	.22**	.53**	.37**	.27**	.32**	.64**	.27**	-			
9. Hedonism	-.07	.36**	.18**	.22**	.30**	.39**	.27**	.49**	-		
10. Universalism	.04	.37**	.62**	.44**	.10*	.61**	.51**	.43**	.26**	-	
11. Tradition	-.18**	.40**	.58**	.67**	.25**	.34**	.27**	.27**	.09*	.49**	-

Relationship of GNS and the Three Work Values Domains

Table 12 Intercorrelations for GNS and Work Values Scale Scores

	1.	2.	3.	4.
1. GNS 'job choice'	-			
2. Affective Work Values	-.12**	-		
3. Cognitive Work Values	.31**	.56**	-	
4. Instrumental Work Values	-.35**	.47**	.39**	-

No prediction was made regarding the relationships between GNS and Work Values. GNS was negatively correlated with affective and instrumental work values with the highest correlation being between instrumental work values and GNS. GNS was positively correlated with cognitive work values. Again, these correlations were modest.

In order to summarise the relationships between GNS, Openness to Experience, the three general values and the three work values, a standard multiple regression was performed between GNS 'job choice' as the dependent variable and Openness to

Experience, Achievement, Self-direction, Stimulation, Affective, Cognitive and Instrumental work values as independent variables. It is important to note that relatively high intercorrelations between some of the scaled scores. The results of this regression are given in Table 13.

Table 13

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		13.108	.000
Openness to Experience	.059	1.476	.141
Achievement	.099	1.881	.061
Self-Direction	.148	2.639	.009
Stimulation	.024	.519	.604
Affective Work Values	-.235	-5.252	.000
Cognitive Work Values	.473	8.965	.000
Instrumental Work Values	-.484	-11.512	.000
F (7,477) = 51.737 p~0		Adjusted R ² = 0.42	

Four of the independent variables contributed significantly to the prediction of GNS though as can be seen, the negative betas for affective and instrumental work values indicate that **low** scores on these variables predict high GNS. The 'best' predictors of GNS for this sample were the work values followed by self-direction. Altogether, 43% (42% adjusted) of the variability in GNS was predicted by knowing the scores on the seven independent variables.

Test of the Moderating Effect of Growth Need Strength on the Job

Characteristics-Job Satisfaction Relationship

The job characteristics model specifies that individual GNS can moderate employees' reactions to their work. The model predicts that the relationship between job characteristics and work outcomes will be stronger for high than for low GNS individuals. The moderating effect of GNS on the above relationship was tested using moderated regression analysis using multiplicative interactions terms was used. This method includes *all* scores and the results are given in tables below and are presented according to the guidelines set out by Tabachnick and Fidell (1996). Moderated multiple regression analysis was performed between job satisfaction as the dependent variable and GNS, MPS, MPSA and MPSM and their interactions as independent

variables. The analysis was performed twice more for the simple additive and full multiplicative versions of the MPS. As a result of the evaluation of assumptions no transformations were made.

Table 14

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		141.260	.000
GNS	.048	1.182	.238
MPS	.458	11.222	.000
Interaction	.074	1.828	.068
F (3,479) = 46.211 p~0		Adjusted R ² = 0.22	

Dependent Variable: Job Satisfaction

The interaction between growth need strength and job characteristics (combined as the MPS) was not significant ($\beta = .074$, $p > .05$). However, the level of significance was 0.068 indicating some weak support for the moderating effect of GNS on the job characteristics combined as the MPS. The adjusted $R^2 = 0.22$ indicating that only 22% of the variance in job satisfaction could be predicted by knowing the scores for the three independent variables so the interaction between GNS and MPS did not explain as much of the variance in job satisfaction as the MPS alone.

Table 15

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		152.571	.000
GNS	.022	.573	.567
MPSA	.576	15.156	.000
Interaction	.096	2.571	.010
F (3,479) = 81.024 p~0		Adjusted R ² = 0.33	

Dependent Variable: Job Satisfaction

Table 15 shows that the interaction between GNS and job characteristics combined as the MPSA was significant ($\beta = .096$, $p < .05$) providing support for the moderating effect of GNS on the job characteristics (MPSA) – job satisfaction relationship. The adjusted $R^2 = 0.33$ shows that a larger amount of the variance in the scores for the

dependent variable could be predicted by using the simple additive version of the MPS as an independent variable than for the MPS. The interaction between GNS and job characteristics combined as the full multiplicative version of the MPS (MPSM) was not significant ($\beta = .034, p > .05$) so no support was found for the moderating effect of GNS on the job characteristics (MPSM) – job satisfaction relationship.

As the review of the literature regarding the Job Characteristics Model had highlighted some confusion over which measure of GNS should be used to test its moderating effect on the job characteristics-job satisfaction relationship and because only weak support could be found for this effect using the 'job choice' measure, both the 'would like' format and a combination of the two formats were used to test the relationship. No moderating effect was found for either.

Test of the Moderating Effect of Openness to Experience, General Values and Work Values on the Job Characteristics-Job Satisfaction Relationship

The analysis provides some limited support for the moderating effect of GNS on the relationship between job characteristics combined as the MPS and job satisfaction (where as the scores for MPS increased so did the scores for job satisfaction) and stronger support for this relationship using the simple additive version of the MPS, the MPSA. The calculation of the MPS will be discussed at the end of this section, however, as some support was obtained for the moderating effect of GNS using the MPS and MPSA these were used in the regression analyses for Openness to Experience, the three general values and the three work values. As suggested on page 31, an examination of the moderating effect of other variables which might be related to GNS, may help towards defining it. The dependent variable in each case was job satisfaction. The interaction between Openness to Experience and job characteristics (combined as the MPS) was not significant ($\beta = .058, p > .05$), indicating no support for the moderating effect of Openness to Experience on the job characteristics – job satisfaction relationship. No moderating effect was found using the simple additive version of the MPS. No support was found for the moderating effect of any of the three general values or the three work values.

Age, Education and Gender

Hackman and Oldham (1976) observed that that a ‘typical’ high GNS individual was a young, well-educated male. To investigate whether any of these three variable predicted GNS for the current sample a standard multiple regression analysis was performed with GNS as the dependent variable and age, education and gender as independent variables. The results are shown below.

Table 16

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		20.224	.000
Age	.031	.646	.519
Education	.136	2.880	.004
Gender	-.034	-.733	.464
F (3,480) =3.255 p=0		Adjusted R ² 0.014	

Dependent Variable: GNS ‘job choice’

Education contributed significantly to the prediction of GNS. However, only 2% (1.4% adjusted) of the variability in GNS was predicted by knowing the scores on all three independent variables. This indicates that contrary to the suggestion made by Hackman and Oldham (1976), age and gender may not predict GNS but in line with their assumption education does weakly.

Courses of Study Outside Work

Respondents were asked whether they had ever taken up courses of study outside work the response to which was either ‘Yes’ or ‘No’. The correlation between GNS and courses of study outside work was $r=-.157$, $p < .01$. The results of the test of this relationship showed a t-value of 3.487 (df=483) which was highly significant ($p<.001$, 2-tailed), indicating that there was very strong evidence for rejecting the null hypothesis of no difference between the two groups.

Discussion

The aim of this study was to attempt to define the individual difference factor ‘growth need strength (GNS)’, which was specified by the Job Characteristics model to moderate the relationship between job characteristics, and outcome measures.

Analysis of GNS ('job choice') items

The results show that all of the items in the 'would like' format of the measure for GNS load onto one factor but the items in the 'job choice' format load onto two factors. (see Appendix U) The reason for the single factor for the 'would like' format is probably shared attributable to the response format. Richard Hackman has indicated in personal correspondence that he was never happy with this format because '*it seemed so inviting of socially desirable responses*'. The reason he gives for using the 'job choice' format in analysis was because it was harder for the respondents to say in effect 'I'd like just everything'.

Analysis of individual items in the 'job choice' format shows that the larger of the two factors includes mostly items dealing with interpersonal relationships paired with GNS items, whereas the second factor consists of items which deal with job security and pay and fringe benefits paired with GNS items. This indicates that in answering questions about their feelings for their employment, employees' responses are affected by organisational elements of the job. For example, when an employee is asked whether he or she would prefer a job which offers high respect and fair treatment from an employer, over one which offers the chance to use his or her skills to the maximum, if the employer is not perceived as fair at that time, the desire for respect and fair treatment may override the desire for personal growth. Similarly, issues of pay and fringe benefits may override the desire for personal growth if an employee is suffering financial difficulties. Oldham (1976) found that interpersonal relationships moderated the relationship between job characteristics and internal motivation and suggested that such relationships were important to employees' satisfaction. Oldham suggested that his results showed that when employee satisfaction with supervisory and co-worker relationships was high no 'interpersonal barrier exists between the job and the employee's motivational state' (p.567).

As discussed in Chapter 1 (page 25), the 'state-trait' personality theorists such as Cattell and Eysenck shared the view that '*the interactive influence of traits and situations produces transient internal conditions known as states*' (Eysenck and Eysenck, 1980, p.191) so a question which arises from the results so far is whether growth need strength rather than being an individual trait, is a 'transient internal condition' – in other words a

‘state’. It would certainly seem that the intrinsic need for personal growth can only be considered by the individual when other extrinsic needs have been satisfied. Thus, whatever growth need strength is it’s satisfaction can be ‘set aside’ whilst other, more pressing life event are dealt with. If this assumption is correct then it would confirm Maslow’s theory that higher order needs, such as growth needs, can only be considered once other more basic needs have been satisfied.

Growth Need Strength (GNS), Higher Order Need Satisfaction (HONS) and Self Actualisation Need Strength (SANS) – are they all the same construct?

Research by Hackman and Lawler (1971) from which the Job Characteristics Model evolved, used items for the measurement of ‘Higher Order Need Strength’ which were *‘judged on an a priori basis to measure desire for higher order need satisfactions’*. The items included in GNS ‘would like’ format, HONS and SANS measured in the current study are very similar in their wording, so the question remains as to whether they are all measuring the same construct or whether GNS as specified by Hackman and Oldham is indeed a separate construct. The results of this analysis in terms of the ‘would like’ format would suggest that they are measuring a similar construct, however it remains that the ‘job choice’ format was found to moderate the relationship between job characteristics and job satisfaction as predicted by the model. The evidence from research into individual needs does suggest that ‘higher order needs’ such as the need for personal growth exist to a greater or lesser degree in the majority of individuals and that they may predict behaviour. Further, the work of Digman (1997) suggests that such needs might indeed form part of the Big Five personality model. The similarity between the items and the reliability of the scales used to measure GNS in the ‘would like’ format, and the measures for HONS and SANS and their differences from the ‘job choice’ measure for GNS do suggest that the confusion lies not in whether there is such a thing as GNS but how it is to be measured. Although the different factors obtained in analysis of the items from the ‘job choice’ format may partly reflect measurement method, they were factorially distinct from the ‘would like’ format and thus may well be measuring a slightly different construct. Is this construct the ‘true’ growth need strength? The analysis of the items comprising the scales for Higher Order Need Strength and Self Actualisation Need Strength showed that these two scales were

distinct from the scale for GNS and so further examination of these two measures was not felt to be necessary. They were therefore not included in the later studies.

Is there a relationship between GNS, Personality, Values and Work Values?

GNS and Openness to Experience

Intercorrelations among GNS and personality domain scale scores show a higher positive correlation between GNS and Openness to Experience than between GNS and any of the other personality domains. This was as predicted but analysis shows that GNS is distinct from any of the personality domains when included in factor analysis of the NEO items. GNS is therefore, not the same as Openness to Experience but there is a moderately strong correlation between the two. (see Appendix D)

GNS and the values – achievement, self-direction and stimulation

Intercorrelations among GNS and Values scale scores show a higher positive correlation between GNS and achievement, self-direction and stimulation than between GNS and any of the other values domains. It was expected that individuals high in GNS would also score highly on these three value domains and the results support this expectation. The table on page 55 shows that self-direction predicted GNS for this sample. If growth need strength includes a desire for worthwhile accomplishment and an opportunity to learn new and interesting things, for example, it was reasonable to expect that high GNS individuals would indicate that they found certain values important as guiding principles in their lives. GNS was negatively correlated with tradition and conformity and to a lesser degree with security and hedonism indicating perhaps, that such individuals are not interested in following accepted patterns of behaviour. (see Appendix D). Although factor analysis shows that GNS is a distinct construct from the values domains, the correlations and regression analysis show that there is a relationship between it and the three values domains discussed which would benefit from further examination.

GNS and Work Values

The intercorrelations among GNS and work values scale scores are perhaps some of the most interesting. GNS was positively correlated with cognitive work values which include items relating to use of abilities, personal growth and meaningful work, negatively correlated with affective and instrumental work values. The correlation was

highest between instrumental work values and GNS. Affective work values include items relating to relations with supervisors and co-workers, recognition and esteem, while instrumental work values include items relating to pay, hours of work, security and benefits. However, as shown on factor analysis of the work values domains revealed 3 factors for cognitive work values which were clearly interpretable (see page 50). This casts some doubt over Elizur et al's model of work values, as there may be more domains than the model specifies. However, affective and instrumental work values are described as 'extrinsic' and the items which make up cognitive work values can all be described as 'intrinsic'. Thus, the need for personal growth relates negatively to the more extrinsic work values and positively to the more intrinsic ones, indicating that people who desire personal growth in their jobs do not place such a high value on pay and benefits.

The question that arises from the results concerning work values is do the values people hold about their working life affect the attitudes they have about the importance of higher order needs? The summary table on page 55 shows strong support for all three work values domains as predictors of GNS this sample indicating that there may be a relationship between these two variables which warrants further investigation.

Is there a relationship between job characteristics and job satisfaction?

An important result, which has emerged from the current study, is that similar relationships have been found between job characteristics and job satisfaction, to those found by Hackman and Oldham (1976), using measures which are independent of the framework of the Job Diagnostic Survey. The results show that for this sample, as expected, there is a significant positive correlation between the job characteristics combined as the MPS and job satisfaction ($r = .47, p < .01$). Employing the two other methods to combine the job characteristics produced the same correlation for the full multiplicative version (MPSM) of the MPS, but a higher correlation ($r = .57, p < .01$) for the simple additive version (MPSA). As discussed earlier in this chapter, Hackman and Oldham (1976) had been concerned about the presence of the two multiplicative terms in the MPS formula which they felt could compound the effects of measure unreliability. The simple additive version certainly produces a higher correlation than

either of the other two formulae and moderating effects of GNS were found in the current sample using the additive formula which were not found using the MPS.

The moderating effect of growth need strength

Support was found for the moderating effect was found using the simple additive version of the MPS (MPSA), a weak moderating effect was found using the MPS and no support for the moderating effect using the MPSM. There are two possible reasons for the lack of strong support for the moderating effect of GNS. The first is that the sample used in the current study was smaller than that used by Hackman and Oldham (1976) and therefore it lacked the power to detect a weak interaction. The second potential reason for the lack of support is the homogeneity of the sample. Further examination of the sample showed that although a large number of questionnaires was distributed to individuals in a wide variety of jobs, the greater percentage (68%) of responses came from staff in nursing and midwifery, professions allied to medicine and professional and technical staff within a National Health Service Hospital Trust. Thus, the sample did not approach that of the original study in terms of job-heterogeneity. One way of assessing the level of heterogeneity was examination of the scores for the individual job characteristics. The mean MPS for the pilot study was 147 but this was reduced to 126.43 in the main study, which was in line with the mean MPS found by Hackman and Oldham (1976). The earlier high mean was certainly due to the way the pilot sample was selected. An examination of the individual job characteristics showed that the sample for Study 1 was made up of respondents whose jobs were higher than would be expected in 4 of the 5 characteristics. Also, taking the job characteristics combined as a mean score, the percentage of respondents with a mean score of below 4 on a seven-point scale was only 9.5%. It was felt that this lack of heterogeneity amongst the characteristics of the jobs should be addressed prior to making any further assumptions about the relationship between job characteristics and job satisfaction.

The moderating effect of Openness to Experience, Achievement, Self Direction, Stimulation and Work Values

None of these individual difference factors moderated the relationship between job characteristics and job satisfaction although there were differences between the high and low groups in each factor and these differences were all in the predicted direction. This

indicates that although there is a relationship between GNS and the Values and Work Values none of them appears to have the same role as GNS in the job characteristics-job satisfaction relationship for this sample.

Age, Education and Gender

Hackman and Oldham (1976) had observed that a high GNS individual was more likely to be a young well-educated male. This study examined demographic properties of the respondents to see if there was any indication that this was so for a sample of employees 30 years on. The multiple regression for age, education and gender (see table 16) shows that only education contributed significantly to the prediction of GNS. This relationship between GNS and education is consistent with the findings of Hackman and Oldham (1976) and Evans et al (1979) where education correlated more highly with GNS than other demographic variables. Evans et al suggested that employees with a relatively high educational background were more likely to have been exposed to 'growth need oriented environment' (p.378). Evans et al also suggested that this had implications for employee development as it implied that GNS scores could be raised through training or environmental stimuli (p.378). The results of the current study may indicate that Growth Need Strength is something which develops through the early learning experiences to which individuals are exposed. This may provide some evidence for GNS as a personality trait as discussed earlier in the section on the analysis of the GNS items, although no firm conclusions can be drawn at this time because the study also shows relationships between GNS and General and Work Values.

As suggested earlier in this discussion there may be reasons for the lack of a moderating effect for GNS for the sample used in Study 1 and these are addressed in the following chapter.

To summarise the findings from Study 1 the data thus far show:

- The measures for the GNS 'would like' and 'job choice' formats are psychometrically distinct from each other but analysis of the items comprising the 'job choice' scale revealed two factors rather than one.
- The GNS 'job choice' measure is psychometrically distinct from those for HONS and SANS.

- GNS is more highly correlated with the personality factor Openness to Experience than any of the other personality factors measured.
- GNS is more highly correlated with the values domains Achievement, Self-direction and Stimulation than any of the other general values domains measured.
- GNS is positively correlated with Cognitive Work Values but negatively correlated with Affective and Instrumental Work Values.
- Education contributed significantly to the prediction of GNS.

CHAPTER 3

Study 1a

Introduction

The reliability of the results from Study 1 could not be assured without investigating further the issues of heterogeneity amongst the jobs of the respondents, and overall sample size. The results of Study 1 had shown that the scores for some of the individual job characteristics were skewed (see page 63) and so if the new participants were also working in jobs which were lower in these characteristics a more representative distribution of scores for MPS might be achieved at the same time as increasing the sample size.

After extensive investigation into various types of contemporary work processes a group of jobs was isolated as fitting the criteria required for the extension to Study 1, Study 1a. Anecdotal evidence indicated that work within call centres may be low in the job characteristics Task Significance, Task Identity, Skill Variety and Autonomy and in 2000 the Health and Safety Executive estimated that there were between 1% and 1.7% of the total UK workforce employed in call centres. This, they reported, was more than the combined workforce of coal mining, steel and vehicle production, and it was predicted that there would be continued expansion to just over 2% by 2002/2003. The same report suggested that there were between 900 and 1300 call centres in the UK in 2000. Current figures from the same source now say that 1.6% to 2% of the total UK workforce is indeed currently employed in call centres but the number of call centres has grown to a staggering 3000-5000. The call centre industry is concentrated in specific areas around the UK including Glasgow, Newcastle, Manchester, South Wales and London and similar types of call centre are beginning to cluster in these areas.

The Health and Safety Executive defines a call centre as '*a work environment in which the main business is conducted via the telephone whilst simultaneously using display screen equipment (DSE)*' and a call handler as '*an employee whose job requires them to spend a significant proportion of their working time responding to calls on the telephone whilst simultaneously using DSE*'. (Health and Safety Executive, 2001)

90 companies involved in call centre work were contacted and discussions with those who expressed an interest in participating in the current research, indicated that the jobs

within call centres were indeed highly restricted in terms of job characteristics. Human Resources professionals in these organisations said that they regarded the jobs as low in variety and autonomy in particular so a decision was made to target this particular area of work.

Method

Pilot Study 1a

The purpose of Pilot Study 1a was to find out the percentage of jobs within a number of organisations which might be low in job characteristics with a view to replicating the whole of Study 1 after adding the new respondents to the database. Of the 90 call centre organisations contacted 10 expressed an interest in taking part. Because of time constraints it was decided to run the pilot study and accept those organisations for the further research that both met the requirements and returned the details required in the shortest time. A short, one page, questionnaire was prepared (see Appendix E) which contained only the information regarding job characteristics from section 5 of the main questionnaire. 20 short questionnaires were sent out to a random sample of staff from each of the 10 organisations along with instructions on how to complete these, endorsed by a senior member of the management team of each organisation. It was agreed with each participating organisation that a report would be provided setting out the findings of the study. Letters were written to two companies which returned questionnaires outside the time scale required, thanking them for their interest in the research.

Results of Pilot Study 1a

6 sets of results were returned (all 20 questionnaires were completed in each case) and of these 3 were lower in job characteristics (30.4%; 39% and 40% respectively) than the other three. As a result of discussions with the 3 organisations with the lowest job characteristics scores, they were able to take all of the remaining full questionnaires and so these organisations were used in the main Study 1a.

Main Study 1a

Method

The method used in Study 1a was the same as in the Study 1 (see page 44) except that only 424 questionnaires were distributed to the three organisations. All participants in

this study were 'call handlers' according to the definition given on the previous page, therefore all respondents both within and between the organisations were employed on similar jobs.

Results

206 completed questionnaires were received from employees working in 3 call centre organisations, a response rate of 48.6%. Of these, none had large amounts of missing or illegible responses so all were included in the analyses. These were added to the existing database containing the scores for 486 respondents. The total number of respondents included in Study 1a was therefore 692. Prior to analysis all data were again examined for accuracy of data entry, missing values and distributions. There were 3 more cases where all responses to the 'would like' format of the GNS scale were missing making a total of 11 in total sample. These were all replaced with the mean for the new sample. To detect univariate outliers among continuous variables z scores were again calculated and the number of extreme standardised scores was less than 0.001%. Scores among the variables to be tested were all reasonably distributed.

Scale Validity and Reliability

Growth Need Strength (GNS)

Principle components analysis with direct oblimin rotation of all GNS items revealed the same 3 factors.

Factor 1 All items from the 'would like' format accounting for 23.6% of the total variance.

Factor II Items 2, 5, 6, 7, 9, 11 & 12 from the 'job choice' format. The Alpha reliability for these items was .69

Factor III Items 1, 4, 8 & 10 from the 'job choice' format. The Alpha reliability for these items was .58.

Factors I and II together accounted for 19.2% of the total variance.

Item 3 from the 'job choice' format, again, did not load onto any factor.

Psychometric Properties of the Scales for JDS, HONS, SANS, Job Satisfaction and Importance of Job Characteristics

Table 17

	Study 1 (n=486)			Study 1a (n=692)		
	Alpha	Mean	SD	Alpha	Mean	SD
1. Job Dimensions						
Skill Variety	-	5.06	1.44	-	4.51	1.70
Task Identity	-	5.01	1.61	-	4.75	1.73
Task Significance	-	5.95	1.37	-	5.45	1.66
Autonomy	-	5.16	1.41	-	4.64	1.71
Feedback from the job	-	4.30	1.51	-	4.38	1.55
GNS 'would like'	0.85	5.63	1.01	0.85	5.68	1.00
GNS 'job choice'	0.69	3.04	0.48	0.69	3.00	0.48
MPS	-	126.43	71.34	-	109.17	72.72
2. HONS	0.88	5.55	0.80	0.88	5.56	0.82
SANS	0.85	6.05	0.74	0.86	6.00	0.80
Job satisfaction	0.89	27.51	4.78	0.91	25.85	5.72
Importance of JCs	0.71	5.62	0.80	0.77	5.38	0.93

The mean Motivating Potential Score for Study 1a was reduced to 109.17 with the inclusion of the participants whose jobs were lower in the job characteristics and the variance now matched that of Hackman and Oldham (1976) (see page 43). The percentage of respondents with a mean score for job characteristics of below 4 increased from 9.5% to 21.2%. There was no change in the variance in GNS scores for the larger sample.

Psychometric Properties of the Scales for Personality, Values and Work Values

Table 18

	Study 1 (n=486)			Study 1a (n=692)			No. Of Items
	Alpha	Mean	SD	Alpha	Mean	SD	
Personality Traits:							
Neuroticism	.76	31.73	7.73	.81	31.42	7.58	12
Extraversion	.76	42.03	5.89	.74	42.15	6.04	12
Openness	.66	39.27	5.62	.67	38.85	5.98	12
Agreeableness	.70	45.04	5.20	.71	44.48	5.32	12
Conscientiousness	.80	46.75	5.95	.82	46.35	6.00	12
Value domains:							
Achievement	.80	6.58	1.12	.82	6.63	1.15	4
Benevolence	.66	6.98	0.90	.79	6.69	0.95	7
Conformity	.62	6.83	1.07	.72	6.88	1.06	4
Hedonism	.60	7.00	1.13	.61	7.13	1.14	2
Power	.66	4.91	1.31	.68	5.08	1.37	3
Security	.65	6.62	1.06	.65	6.70	1.04	5
Self-Direction	.72	6.76	0.94	.73	6.85	0.94	5
Stimulation	.72	6.18	1.23	.74	6.39	1.26	3
Tradition	.72	5.37	1.18	.73	5.42	1.17	6
Universalism	.77	6.79	0.97	.80	6.80	0.98	9
Work value domains:							
Affective	.71	4.19	0.52	.75	4.20	0.51	5
Cognitive	.87	3.92	0.50	.88	3.91	0.51	14
Instrumental	.78	3.94	0.66	.79	4.02	0.66	5

In general, the results suggest that the internal consistency reliability of the scales is similar to those found in Study 1 but the Alpha coefficients improved with the increase in sample size.

Relationship Among the Questionnaire Scales

Correlations among the questionnaire scales scores are presented in Appendix F.

The correlations of the job characteristics with job satisfaction are shown below. The results are again consistent with expectations, are in the predicted direction and achieve acceptable levels of significance. The correlations for skill variety, task identity, task significance and autonomy are higher than those for Study 1 and the correlations for feedback are lower.

The Relationship of Job Characteristics with Job Satisfaction

Table 19 Correlations of Job Characteristics with Job Satisfaction

	Job Satisfaction Study 1a	General Satisfaction Hackman & Oldham 1976 (n = 658)
Skill Variety	.60** (.51**)	.32**
Task Identity	.38** (.36**)	.22**
Task Significance	.45** (.25**)	.21**
Autonomy	.49** (.30**)	.38**
Feedback	.26** (.37**)	.38**

n = 692 ** = $p < .01$

The numbers in brackets show the results from Study 1 n=485.

Table 20 Comparison of Three Models for Combining the Job Characteristics

	Job Satisfaction Study 1a	z (for difference between rs for studies 1 and 1a)	General Satisfaction Hackman & Oldham 1976 (n = 658)
MPS $[(SV+TI+TS) \times A \times F]$ 3	.55** (.47**)	1.82 (ns)	.43**
Full multiplicative (MPSM)	.53** (.47**)	1.35 (ns)	.45**
Simple Additive (MPSA)	.66** (.57**)	2.44 (sig.)	.52**

n = 689 ** Correlation is significant at the 0.01 level (2-tailed)

The numbers in brackets show the results for Study 1 n=486.

The summary motivating potential score (MPS) relates more strongly to job satisfaction than does any of its component job characteristics except skill variety and the

correlation is in the predicted direction. The simple additive motivating potential score (MPSA) produces a higher level of correlation than the motivating potential score (MPS) and the full multiplicative motivating potential score (MPSM) now produces a slightly lower correlation than the MPS.

Relationship of GNS and the Five Personality Domains

Table 21 Inter correlations Among GNS & Personality Domain Scale Scores

	1.	2.	3.	4.	5.	6.
GNS (job choice)	-					
Neuro.	-.13**	-				
Extra.	.07	-.36**	-			
Openness	.30**	.03	.07	-		
Agree.	-.13**	-.14**	.15**	.02	-	
Conscien.	.08*	-.32**	.30**	-.01	.22**	-

n = 692 * Correlation is significant at the 0.05 level (2-tailed)

** Correlation is significant at the 0.01 level (2-tailed)

Again, as predicted, the highest correlation between GNS and the personality domains was with Openness to Experience.

Relationship of GNS and the Ten General Values Domains

The Table 22 shows the intercorrelations between GNS and the Values scales scores. As predicted the GNS correlated most highly with the values domains achievement, self-direction and stimulation, however the correlations are all fairly small and again, some of the values domains are quite highly correlated with one another.

Table 22 Inter correlations Between GNS and Values Scale Scores

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1. GNS 'job choice')	-									
2. Achievement	.22**	-								
3. Benevolence	-.01	.48**	-							
4. Conformity	-.11**	.50**	.60**	-						
5. Power	.10*	.54**	.13**	.28**	-					
6. Self Direction	.24**	.64**	.50**	.42**	.38**	-				
7. Security	-.07	.44**	.48**	.60**	.35**	.50**	-			
8. Stimulation	.16**	.58**	.37**	.31**	.40**	.64**	.31**	-		
9. Hedonism	-.07	.41**	.23**	.27**	.35**	.45**	.30**	.56**	-	
10. Universalism	.04	.38**	.59**	.45**	.12**	.60**	.52**	.42**	.27**	-
11. Tradition	-.17**	.42**	.55**	.64**	.31**	.34**	.51**	.32**	.14**	.49**

Relationship of GNS and the Three Work Values Domains

Table 23 Intercorrelations for GNS and Work Values Scale Scores

	1.	2.	3.	4.
1. GNS (job choice)	-			
2. Affective Work Values	-.14**	-		
3. Cognitive Work Values	.26**	.59**	-	
4. Instrumental Work Values	-.37**	.55**	.43**	-

No prediction was made regarding the relationships between GNS and Work Values but the results from Study 1 showed that GNS was negatively correlated with affective and instrumental work values and positively correlated with cognitive work values. The results for Study 1a showed the same correlations although those between GNS and affective and instrumental work values were slightly higher than before and that for GNS and cognitive work values was slightly lower.

In order to summarise the relationships between GNS, Openness to Experience, the three general values and the three work values, a standard multiple regression was again performed between GNS as the dependent variable and Openness to Experience, Achievement, Self-direction, Stimulation, Affective, Cognitive and Instrumental work values as independent variables.

Table 24

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		16.032	.000
Openness to Experience	.108	3.227	.001
Achievement	.086	1.845	.065
Self-Direction	.124	2.683	.007
Stimulation	.005	.128	.898
Affective Work Values	-.221	-5.584	.000
Cognitive Work Values	.461	10.022	.000
Instrumental Work Values	-.491	-13.374	.000
F (7,683)=69.744 p~0			
Adjusted R ² = 0.41			

Dependent variable: GNS ‘job choice’

Five of the independent variables now contribute significantly to the prediction of GNS though as before, the negative betas for affective and instrumental work values indicate

that low scores on these variables predict GNS. The ‘best’ predictors of GNS for this sample were the three work values followed by Openness to Experience and the general value Self-direction. Altogether, 42% (41% adjusted) of the variability in GNS was predicted by knowing the scores on the seven independent variables which was similar to the results for Study 1.

Test of the Moderating Effect of Growth Need Strength

The table below shows the distribution of scores for the variables tested and also the scores for Study 1.

Table 25

	Job Satisfaction		MPS		GNS ‘job choice’	
	n = 486	n = 692	N = 483	n = 689	n = 486	n = 692
Mean	27.51	25.85	126.43	109.17	3.04	3.00
Std. Deviation	4.78	5.72	71.34	72.72	.48	.48

The variance in the scores for job satisfaction and MPS was increased in Study 1a and the variance for GNS remained the same.

The job characteristics model specifies that individual growth need strength (GNS) can moderate employees reactions to their work. The model predicts that the relationship between job characteristics and work outcomes will be stronger for high than for low GNS individuals. The test of moderation used in Study 1 was again used to compare with the results of Hackman and Oldham (1976). Variables entered were the 3 versions of the motivating potential score, growth need strength with the dependant variable job satisfaction and their interactions. Tables 26 – 28 show the results of the moderated regression analyses.

Table 26

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		141.693	.000
GNS	-.004	-.124	.902
MPS	.552	17.318	.000
Interaction	.072	2.282	.023
F(3, 685) = 104.139 p ~ 0 Adjusted R ² = 0.31			

Dependent Variable: Job Satisfaction

The interaction between growth need strength and job characteristics (combined as the MPS) was significant ($\beta = .072$, $p < .05$) so support was found for the moderating effect of GNS on the job characteristics (MPS) – job satisfaction relationship. Adjusted $R^2 = 0.31$ indicating that a greater proportion of the variance in the scores than for Study 1 can be attributed to the independent variables.

Table 27

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		157.934	.000
GNS	-.014	-.490	.624
MPSA	.664	23.174	.000
Interaction	.065	2.269	.024
F(3, 685) = 184.285 p ~ 0 Adjusted $R^2 = 0.44$			

Dependent Variable: Job Satisfaction

The interaction between growth need strength and job characteristics (combined as the MPSA) was significant ($\beta = .065$, $p < .05$) so support was found for the moderating effect of GNS on the job characteristics (MPSA) – job satisfaction relationship. The adjusted $R^2 = 0.44$ indicates that even more of the variance in the scores than for Study 1 can be attributed to the independent variables.

Table 28

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		138.954	.000
GNS	-.011	-.332	.740
MPSM	.532	16.319	.000
Interaction	.053	1.647	.100
F(3, 685) = 92.923 p ~ 0 Adjusted $R^2 = 0.29$			

Dependent Variable: Job Satisfaction

The interaction between growth need strength and job characteristics (combined as the MPSM) was again not significant ($\beta = .053$, $p > .05$) so no support was found for the moderating effect of GNS on the job characteristics (MPSM) – job satisfaction relationship. In line with Study 1, as support had been found for the moderating effect

of GNS on the job characteristics – job satisfaction relationship using moderated regression with multiplicative interactions terms, only this method was used to test the moderating effect of the remaining variables.

Test of the Moderating Effect of Openness to Experience

Table 29

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		141.892	.000
Openness to Experience	-.029	-.920	.358
MPS	.558	17.514	.000
Interaction	.023	.733	.464
F(3, 685) = 102.327 p ~ 0 Adjusted R ² = 0.31			

Dependent Variable: Job Satisfaction

The interaction between Openness to Experience and job characteristics (combined as the MPS) was not significant ($\beta = .023, p > .05$) indicating no support for the moderating effect of Openness on the job characteristics (MPS) – job satisfaction relationship.

Test of the Moderating Effect of the Values – Achievement, Self-Direction and Stimulation.

Achievement

Table 30

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		141.497	.000
Achievement	.024	.742	.458
MPS	.558	17.304	.000
Interaction	-.061	-1.914	.056
F(3, 685) = 103.446 p ~ 0 Adjusted R ² = 0.31			

Dependent Variable: Job Satisfaction

The interaction between Achievement and job characteristics (combined as the MPS) was not significant ($\beta = -.061$, $p > .05$). However the level of significance was 0.056 indicating some support for the moderating effect of Achievement on relationship between job characteristics and job satisfaction. No support was found for the moderating effect of any of the three work values on the job characteristics – job satisfaction relationship.

Age, Education and Gender

A standard multiple regression analysis was performed as for Study 1 and the results of this are shown in the table below.

Table 31

Standardised Coefficients			
Model	Beta	t	Sig.
(Constant)		31.039	.000
Age	.072	1.911	.056
Education	.169	4.484	.000
Gender	-.040	-1.051	.293
F (3,480 = 3.255 p=0		Adjusted R ² = 0.01	

Dependent variable: GNS 'job choice'

Education again, contributed significantly to the prediction of GNS and as can be seen there is also some support for age as a predictor. Again, only a very small amount of the variability in GNS was predicted by knowing the scores on all three independent variables.

Although Hackman and Oldham (1976) had suggested that there might be relationships between job characteristics, human needs and demographic variables, the strength of the relationship between education and GNS was unexpected. As there was a possibility that education alone might moderate the relationship between job characteristics and job satisfaction this was tested using Fisher's R-Z transformation. It was not possible to use moderated multiple regression using multiplicative interactions terms on this occasion because the scale for education was not continuous. The results of this are given in Table 32. The high and low education groups were defined by taking those respondents who were not educated to degree level and those who were.

Table 32

	Low Education	High Education	Z (for difference between rs)
Motivating Potential Score (MPS) with Job Satisfaction	.48**	.62**	1.96*

* $p < .05$ ** $p < .01$

Totals $n = 408$ (228 and 180 respectively, in the high and low education groups; ns are unequal because of tied scores).

The difference between the correlations for the high and low education groups is both in the predicted direction and significant.

Courses of Study Outside Work

There was again a significant relationship between GNS and participants who reported taking up courses of study outside work ($t = 4.106$, $df = 682.413$, $p < .001$). However, the correlation between GNS and outside study was in a negative direction ($r = -.153$) indicating that the higher the GNS the less likely it is that individuals will take up courses of study outside work.

Discussion

The aim of Study 1a was to extend Study 1, to see if by doing so the results obtained might be more consistent with the findings of the original study by Hackman and Oldham (1976). It was felt that if respondents could be found who were working in jobs low in the job characteristics thus increasing the heterogeneity of the jobs sampled, this might help to obtain support for the hypothesis that growth need strength had a significant moderating effect on the relationship between job characteristics and job satisfaction. To this end respondents were sought from the call centre industry as Pilot Study 1a had shown that jobs in this area would be lower in job characteristics than those used in Study 1. The results showed that the scale validity and reliabilities for Study 1a were similar to those for Study 1.

Analysis of GNS ('job choice') items

Principle components analysis with direct oblimin rotation of all the GNS items still revealed three factors which together explained 42.8 % of the variance amongst the

items. This was the same as for Study 1 (see Appendix U). The items from the 'job choice' format loaded onto two factors which were comprised of the same items as the results from Study 1. As before the largest of the two factors was comprised of items measuring GNS paired with items which dealt with workplace issues concerning human relationships. This would suggest that the presence of individual growth need strength is affected by such relationships and provides support for the conclusions drawn from Study 1, that employees responses regarding GNS are affected by organisational elements of the job, particularly those between employees and co-workers and supervisors (see page 59). The nature of these two factors suggests that employees attach priority to different aspects of their working life such as interpersonal relationships, and that the GNS 'job choice' scale is actually measuring individuals' desires for other features. In this sense it is possible that GNS is the anchor point from which the other features of the job are assessed. Variance in the scores for GNS was again low, indeed it was unchanged by the greater heterogeneity of the sample. The normative data for the Job Diagnostic Survey (JDS) reported on page 50 suggests that although the variance in GNS scores for the current sample was low, it was not a great deal lower than the data reported by Oldham, Hackman and Stepina (1979). In their 1976 paper Hackman and Oldham report using the 'job choice' format of the GNS measure only, to test the moderating effect of this individual difference factor on the job characteristics-job satisfaction relationship, but when describing the measures employed in the JDS they suggest that *'the scores for both of these sections are averaged to form a total growth need strength index'* (p. 6). In Chapter 1 (p. 15) it was suggested that there was some confusion over which measure of GNS should be used to test the moderating effect and so in addition to the analyses carried in Studies 1 and 1a both the 'would like' format and the combined formats of the GNS measures were used to see whether this made a difference to the results. No effects were found for either measure.

The Relationship Between Job Characteristics and Job Satisfaction

The percentage of respondents who had a mean score of 3.5 or less for the job characteristics rose from 4.9% to 13.6%, indicating that the collection of data from the call centres had indeed provided more scores at the lower end of the scale for the combined job characteristics.

The results of the correlations of the individual job characteristics with job satisfaction were again consistent with expectations. However, although the correlations for skill variety, task identity, task significance and autonomy were higher than those for Study 1, the correlation for feedback with job satisfaction was lower. As with the original study by Hackman and Oldham (1976), the current study only asked respondents about feedback from the job itself and feedback is received by employees from many other sources. It may be difficult, psychologically, for employees to filter out the feedback they receive from the job itself, from the influences of those other sources of feedback such as managers and co-workers. Hackman and Oldham (1976) p. 272 suggest that any weakness in the relationship between feedback and job satisfaction may be due to the fact that feedback from other sources was not accounted for in the questionnaire, which is the same for the current study.

The results for the correlations between job satisfaction and the three methods for combining the job characteristics followed the same pattern as for Study 1 but all three correlations were higher than before, as might be expected with the increase in sample size. Although the correlations were higher for Study 1a the only significant difference was between the correlations for the simple additive version of the motivating potential score (see Table 20 on page 70).

The Relationship Between GNS, Personality, Values and Work Values

Correlations among GNS and personality scale scores again showed a higher positive correlation between GNS and Openness to Experience than between GNS and any of the other personality domains. Intercorrelations between GNS and the Values scale scores were again consistent with the findings for Study 1 in that the correlations were higher for GNS and achievement, self-direction and stimulation than any of the other values domains. The same consistency was found with Work Values, where cognitive work values were more strongly correlated with GNS than either affective or instrumental work values. The fact that these results were similar to those found in Study 1 is hardly surprising since two thirds of the respondents were the same. However, in addition to the support found for Openness to Experience, the value Self-direction and the three Work Values as strong predictors of growth need strength found in Study 1, there was weak support for the value Achievement as a predictor of GNS. Examination of the items which are used to measure these variables shows that they all

have some similar properties. One of the items measuring Openness to Experience, for example, asks individuals whether they have 'a lot of intellectual curiosity', the measure for Self-direction asks individuals how important 'curious (interested in everything, exploring)' is as a guiding principle in their lives, the measure for Cognitive Work Values asks individuals how important it is to them that they 'do work which is interesting' and the measure for GNS asks respondents whether they would prefer a job which provides 'constant opportunities to learn new and interesting things'.

Elizur et al (1991) discuss how research into needs and values has shown that both have the same conceptual property of the ability to motivate goal directed behaviour in individuals, by inducing valence on certain issues (p.22). Cognitive work values, which include aspects such as achievement, advancement and personal growth, are described by Elizur et al (1991) as being 'intrinsic', whilst affective and instrumental work values are described as 'extrinsic'. The psychological process involved in the value attached to certain work issues is described by Elizur et al as the extent to which an employee assesses the importance to him or her, of having each component of their job (e.g. achievement in work), is of importance to that individual for a sense of well-being at work. It may be that the psychological mechanism involved is the same for all of the above variables, in that if employees feel that certain things are either important or of value to them this will affect the relationship between the characteristics of their jobs and the work outcomes such as well-being or job satisfaction.

The work by Luk and Bond (1993) mentioned in the introduction to this thesis, suggests that people have different needs according to their personality. These needs define what is desirable and what should be valued and their results showed that with the exception of neuroticism (as measured by the NEO-PI), personality and values endorsement are linked through peoples' needs. Of particular interest to the current research is that Luk and Bond (1993) found that Openness to Experience was related to Self-direction and they explain this by suggesting that an open person has a higher need for diversity and is therefore motivated towards self-direction.

The Moderating Effect of Growth Need Strength

Moderated regression analyses revealed a significant interaction between growth need strength and job characteristics combined as the MPS, so support is now found for the

moderating effect of GNS on the job characteristics and job satisfaction relationship, which is consistent with the findings of the study by Hackman and Oldham (1976). A similar result was found for the combination of the job characteristics as the MPSA.

The Moderating Effect of Openness to experience, Achievement, Self Direction Stimulation and Work Values

There was weak support for the value Achievement as a moderator of the job characteristics – job satisfaction relationship indicating that there may be a link between GNS and this variable. People who value achievement in life in general would be more likely than those low on achievement to seek opportunities for personal growth and development, as such opportunities would help them to achieve their desired life goals. It is therefore unsurprising that Achievement and GNS both moderate the same relationship.

None of the remaining individual difference factors moderated the relationship between job characteristics and job satisfaction although there were differences between the high and low groups in each factor, and these differences were all in the predicted direction.

Demographics

Demographics refer to the ascribed or achieved characteristics of individuals (Colquitt et al, 2000 p. 680) and have been the focus of much empirical research into the differences between people which cause them to behave in certain ways. This study found no gender effects on the level of GNS which is hardly surprising given the paucity of theoretical rationale for such effects. Hackman and Oldham (1976) described a ‘typical high GNS individual’ as a ‘young well-educated male’ (p. 275). One of the reasons for this may be that their research was carried out in the 1970s when there was a larger number of males than females in employment than now. Figures from the U.S Department of Labour, Bureau of Labour Statistics (www.bls.gov) show that in 1976 55.26% of the total civilian labour force in the US was male and 37% female. By 2002 this had changed to 50.18% male and 44% female. Figures for educational attainment for 1976 are not available on the BLS database. The UK Labour Force Survey reports data from the 2001 Census, which show that for that year there were approximately 13 million women and 15 million men working in the UK. This is compared to just fewer

than 15 million men and 9.25 million women in 1975. A further reason for the lack of gender effects may be the change in attitudes of women to work in contemporary society – an issue which does not form part of the current research but which may be interesting to pursue in the future. In their test of the Job Characteristics Model, Hackman and Oldham (1976) suggest that ‘Knowledge and Skill’ and ‘Context Satisfactions’ may also moderate the relationship between job characteristics and job satisfaction but these two factors have been largely ignored, Johns et al (1992), however, examined the mediating and moderating effects in job design and tested the moderating effect of employee skills and knowledge. They suggested that a *‘good case can be made that educational level and tenure in the job might be reasonable surrogates for skills and knowledge’* (p.659) and said that the predicted effect for education was simply that employees with a higher education would be better able to cope with jobs high in motivating potential. Johns et al’s results found that educational level exhibited the strongest moderating effects but that they were counter to expectations in that the lower the educational attainment the greater the effect. The results of Study 1a showed that when the top and bottom quartiles of educational level were taken, education was found to have a significant moderating effect on the job characteristics-job satisfaction relationship, again an issue that would provide an interesting area of research for the future.

The Nature of Growth Need Strength

The support found for education as a predictor of GNS for the larger sample, confirms the findings of Study 1 that GNS may be a trait which develops through the educational/life process. There are two possible explanations for these findings the first of which may be connected to individuals’ value systems. Research into social values, as discussed by Furnham (1997, p. 262), has described them as a system of beliefs which are derived in part from social demands. Such research has linked these value systems to culture, religion, political persuasion, family and educational background. Feather (1975) suggests that the value systems which evolve throughout life help to determine behaviour in a variety of settings, including work. It is possible that GNS itself is a trait that develops through the socialisation process and gives rise to certain belief systems which give meaning and consistency to peoples’ behaviour in any given situation. Thus, high GNS individuals will attach greater value to educational attainment as it helps to satisfy their need for personal growth. The second explanation

as suggested in the discussion for Study 1, is that it is likely that employees with a relatively high education have been socialised and exposed to growth need oriented environments (Evans et al, 1979 p. 378) and so their need for growth has been reinforced by success within the education system. In this way educational attainment both stimulates and perpetuates the need for personal growth.

Personality creates differences between people in terms of the cognitive construction of their environments and in the goals they set themselves throughout life. This means that knowing which traits make up an individual's personality, also 'permits a prediction of what a person will do in a given situation' (Cattell, 1950, p.2) and a later study in this research (Study 3) will examine whether GNS does indeed predict people's behaviour. If, as Digman (1997) suggests, there is evidence for the existence of 'higher-order factors' – one of which relates to personal growth, then it is certainly a possibility that GNS is part of personality i.e. a trait rather than a state. Eysenck and Eysenck (1980) describe a state as a transient internal condition produced by the interaction of traits and situations, and it is suggested here that Growth Need Strength may be a trait but the *level* of that trait can change over time, creating a series of 'states' which are dependent upon life events that may either enhance or override the need for personal growth.

Finally, the relationship found between GNS and participants who had taken courses of study outside work indicates that GNS may be predicting peoples' tendency towards undertaking learning experiences and as such provides some support for investigating this relationship further. This issue will be explored in Study 3 but for the moment the results of Study 1a provide evidence of a link between growth need strength and Openness to Experience, but also links between GNS and values, work values, educational attainment and the taking of courses of study outside work. There has been confusion in past research about which measure of GNS actually moderates the relationship between job characteristics and job satisfaction and it is suggested that serious consideration be given to how this factor should be measured. However, before making any decisions about *how* to measure the factor more work needs to be done in attempt to define the construct for without a reasonable definition the issue of how to measure it is irrelevant. The following chapter reports on Study 2, which was designed to obtain further evidence which might help towards the definition of GNS.

CHAPTER 4

Introduction and Overview

As has been discussed in the previous chapter, although GNS was found to moderate the relationship between job characteristics and job satisfaction the results did not provide a clear definition of the individual difference factor 'Growth Need Strength' (GNS). Thus far, it appears that GNS is similar to some of the other factors examined in the earlier studies but no single factor has been found which defines it. For example, Study 1a showed that people who were high in GNS were also high in the personality trait Openness to Experience and that Openness to Experience was a significant predictor of GNS, but analysis of the items from the measures for GNS and Openness to Experience revealed that the two constructs are factorially distinct.

In order to define the boundaries of the construct it was decided to look at what might be included in the factor GNS, whether the measure designed by Hackman & Oldham (1975) covers the whole domain of interest and whether there are other attributes which might be expected to be exhibited by people said to be high in Growth Need Strength. To attempt to answer these questions it was decided to collect and analyse qualitative data from experts in the field of Human Resources (HR). The reason for selecting this group was that it was hoped that such experts, with their experience of recruitment and selection, might be able to shed more light on the meaning of Growth Need Strength, and that the results of the interviews would make it possible to generate a list of items which would help identify those people who are high in GNS. Questions relating to GNS would therefore form a large part of the interviews. The work of Hackman and Oldham (1974; 1975; 1976) and the results of Study 1a showed that there was a positive correlation between job characteristics and job satisfaction which was moderated by GNS, so questions regarding job satisfaction would also form part of the interviews. The results of Study 1a also showed a positive correlation between people high in GNS and those high in the personality factor 'Openness to Experience'. In their work on the five factor model of personality, Barrick and Mount (1991) had suggested that people high in Openness to Experience might be those who showed a readiness to take up learning experiences. If, as Study 1a has shown, there is a relationship between Openness to Experience and GNS, people high in GNS might also share some individual difference factors with people who take up training opportunities in work,

more readily than others. To this end, questions regarding the types of people who take up training opportunities were included in the interviews. Specific detail about the interview schedule is given later in this section.

Why Qualitative Data Analysis?

In Studies 1 and 1a methods of quantitative data collection and analysis were utilised but as the results regarding the nature of GNS were inconclusive, it was important to find an alternative methodology, which would allow for further exploration of the construct. Qualitative data collection and analysis was historically employed in fields such as social sciences, history and anthropology but Miles and Huberman (1994) note that qualitative studies are now used in other disciplines such as psychology, business and organisational studies, health care and urban planning. It is not the purpose of this review to provide a lengthy critique of the ways qualitative data analyses have been conducted over the years. Rather, its purpose is to show how such methods could help the current researcher to answer the question 'What is Growth Need Strength?'

Miles & Huberman (1994) state:

'Qualitative data are sexy. They are a source of well-grounded, rich descriptions and explanations of processes in identifiable local contexts. With qualitative data one can preserve chronological flow, see precisely which events lead to which consequences, and derive fruitful explanations. Then, too, good qualitative data are more likely to lead to serendipitous findings and to new integrations; they help researchers get beyond initial conceptions and to generate or revise conceptual frameworks'. (p.1)

Qualitative data analyses are concerned with how human behaviour can be explained within the framework of the social structures in which that behaviour takes place. They are conducted by means of in depth studies of 'real life' situations, which reflect the 'everyday life of individuals, groups, societies and organisations' (Miles and Huberman (1994, p.6). The crucial role of qualitative data in this study was to try to illuminate the meanings of human behaviour, which might have been missed by conducting quantitative research alone.

Grounded Theory versus Interpretative Phenomenological Analysis

Willig (2001) discusses the relationship between two approaches to qualitative research – Grounded Theory (Glaser and Strauss, 1967) and Interpretative Phenomenological Analysis (IPA) (Smith, 1999). She suggests that whilst there are differences between the two approaches both *'aim to produce a cognitive map that represents a person's or a group's view of the world'* (p. 68). Willig (2001) further suggests that Grounded Theory was developed specifically for use in the study of basic social processes, whereas IPA was designed to gain an insight into *'individual participants' psychological worlds'* (p. 69). Thus, Willig expresses the view that IPA is a specific *psychological* research method. Central to IPA is how participants experience a particular event or situation and is associated with the work of Jonathan Smith (1995). Smith's work was mainly in the field of health psychology and he suggests that the IPA approach can help to illuminate a person's behaviour and experience. Smith argues that qualitative methods can be used to access individuals' beliefs and attitudes and it is hoped that by accessing participants' experiences in the world of work, their perceptions of the constructs under discussion in this thesis, might help to illuminate those constructs and thus help to define 'Growth Need Strength'. Further, it was felt that IPA was best suited to the needs of Study 2 in that it recognises that the exploration of participants' experiences involves the interaction between the researcher's view of the world as well as that of the participants, whereas grounded theory would require the current researcher not to have a starting assumption of the existence of GNS. The results of Study 1a had already provided support for the existence of Growth Need Strength as a moderator of the relationship between job characteristics and job satisfaction, so a sequence of events had already been documented. The researcher was entering Study 2 having conducted extensive reviews of the literature relating to the Job Characteristics Model and thus could not easily set aside pre-conceived ideas about what Growth Need Strength might be. So, in this case the researcher's view of the world played a key role in the design of the study.

The current research begins with human behaviour in a work setting so the collection of data for Study 2, in the setting in which behavioural phenomena appear, might help to illuminate them. Hackman and Oldham (1975) described people who were high in GNS as being those who *'strongly value and desire personal feelings of accomplishment and growth'* (p.160). The words they use to tap this factor in their questionnaire include

‘stimulating’, ‘challenging’, ‘creative’ and ‘imaginative’. With such rich descriptives it is possible that GNS is something which cannot be defined by the use of self reports alone but that its meaning needs to be sought within the framework of its social setting. The collection of data in Study 2 might provide supportive evidence for the findings of the previous study, of a link between growth need strength and job satisfaction and also move towards answering the research question ‘What is Growth Need Strength?’ It was decided that the most effective means of collecting such data was to ask individuals questions about the components of GNS, job satisfaction and the identification of characteristics which might relate to them. IPA works specifically with transcripts of semi-structured interviews so this method of data collection was used in the current study.

Conceptual Framework

Miles and Huberman (1994) give a brief description of a conceptual framework:

‘A conceptual framework explains, either graphically or in narrative form, the main things to be studied - the key factors, constructs or variables - and the presumed relationship among them. Frameworks can be rudimentary or elaborate, theory-driven or commonsensical, descriptive or causal’ (p. 18).

Appendix G shows a conceptual framework, which summarises the relationships found in the research so far, and the issues which are to be investigated in Study 2 and helps by focussing and bounding the collection of qualitative data. Each item in each box relates to different questions in the questionnaire and setting them out in this way ensures that important information, which may emerge from the data, is not missed. By retaining all of the main concepts described in the earlier quantitative study the two studies may be linked together at the end. The two questions to answered by this study were a) Do Human Resources Professionals (HRPs) have a notion of GNS when selecting people for employment? and b) Does that concept map on to Hackman and Oldham’s idea of GNS. Further, any similarities between high GNS people and people who take up training opportunities more readily, could be explored to aid with the design of the third study. With reference to the conceptual framework, each proposed relationship is numbered and the key along with the questions to be answered is as follows:

1. The results of Studies 1 and 1a showed that there was a relationship between GNS and job satisfaction. Are people who are perceived as high in GNS, also perceived as experiencing a higher level of job satisfaction? Is GNS perceived to affect job satisfaction in that people 'bring it with them' to the job and because they are inherently high in the factor, they would derive high job satisfaction whatever they were doing?
2. The results of the previous studies showed that there is a relationship between job characteristics and job satisfaction. Do the participants perceive people as having a higher level of GNS as also responding 'more positively' to jobs with high motivating potential?
3. There may be particular characteristics of jobs which might be perceived as attracting people with a higher level of GNS, are these the people that participants would select for training opportunities?
4. Do the participants think that the fact that people are satisfied with their jobs make them more likely to seek training opportunities or do the training opportunities offered by the organisation lead to increased job satisfaction?
5. Is there a perceived relationship between job characteristics and job satisfaction?
6. Are the characteristics of peoples' jobs perceived to affect their feelings about work in general, do their feelings about work in general lead them to seek certain jobs and do people impute characteristics into their jobs to make them more satisfying?
7. Do the participants perceive the characteristics of the job as affecting turnover?
8. Do the participants perceive job satisfaction as affecting turnover?
9. Do the participants think that job satisfaction affect peoples' feelings about work in general and vice versa
10. Do the participants think that peoples' feelings about work in general affect turnover?
11. What is the participants' perception of GNS? Do they feel there is a more appropriate way of describing it and does examining the wording used to measure GNS help them with its definition?

12. Does the definition of GNS lead to increase ability of participants to predict job satisfaction and the tendency to take up training opportunities?
13. Are people perceived as high in GNS also perceived as actively seeking training opportunities to either help improve their satisfaction with the job or increase their career prospects?

Method

The Interview Schedule

The interview schedule (see Appendix H) was designed to answer the questions set out in the conceptual framework. Although the interview schedule was semi-structured, particular care was taken with the wording of the questions to try to ensure that they were unambiguous, did not include jargon words, did not lead the participants to give a particular answer and did not ask for so much information that participants would find them difficult to answer (Breakwell, 1995 – p.232). Questions 1 and 2 ask the interviewees to describe what gives them personally, satisfaction with their jobs and how the achievement of job satisfaction made them feel. This was to enable them to focus their minds on issues surrounding job satisfaction by relating it to their own experiences. Questions 3 to 9 require interviewees to draw on their past experience in their work in human resources in order to think about and describe people they know or have known in the past, who they perceived as deriving high and low satisfaction from their jobs. Particularly, the questions ask the interviewees to describe any differences in those people in terms of personal attributes or characteristics about them as individuals. Question 10 asks interviewees whether they conduct exit interviews and if so to list the reasons people give for leaving the organisation. Questions 11 to 15 relate to training opportunities offered by the organisations in which the interviewees have worked, whether there are people who take up training opportunities more readily than others and if so what personal attributes or characteristics do they share.

At this point, the construct Growth Need Strength is introduced by giving a brief description of it as an individual difference factor which affects the way people respond to their jobs and how much satisfaction they derive from them. Questions 16 to 27 all relate to GNS and ask interviewees whether, in their opinions, it is a useful construct to use in their work, whether they can think of others ways to describe it and what

attributes or characteristics people high in GNS would share. Questions 21 to 25 are designed to reflect those aspects of work that were purported by Hackman and Oldham (1975) to be those that a person high in GNS would be attracted to. These aspects were covered in the 'would like' and 'job choice' formats of the Job Diagnostic Survey and reproduced in sections 1 and 2 of the questionnaire used in Studies 1 and 1a (see Appendix B). They are: 'stimulating and challenging work', 'chances to exercise independent thought and action', 'opportunities to learn new things from my work', 'opportunities to be creative and imaginative in my work', 'opportunities for personal growth and development in my job' and 'a sense of worthwhile accomplishment in my work'. The first five of these areas were used in the interview schedule and framed in questions that asked 'If in a job description, the work was described as..... what would you look for in person in order to fill that role?' It was felt that the sixth area, 'a sense of worthwhile accomplishment in my work', could not be framed easily within the schedule. It was hoped that if the desire for a sense of worthwhile accomplishment in the job indicated that people were high in GNS, it would emerge from the data from the descriptions the interviewees gave of people who were high in the construct.

The results of Study 1 showed that the items in the 'job choice' format used by Hackman and Oldham (1975) to measure GNS did not all load onto the same factor (see page 78). Examination of the wording of the items themselves suggested that two of the items might be measuring 'autonomy' rather than GNS, so questions 28 and 29 ask interviewees whether they think there is a difference between GNS and autonomy and what they think that difference is.

Discussions in chapters 1 and 2 question whether GNS is a 'state' or a 'trait'. Questions 30, 33 and 36 ask interviewees whether they think GNS is part of peoples' personality or whether it emerges as a function of the environment people live and work in. Further, questions are asked as to whether peoples' level of GNS changes at different times of their lives and what might bring about that change. If GNS is a personality trait then it would be demonstrated in peoples' lives outside work. Question 35 asked interviewees how people high in GNS might behave outside work in terms of the hobbies or pastimes

they might have. Finally, question 34 asks interviewees to think about what characteristics of jobs people high in GNS might be attracted to.

The questions in the schedule will enable the generation of lists of items relating to job satisfaction, GNS, and training. Comparison of these should enable the researcher to see if there are any personal characteristics which distinguish people who are high in GNS from those high in job satisfaction and those who take up training opportunities more readily than others. This in turn may enable the development of items which can be added to the existing measure for GNS and will also give some idea as to whether there are similarities between people high in GNS and those who take up training opportunities – both of which will help in the design of Study 3.

Participant Selection

A range of sampling methods was available for use in qualitative data collection including random sampling and theoretical sampling, as described in detail by Miles and Huberman (1994). However, the latter authors suggest that purposive sampling is the most appropriate method for qualitative data collection as it allows for the selection of participants based on the researcher's knowledge of the domain of interest. This method was thus chosen for the current study as firstly, the researcher already had a substantial knowledge of the domain of interest and secondly, the participants needed to have a reasonable understanding of the processes involved the world of work. As Cunliffe (2000) suggests '*if an individual has no experience of the social and psychosocial processes, how can they comment on it?*' (p.1478). Purposive sampling would allow for the selection of participants who were considered most suitable for identifying and exploring issues such as job satisfaction, training and growth need strength, by virtue of their operational knowledge and experience of such work related factors.

In choosing participants based on the researcher's knowledge of the domain of interest there was however, the possibility that the sample would be biased. Also, the willingness of human resources professionals to take part in the study may have meant that they had personal reasons for wanting to share their experiences, such as their own satisfaction or dissatisfaction with their jobs. Further, the length of tenure of participants

within their jobs was felt to be an important issue as there was the possibility that years of experience of working in HR, rather than providing the participants with a greater knowledge and understanding of the world of work, may have left them with a somewhat cynical view, particularly in the area of job satisfaction. Conversely, participants who had a limited tenure in HR may not have had the experience to speak confidently on areas such as recruitment and factors affecting job satisfaction. Based on these observations, a number of criteria were used to select participants. Firstly, organisational issues included finding senior managers who would allow their staff to take part in the study, given that they would need to be able to give a maximum of two hours of their time and finding a diversity of organisations in terms of the type of work carried out. Secondly, a range of individual criteria were used including gender, age, length of service and experience in the field of human resources.

The number of participants deemed appropriate for the study was also important. Miles and Huberman (1994) suggest that when conducting a multiple-case study the number of participants required can only be decided on conceptual grounds according to how many 'cases' provide the researcher with confidence in his or her 'analytic generalisations' (p.30). For the current study it was decided to continue interviewing until themes continuously appeared and the researcher was as certain as possible that no new insights into the domains of interest were being manifested. Employees from an organisation known to the researcher provided the first participants for the study and gradually other participants were found from a variety of organisations.

Refining and Testing the Interview Schedule

The interview schedule was refined and tested following the guidelines set out by Breakwell (1995) and the procedure was as follows:

1. A pilot study was carried out to test whether the explanation of the interview was understood by a small sample of 2 interviewees. They were also asked whether they had any doubts or queries about the interview.
2. The two interviewees then answered the majority of the questions in the schedule in order to test the comprehension of the questions.
3. The explanation and questions were amended as a result of 1 and 2 above.

4. The revised schedule was tested using the revised explanation and all of the questions for comprehension and to ensure that the answers obtained fulfilled the requirements of the study.
5. The timing of the interviews was also checked.

Pilot Study

Stage 1

Four members of staff from the first organisation volunteered to be interviewed. All were females as there were no males employed in the HR department, their ages were between 26 and 45 years and they each had different employment histories with between 2 and 15 years experience in HR. It was decided that two of these would be asked to participate in testing the explanation of the interview and the comprehension of the questions. Prior to testing the interview schedule, each participant was given a separate explanation (see Appendix I) making it clear that anything they said was of interest to the interviewer. Permission was sought and obtained from the participants to record the interview and they were told that the draft explanation being tested also applied to them with regards to the confidential nature of the contents of the tape. Following the interview the tape recorder was turned off and the researcher talked informally with each participant to find out their feelings about the questions, to tell them more about the research and to take them from the formal setting of the interview before they returned to work. Each participant was told that they could contact the researcher at any time to discuss any issues regarding the interview, which might arise subsequently.

Results

The participants both agreed that the explanation was easily understood by them and therefore no changes were made to this part of the schedule. A number of changes were made to the questions and others were either omitted or combined. For example, following Question 1 a question was added which asked participants how deriving satisfaction from their jobs made them feel. The draft for Question 3 asked 'Thinking of people you know who derive high satisfaction from their jobs, are there any personal attributes which they share?' This was changed to 'Can you think of two people you know in similar jobs – one who seems to get a lot of job satisfaction and one who does not? Can you compare them and tell me what's different about them? Such changes

were made because following the pilot interviews, it was felt that participants needed to be able to frame their answers within their own real life experiences and to be given time to think about the issues involved. It was also apparent that the researcher should not attempt to adhere rigidly to the schedule, rather, to let the participants discourse flow from one area to another in order to obtain the richest possible data from the interviews.

Stage 2

The remaining two volunteers from the same organisation agreed to be interviewed using the revised schedule and all of the questions. Time was allowed at the end of the interviews for the participants to be debriefed, with the tape recorder turned off (see later section regarding Ethical Issues).

Results

The final two pilot interviews ran smoothly and the participants both agreed that they enjoyed the process. However, one participant said that the interview had caused her to re-think her career to date and to have serious considerations about her future, as she felt that she was high in 'Growth Need Strength' and did not feel that her current position stretched her enough. As the researcher was employed by the organisation at the time, it was possible to follow up these issues with her. She said in later discussions that she had felt stimulated by the experience of the interview and it, and the subsequent discussions, had helped her to clarify her thoughts. Six months later she left the organisation and took up a post which gave her promotion and, in her opinion, greater chances of progression within her chosen career. This highlighted the importance of being mindful of how questioning individuals' about their feelings has the potential to affect their behaviour in the future. As a result of this participant's report two questions were added at the end of the schedule asking whether participants felt that they had advanced as much as they had hoped at the start of their careers and whether they felt rewarded enough for their work. The purpose of these questions was to allow participants to think about themselves and their feelings. It was hoped that this would prompt further informal discussion once the tape recorder was turned off so that participants left feeling comfortable about their own situation. Also, it was felt that the participants' own attitude towards their careers might be relevant to the way in which

they answered the questions. Each interview took 90 minutes so it was decided to ask future participants to allow two hours for their interviews.

Main Study 2

Continuing Participant Selection

The pilot interviews indicated that selecting participants from the field of Human Resources would provide the appropriate data for Study 2. The researcher then made contact with a firm of consultants in HR and received substantial support from one of the directors. He himself agreed to participate and kindly provided details of colleagues who he believed would also be prepared to take part. He first made contact with them himself, to ensure that they would be happy to be approached and the researcher then wrote to the contacts to formally ask for support. One contact was also an independent consultant and agreed to be interviewed. All of the remaining the contacts agreed either to take part themselves or to ask for volunteers from their departments. Six interviews were arranged with individuals working in 3 organisations, in different geographical locations. The range of tenure in the field of HR management ranged from 10 years to over 20 years. All had worked in more than one organisation and had significant experience in all areas of HR management. All were asked to set aside 2 hours for the interviews.

Procedure

The researcher conducted one interview with each participant, lasting 60 to 90 minutes. With permission from each participant (given verbally and recorded at the beginning of each interview) the interviews were audio-recorded and transcribed to help ensure preservation of both the context of their responses and the questioning of the researcher. This method would also help with the analysis of their reported experiences. The purpose of the interview and the procedure to be followed were explained to participants and they were assured that the details of the interview would only be seen or heard by Surrey University staff and would be held in confidence (see Appendix J). The interview schedule guided the interview but did not dictate its exact course, so that questions could be adapted according to the participants' narratives to explore interesting areas of discussion. As the time for the ending of the interview was approaching, the interviewer indicated this to the participants and began closing the

interview by asking some questions about the participants' careers to date. The tape recorder was then switched off and the interviewer gave each participant a brief overview of the research being carried out and answered any questions they had about their experience of the interview. Generally, participants said that they had enjoyed the interview and found it interesting.

After each interview, the interviewer made notes about any thoughts she had had about the process and general demographic details which the participants had provided (see Appendix K). The tapes were then professionally transcribed and the researcher checked each one by listening to the tape and reading the transcription at the same time. At that time, minor alterations were made such as, adding or correcting words that the transcriber had not heard clearly and correcting any typing errors. The researcher also added into the transcriptions some of the details she had noted during the interviews, any pauses and why they had occurred.

It had been thought that it would be necessary to interview a large number e.g. up to 20 HR professionals, however, after the eighth interview it was felt that no new information would be gained by further collection. The volume and richness of data collected from the participants was deemed to be enough for the purposes of the analysis - to give some support to the findings of Studies 1 and 1a.

Data Analysis

As Willig (2001) suggests, IPA provides researchers with clear guidelines which allow for the identification and progressive integration of themes. This method of integration of themes across cases employed in IPA was particularly in the current research as both within-case and cross-case analysis was used. In this way the analysis of each participant's unique view of the issues covered in the interview could be compared with the views of the other participants to ascertain similarities and differences in perception of the areas under analysis. Analysis began with an examination of individual transcripts using a computer software package to help sort the interview data into sections, for example, collating all answers to each question, and coding of the transcriptions into themes. Although a number of programs was available including ATLAS/ti and HyperResearch (Miles and Huberman, 1994, Weitzman and Miles,

1995), which would allow for both coding and theory building, following advice from colleagues involved in the CAQDAS Networking Project, QSR NUD*IST VIVO (NVIVO) was chosen for the following reasons. Firstly, the programme was easy to use and would save time given the large amount of data generated by the interviews and secondly, it was compatible with Microsoft Word for Windows, enabling the copying and pasting of text from one computer application to another. The transcriptions were formatted and copied into NVIVO to allow for coding directly onto the software.

As some themes had already been decided upon as a result of Studies 1 and 1a, these were used as themes in the initial coding of the transcriptions (see Appendix L). New themes or phrases which could not yet be categorised were also saved using the software. This initial coding enabled the researcher to collate 'good explanatory exemplars' (Miles & Huberman, 1994 p.65) of what each participant had said. For example, "They like talking to people and dealing with people issues all the time" was included as an exemplar of what a participant had said about people who seem to derive high satisfaction from their jobs. Three sets of data which resulted from initial analysis formed the basis for the IPA These were the words and phrases used by the participants when describing individuals who were high in job satisfaction, high in growth need strength and those who took up training opportunities more readily than others.

Theming

This phase of the analysis began using the second stage of IPA (Smith et al, 1999) where themes were identified which characterised each section of the text (see Appendix M). The first set of text containing quotes regarding high job satisfaction people was coded with themes which arose from the participants' transcripts so that a summary could be produced which included the theme, the participant and the page/line number containing the quote. The sets of text regarding high growth need strength people and people who take up training opportunities were then coded using the themes used in the first set of text and any new themes which emerged were also coded. 132 themes emerged in total and these are shown in the form of a list in Appendix N.

Clustering

This stage of IPA involves the introduction of structure into the analysis (Willig, p.55) and the researcher returned to the data, re-examined the first set of themes and thought about them in relation to one another. During this process the researcher re-read transcripts and listened to the tapes whilst combining and renaming themes, a process described by Willig as 'cyclical' (p.58). It was important during this stage of refining the data that the essence of what was being said by participants was not lost by combining themes without reference to the participants' own stories. As this refinement progressed patterns in the words and phrases began to emerge which were common to participants' discourses and to the 3 sets of data relating to the three groups of people. The themes were renamed as 'personal characteristics' such as energetic and forward thinking and were then placed under Master Themes such as 'Achievement', 'Change' and 'Development'. To begin with there were a number of themes which could not be categorised but with this gradual refinement came more emerging themes, until eventually after several repetitions of the process, 46 themes were produced which could be fitted into 12 Master Themes.

Following further examination of the transcripts, the number of themes was reduced to 35 (see Appendix O). Finally, the Master Themes were renamed to encompass the personal characteristics described by the themes. There were 8 final Master Themes identified and these form the basis of the next section. For example, the master themes 'Change' and 'Development' were combined and included a refinement of the two under which there were 13 personal characteristics. Thus the master theme became 'The Importance of Change and Development' and included 5 constituent themes. Separate summary lists of codes were made for each of the three sets of data so that the personal characteristics of people who are perceived as high in GNS, high in job satisfaction and those who take up learning opportunities more readily, could be compared for similarities and differences. An example of such a table is given in Appendix P.

Relating the Findings to other Research

The findings of the current study were related to Studies 1 and 1a and to the research described in Chapter 1, with particular reference to the development of the Job Characteristics Model and the descriptions given by Hackman and Oldham (1974, 1975,

1976) of GNS as an individual difference factor, and the characteristics of people who they believed were high in that factor. The researcher then sought evidence from literary sources on personality and values and similarities and differences in these and the findings of Study 2.

Ethical Issues

This study was conducted with due regard for the ethical principles and guidelines laid down by the British Psychological Society (2000) which are described on page 46. In addition to this there were certain additional considerations, given the nature of the study, which asked participants to examine their own personal worlds and describe them to the researcher. Qualitative data analysis is by nature an interactive process between the researcher and the participants and may heighten the self-awareness of both parties (van Manen, 1990). Participants gave their informed consent for the interviews to both take place and to be tape-recorded. They were told that their anonymity would be assured but that some of the content of the tapes would be reported verbatim, changing names and demographic details where necessary. Participants were given time at the end of the interviews to discuss the process and to voice their feelings about what had been said. They were also told that they could contact the researcher at any time in the future should they have any further issues they would like to discuss.

Findings of Study 2

In this section the findings of the study are reported as descriptions of three groups of people, those who were perceived by the participants as deriving high job satisfaction, taking up training opportunities more readily and having high GNS. Here, elements from the data which relate to the three groups are organised in such a way that the interviews with participants are re-created and their meanings and recurring themes are located within an analytical framework (Denzin, 1989).

Each section follows the same format by outlining the areas to be addressed and begins with a brief overview of the consensus of opinion, expressed by the participants, concerning each area. The 8 Master Themes form the next paragraphs and within these, some of the constituent themes which emerged for each of the three main groups are shown in *emboldened italics* in the text. The personal characteristics described are given

here as narrative extracts from the interviews which were felt to be appropriate for representing the reports of the participants' experiences. For ease of reading the number of extracts is restricted to one or two for each constituent theme and the number following the participant's name indicates the first line in the transcription from which the extract is taken. A summary of the constituent themes is given at the end of section 3 and the links between the three groups of people are set out in the 'Discussion' section of this chapter.

1. High Job Satisfaction People

Six out of the eight participants said that they could predict at interview those people who were going to derive high satisfaction from their jobs. All were asked what it was about people that lead them to think that and words used to describe people were, approachability, confidence, enthusiasm, passion, interest and energy. Whatever words participants used, most said high satisfaction people had more than just ability to do the job. The six participants who said they could predict high job satisfaction said there was just 'something about people' for example, Nigel said *'you shouldn't say "gut reaction" but sometimes it's the persona they give off'* (210).

All participants said that the way people felt about work in general contributed to the level of job satisfaction they derived. To illustrate this John talked about the mining communities in South Wales or Nottingham, and said *'mining wasn't just a job it was a whole way of life to these people'* (113). Sue also referred to this work ethic implying that it holds true in contemporary society *'If you have come from a family where there is a strong work ethic then I would imagine that would filter down'* (93).

Other participants referred to peoples' attitudes towards work in terms of whether they just came to work for the financial rewards it could bring. However, all participants agreed that job satisfaction meant different things to different people, and the factors affecting satisfaction in work included working conditions such as the physical working environment, salary and other rewards, areas such as the level of communication between staff and management and being valued for the work in terms of the amount of respect received. The paragraphs which follow show the master themes and their constituent themes which emerged from the interview data.

1.1 The Role of Achievement

Participants described people who they perceived as deriving high satisfaction from their jobs as having *a desire and capacity for achievement*. They said that the desire to achieve was something they themselves had observed in others who seemed to thrive on the achievement of results. They also talked about peoples' capacity for achievement in terms of taking on high workloads and meeting deadlines. Gemma talked about peoples' attitude to work in general and how much satisfaction they derive *'somebody who works because they genuinely love the job or want the stimulation would get the satisfaction from what they achieve and how they achieve it'* (101).

Participants also talked of achievement in terms of success and said that people showed a high level of *determination to succeed*. Brenda said *'Yes, I think if you look at somebody who knows they want to progress in whatever field they are doing, then they will have a completely different attitude to somebody who is just doing something for the sake of the job. Their determination to succeed, the amount of effort they put into it I think is all affected by it'* (108). People who derived high satisfaction from their jobs were also described as having *ambition* in that *'they are very ambitious and want what they are doing to be seen by people to progress their careers'* (Anne, 192).

1.2. The Importance of Change and Development

Clarity of sense of direction refers to an individual's knowledge of where they are going in terms of their career, the ability to set goals for themselves and to be able to visualise how those goals are to be met. John describes a person as *'has a purpose, has enjoyed a defined career'* (64) and Trisha says *'they are very clear about where they personally want to go, and I think they are very clear about where the company wants to take them, and whether those two things go hand in hand because it's about "either I'm going to stay here until I have completed that part of my career progression or in actual fact I recognise I have to move to satisfy all my career needs"'* (62).

Participants also referred to people who derive high job satisfaction as having a *desire or need for personal development*. Bob described such people as those who are *'keen on developing their talents, their careers and earning reasonable amounts of money'* (160). Participants also talked about the importance of *making a difference to the*

business, in other words being able to change things which would be of benefit to the organisation. Anne said that one of the similarities shared by people who derive high job satisfaction is that *'they are driven, that drive – that want to make a difference to the business'* (191).

1.3. The Role of Learning

The constituent themes included in this master theme were the desire/need for learning, individuals' receptivity to learning, curiosity and seeking new experiences. However, none of these themes emerged from the participants' accounts of people who derive high satisfaction from their jobs.

1.4. The Importance of Personal Knowledge and Skills

Participants talked about the level of *confidence* which individuals exhibited as being a key factor in high job satisfaction people for example Brenda said *'I think sometimes it's a confidence thing, not just confidence on its own but that's certainly a factor to consider. Certainly if somebody comes to you [and] they have got that confidence in them and you think "well, yes, I think they will be quite satisfied in that role" '* (149).

Participants also talked about *competence* both in specific areas of peoples' jobs and as an overall impression of people who derive high satisfaction. Anne said *'I think xxx is more competent and she's not afraid to pick up the phone and talk to people'* (81).

All participants said that *experience and knowledge of the job* were important, as if an employee felt unsure how to actually perform their role or had little experience in their job, they could not be expected to derive high satisfaction from it. John compared people who he knew, one who derived high job satisfaction and one who did not. He said that the one who had high satisfaction had *'enjoyed a defined career, had defined a career at the top of the company that he is in and also he enjoys a lot of authority'* (64). Later he said *'I can look at their background, the way they are coming across that way when I am talking to them, they are answering the technical questions'* (164).

Interpersonal skills were described by participants as being something they had observed about people who derived high satisfaction. This included communicating with others on both a one-to-one basis and making presentations to groups. Anne said

that such skills came from past experience of dealing with people *'she is not afraid to pick up the phone and talk to people and I guess perhaps some of that is to do with the sort of industry she worked in before ... she worked in a hotel in the leisure industry and I think you're dealing with people all the time'* (81).

Sue was the only participant to mention intelligence as a factor in job satisfaction, when asked if there was anything about people who had high satisfaction which set them apart from others she said *'I always tend to equate it, and it's probably wrong, but probably higher levels of intelligence or a higher degree of common sense in a lot of cases. I don't think you can separate the two out, but if you have the two together it's a very powerful combination'* (114).

1.5. The Importance of Commitment and Loyalty

The rewards derived from work are included in the master theme 'importance of commitment and loyalty' as participants spoke of rewards as being a way in which the organisation demonstrated its loyalty to the individual. For example, when Bob was talking about individuals having the scope to make a difference, his last comment was that people who get high job satisfaction are those who, amongst other things, *'feel what they do is valuable'* (87). To him, that sense of doing something valuable was reflected in the organisation's treatment of the individual in terms of either financial reward or allowing the individual to take responsibility. Bob said he felt people who derived high job satisfaction were those who were *'earning reasonable amounts of money'* (160) but later in the same text he talked about the importance of the organisational climate in how much or how little job satisfaction is derived by staff. He said *'Businesses that give people scope to contribute, make a difference and use the qualities that they [the individual staff members] have got, will be businesses that find those individuals really deliver for them. Organisations that squeeze the life out of people and command and control the environment, typically will not get any long term return'* (100).

Anne talked about a particular team into which she had recruited several people and said that the reason they got so much satisfaction from their jobs was *'they are really passionate about what they are doing and that passion is shown in their commitment to the company, their commitment to the job and to each other really'* (207). This is an

example of how participants felt that *organisational commitment* was something which was shared by people who derived high satisfaction from their jobs.

1.6. Level of Activity

Sue said that the word that described people who were high in job satisfaction was '*energetic*' *'I suppose energy is the word that describes them, lots to do, not enough time in the day'* (49). Ann's account described this in more detail *'I think in common with some of the other people in that particular department, for example where I recruited a few people last year, they have a lot of energy, they are all quite young, the average age is probably 26-27, very young, lots of energy, very competitive, they are driven, that drive, they want to make a difference to the business'* (188) and later *'I think it's passion really and energy'* (210). Anne also mentioned *competitiveness* in the above quote but she was the only participant to do so when talking about people who derive high job satisfaction.

1.7. The Role of Motivation

Participants described people who have high satisfaction as not being afraid to speak their minds or having a level of *self-belief*. When describing people he knew who derived high satisfaction from their jobs Nigel said *'all three of them believe in speaking their minds if they don't agree with something, and they can all use very good counter arguments'* (175) and Sue described people as *'prepared to take a stand in what they believe is right'* (107). Nigel was asked whether there was anything else about people who he felt derived high satisfaction from their jobs and was adamant that *self-motivation* was a common factor, he said *'they have got lots of self-motivation, lots of self-motivation – I can think about the three that I'm thinking about. They are very self motivated and they kind of have the attitude of what you put in is what you get out'* (170). Later he said *'they are self-starters in effect'* (178) and when asked to define a self-starter he said *'They are probably in the group of people, if you have a group of people and you were to bring something new to them, they [the self starters] would probably be in the minority by saying "that's really good, let's get on with it, I can see the benefits to the business" straight away. Or if they didn't see the benefits to the business say "I'm not sure what benefit this has to the business, it's a great idea but what do we get out of it" and they will go on and do it and you know you can just say to*

them “this just needs to be implemented across the area” and you know it will be done and you don’t have to check it because they have gone and made sure and done it to their best, because they have very high standards that they set themselves’ (185).

All participants talked about the importance of *stimulation from colleagues and the work itself* in the attainment of high job satisfaction and Gemma talked about the things which gave her personally, satisfaction in her job, she said ‘*the variety of what I do each day*’ (28) and described how this made her feel ‘*The feeling that when you get home you have achieved something and I also get a bit of a buzz at the end of a really busy day and I have achieved lots and I go home feeling good and I’ll be driving along thinking “Oh, that was a good day”, and equally when I wake up the next day to come to work I switch on and think about what I’m going to do and prepare myself mentally for the day ... I want to come to work*’ (38).

1.8. Attitude Towards Life in General

Some participants said that one thing which characterised people who derive high satisfaction from their jobs was a *positive attitude* towards life and work. Nigel said ‘*to a degree quite optimistic about everything, they can always see the good side of every project they go into*’ (178) and Gemma said that one of the people she thought of as having high satisfaction was ‘*a more positive person*’ (64) and she speculated as to whether this was linked to the fact that that person had a happy home life. Trisha described this positive attitude in a colleague ‘*she is one of those people who can generate that kind of enthusiasm for absolutely everything*’ (160).

In describing people who have high satisfaction, participants said that all of the people they thought about were more *forward thinking* and seemed to have a *vision* about where both they and the organisation should be going. Sue said that people who seem to have high satisfaction are able to ‘*see the bigger picture... they don’t just look at the day-to-day stuff, they think about where the organisation’s going*’ (108).

Participants talked about people who had high satisfaction as being ‘open to ideas’, ‘open to people and why they do things’ and ‘open to change’. Nigel said ‘*I think she’s definitely generally open to people and why they do things*’ (130) and later, ‘*They are*

quite open to new ideas as well' (175). He also said they 'believe in speaking their minds' which implied an *openness* in communication.

Finally, participants described people who derive high satisfaction from their jobs as seeming to be enjoying everything they did. Anne said '*he just really enjoys taking on the high work loads and being able to show what he's done*' (179) and John said '*the one who appears to be enjoying the role has enjoyed a defined career... And he also enjoys a lot of authority*' (63). Brenda said that a colleague seemed to really love his job and she asked him what it was he enjoyed about it. He said that it was because '*it was quite a fast moving job and he made and decisions and he had to make the decisions quickly*' (77).

2. People Who Take up Training Opportunities More Readily

All 8 participants said that, in their experience, there were people who took up training opportunities more readily than others. Gemma illustrated this when she described what she meant when she said there were some people who were always '*hungry for knowledge*' and would always want to be learning (275) '*I think some people are happy to plod along in the same job all the time and it's got it's parameters and they will work within those and they are quite happy to keep doing that. Whereas you have got other people who will always want to advance themselves and go and find out what their colleague is doing and maybe see how their job fits into the bigger picture*' (282).

2.1. The Role of Achievement

Participants all said that one of the things that typified the people who they thought took up learning opportunities more readily, was a *desire and capacity for achievement*. Bob said '*there is a desire not to be in the situation of status quo for any longer than they need to be*' (278). He was saying here that some people have a desire to achieve something more than others and to do certain things and move in certain directions which might set them apart from other people.

Participants also described this group of people as having a certain *determination to succeed* and Ann said they really wanted to be good at what they were doing. Brenda said that she thought that if someone asked about training at interview it would show

her *'how sort of determined they are to succeed'* (175) and Bob elaborated on this by saying that there were people who were happy to *'put themselves in situations where they may look silly for a while, they may even fail before they succeed'* (274).

2.2. The Importance of Change & Development

Sue said that there were people who committed themselves to training and they were the ones who *'want to go somewhere, they are not just going on a course for the sake of going on it, they have got their own individual development plan in mind and so they have a plan of where they will be in 5 years time'* (218). This illustrates the theme *'clarity of sense of direction'*.

Nigel described what he saw as the *malleability* of people who took up training opportunities and said that *'they are at a stage where you know you can mould them to be what you want them to be ... they have got the room for growth and they show the potential to be moulded ever so easily into what we want'* (377)

2.3. The Role of Learning

All participants said that a *desire or need for learning* was common amongst people who took up learning opportunities more readily. This was illustrated by Anne who described such people as *'they want to know more and they want to be good at what they're doing'* (316) and John said *'you will have those who are just interested and who want to do it because they just like learning'* (287). Brenda said they have *'a desire to want to be trained'* (212) and that *'they are the more willing learners, when you get them in a training environment they are the people answering questions, asking you questions'* (215).

Gemma said that some people go into training with the attitude that they don't know everything to start with and that *'we can all learn something however small it might be'* (239). Anne said that people who took up training opportunities were *'more curious and just receptive'* (327) and reiterates this later with *'it's their attitude towards training generally, how receptive they are'* (345). Nigel described this sense of *receptivity* by saying that there were people who *'show that they have room for growth'* (375).

Anne had said that people who took up training opportunities had more *curiosity* and elaborated on this by saying that *'they want to know what the opportunities are for them'* (330). Participants said that people who take up training opportunities are generally the ones who are always *seeking new experiences* and Bob said they were characterised by a certain 'restlessness' and that they didn't want to be in the same situation any longer than they needed to be (278).

2.4. The Importance of Personal Knowledge & Skills

Sue said there was a certain *confidence* about people and that they would have thought about what they were going to *'get out of this course and how they were going to apply it to the job and to move them forward'* (244)

Again, *interpersonal skills* were felt by all participants to be typical of this group of people. Nigel said that such people have good interpersonal skills and described one person he knew saying *'his people skills are an exception to the rule'* (370).

Ann described the characteristics she would look for in people where the organisation offered the opportunity for training *'we were looking for people who could build up relationships with the client, very good communication skills. We were also looking for people who could present information very clearly'* (284).

2.5. The Importance of Commitment & Loyalty

John described the things which set people who take up training opportunities apart from others and said *'you will have those who genuinely believe that it is for the good of the company and themselves so it is a mutual thing'* (285). Here he illustrated the importance of both *personal and organisational commitment*. Trisha said that when choosing people to go on training schemes in her company she would be looking for 'commitment' but that to her this would be demonstrated by what they have either done in the past or what they do in their spare time *'you can see from their background that they might get committed to that type of thing. So you might see that they are committed to a church choir or they are committed to a school rugby team or they are a big fan of Manchester United so you know that people are actually committed to things and so they illustrate it in that way'* (317).

John made an interesting point about how organisations can show their commitment to individuals when he said that in choosing people for training, he would be more likely to choose the one who has not had the opportunity to go on a training scheme before because *'training is also a motivator for people as well, so let's get the one who hasn't had any training and push them up'*. (268).

2.6. Level of Activity

The themes 'energetic' and 'competitive' did not emerge from this set of data.

2.7. The Role of Motivation

Bob illustrated the theme of *self-belief* when he said that individuals who take up training opportunities are those who *'believe that it's important for them to do certain things and move in certain directions'* (277).

3.8. Attitude Towards Life in General

All participants said there were people who took up training opportunities to enhance their *personal image*. John said that this was simply because it would look good on their CVs and described them as *'CV enhancers'* (285).

Finally, all participants indicated that people in this group demonstrated an *openness* to new experiences and Brenda said that the type of person she would look for when selecting for training would be someone who would say that they *'welcome new experiences'* (195). Bob said such people were *'around an openness'* (272) and that *'they are happy to look at doing new things'* (274).

3. High Growth Need Strength People

All 8 participants said that being high or low in 'growth need strength' was a 'good' way of describing people but 2 said that the term needed further clarification. All found it difficult to find other words to describe GNS. Bob said it sounded like *'self-actualisation'* and clarified that by saying *'make the most of yourself in business terms'* (353). John said *'a need to learn, developmental needs'* (359) and Anne said *'self-development'* (389). Nigel said that he would describe it as *'the potential of what they could contribute to the business'* (493) and Sue said *'for me it's about personal*

motivation and growth' (280). All however, said that GNS was quite different to 'autonomy'.

All said that a particular 'type' of person came to mind when they were discussing the meaning of GNS. Gemma said she thought of someone who *'loves coming to work, loves to learn and likes to pass that knowledge on to other people, possibly, and generally enjoys what they are doing'* (381). Sue said *'that is a person who needs to move on, who needs to grow and who needs to be challenged, who needs to grasp skills and constantly up skills and move on to be constantly challenged – to get the level of satisfaction they need'* (286). Brenda said that describing people as high in GNS would tell her that *'they wanted a lot of career development, they needed constantly stimulating with new things for them to keep learning more to move on to bigger and better things'* (280).

All participants said that they thought GNS was part of peoples' personality but that the level of GNS people possessed was affected by upbringing, the environment in which they worked and what else was going on in their lives at any particular moment in time. Brenda said *'The environment can either suppress it or expand it so if you were brought up in an environment where you will succeed, perhaps people around you have succeeded it will go from there and that will develop on as you sort of progress in the adult life'* (425) and later when asked if she thought peoples' level of GNS could change depending on what they were doing she said *'To a certain extent, yes, I think that if you have got a lot of growth needs then I think in a way you will always have a high need, but there will be times in your life when it will not be as great, for example if you are in an environment where you are learning, you're being challenged so therefore you're growing. You'd be sort of content with that in the interim until you make the next big step in your life'* (439).

Sue described GNS as self-motivation and said *'I think other things in peoples' lives may overtake the initial motivation or whatever it is they may have. They may start off with nothing in terms of motivation and end up with being hugely ambitious. I think it can work both ways. I do think it's not something that works in isolation, it must be affected by other areas in your life.... I would say life changing events, it could be for a*

woman, it might be marrying and having a family and that becomes a different priority. Some people might suffer an illness or people may just decide that, like lots of people have done, that it's a rat race, it's not for them. Having been ambitious and very keen to climb the ladder, suddenly it's not important' (458).

3.1. The Role of Achievement

Participants said that people high in GNS had a *desire and capacity for achievement*. Anne said that people high in GNS want to achieve (397) and Nigel said that high GNS people were '*probably very effective in the role that they do but they need something else to go on to*' (501). Sue said that a high GNS person was what she would describe as a 'high achiever' (430). Some participants also said that this group of people had a *determination to succeed* and Brenda said that she believed that she herself was high in GNS and she had '*just a desire to succeed and to prove that I can succeed and that I can do well and to get somewhere that I want to be*' (299). Sue said that a person high in GNS doesn't believe anyone should stand in their way (392). John described this desire and determination as *ambition* and said that he saw a person high in GNS as being career minded and career motivated (430). Trisha said that such people had a high potential in anything '*from being very good at the job that you do to being very good in the next move to make, the next move up the status ladder or succession ladder*' (416).

3.2. The Importance of Change and Development

Some participants said that high GNS people demonstrated a *clarity of sense of direction* for example, Sue said that people high in GNS are very clear about what they want for themselves (438). However, this was also illustrated by participants when they discussed the individual's *desire for personal change & development*. Nigel said that a person high in GNS would be typified by his or her need to move on to work which would '*give them some new challenges or new development or something to feed them with to grow or give them more of a learning curve*' (503). Brenda said that they would want a lot of career development (280) and Anne said that there were people who, in work, really want to develop themselves (677).

Participants were asked what type of people they would look for when recruiting, in order to fill jobs with certain specifications. These were all aspects which were

proposed by Hackman and Oldham (1974) to be the job characteristics which would be sought by people high in GNS. When John was asked what type of person he would look for if the job offered opportunities for personal growth and development he replied that they would be people who wanted to broaden their experiences *'so if they are asked to work and learn a new technology or computing package or something like that they have got to be interested by it rather than it causing them a sleepless night and upsetting them'* (427). Nigel also mentioned this **ability to cope with change** when discussing the type of person he would look for to fill a role which offered opportunities to learn new things from the job *'someone who isn't fazed by change because in this day and age within retail, changes are ever increasing because we have to look to remain more competitive than our competitors in the high street and always have the competitive advantage, so they can't be adverse to change'* (578).

Trisha described the type of person who she would look for if the work were described as stimulating and challenging, which was again a characteristic which Hackman & Oldham (1974) suggested that a person high in GNS might be attracted to. She said *'you can see that they are enthused or that they have moved something on'* (448). Nigel talked about the type of person he would look for if the job offered opportunities to be creative and imaginative and said *'a lot of innovation into how they could make things new and different, how they could make things better'* (601). Anne talked on this subject of **making a difference to the business** and said she would look for someone who could show *'what differences have they actually made, how they delivered them and have improved things as a result'* (446).

Nigel said that he thought that the percentage of people *'brought GNS with them to the job'* but there were others who developed GNS through their working lives. He said *'they may be graduates who show the potential to do really well in management or they may be people who have come from finance or anywhere that have management skills that are transferable but they just haven't got the skills or the things that we want to mould them into, but they are still quite malleable to change them into the way we want them to be'* (646).

3.3. The Role of Learning

Anne said that to her, someone high in GNS would have *a desire or need for learning*, they just *'want to learn, they just want to learn, take on board new information and skills'* (400) and Gemma said such a person *'loves to learn'* (380). Nigel said that the type of person he would look for to fill a job which was stimulating and challenging would be one who *'likes to learn'* (382) and Sue said that there were some young people who came for interviews in her organisation who she knew straight away *'have that something, you know they listen, they learn, they move it on'* (529). Anne talked of a *receptivity to learning* when describing the type of person she would look for to fill a job which offered the opportunity to learn new things from the job. *'I would look for someone who would show me what they had learnt perhaps in the last 6 months of their [previous] job, so some tangible evidence of new skills that they have developed, new information that they had taken on board and how they've actually used it'* (434). Nigel said that people high in GNS were *'the type of people that look at all six people's view points and pick up the best things out of all of them and then become the outstanding performer at that particular aspect'* (756).

Anne said that not only were people high in GNS able to keep focussed they also had a certain *curiosity* and they wanted to learn (399). She said she would recognise such a person outside the work setting as probably quite curious and quite challenging to talk to (672). Nigel said that people he would look for to fill a job which offered opportunities for personal growth and development would not only like to learn but to *'question things as well'* (624).

Some participants said that people high in GNS would constantly be *seeking new experiences* both in and out of the workplace. John talked about someone who might be attracted to a job which offered opportunities for personal growth and development, as wanting to *'broaden their experiences and horizons within the company'* (426) and Nigel said people high in GNS would be looking for new challenges (503). Brenda said from her own point of view, being high in GNS, every time she learnt something she wanted the next stage, *'I want that bit more'* (293).

Participants were asked what sort of hobbies or interests outside work a person high in GNS might have and the majority said that they could not see such a person as 'sitting around watching television'. Gemma said they might 'surf the net' or do some sort of research or perhaps learn photography. Brenda said that they would be very active people who would '*try new and innovative things*' (462) and Nigel said they would either be very academic or very sporty but '*they will probably do things to learn and as soon as they drop one thing after learning it or doing it they will go on to another one again, to do something else*' (788).

3.4. The Importance of Personal Knowledge and Skills

Brenda described a person high in GNS '*they would be a very **confident** person, not direct but they would quite like to tell you what they have done and what they are doing. I think that would sum them up*' (491). Gemma talked about the type of person she would look for when interviewing someone for a job which involved stimulating and challenging work '*I think you would be looking for someone who's very confident and self-sufficient. Somebody who could take command of a situation*' (411).

Anne described the type of person she would look for when interviewing for a job which offered chances to exercise independent thought and action, as having a degree of **competence** in their role. '*I would look for someone who can demonstrate that they have made quite senior level decisions, perhaps they have developed strategies in organisations in the past and that they can justify why they developed those strategies, and how effective those strategies had been*' (425). Nigel said that someone high in GNS would probably be quite effective in their role (500). Sue described this **experience and knowledge of the job** and said that she would see someone high in GNS as somebody who '*has some sort of credibility in terms of your own expertise and who brings other skills*' (435). Participants also said that people high in GNS have good **interpersonal skills** which she felt were important in any job. Sue described these as the 'softer skills' or 'people skills' (437).

3.5. The Importance of Commitment and Loyalty

Nigel saw himself as high in GNS and said that his own **personal commitment** was important '*If there is something I really want to learn that [need for growth] is more*

intense than it would be normally. I would be pulling out all the stops to find out about it, finding out from whoever, what information I could get' (725). Later he talked about a manager he knew who did not work in HR but had completed a higher degree in HR because he wanted to understand the role of HR people (784). Also said that he would recognise at interview someone who was high in GNS because they would be *'someone who does academic study because they have got the commitment and drive and tenacity to do it whilst they are working'* (814). Trisha said that someone high in GNS would be *'committed to life and work'* (651) and Bob implied this personal commitment when he talked about the type of hobbies a person high in GNS might have *'musical instruments, learn a language, maybe get involved in local politics... I don't see these people in local community charity type stuff, it's got to have an individual frame of reference for them'* (562).

Bob talked about the type of person he would look for if the work was described as stimulating and challenging and said that they would need to have a level of **organisational commitment** and described such an individual as *'that would be an individual who is I think ambitious, interested in being successful, maybe committed to being successful, be able to put the work into being successful if we're talking in a work context which I guess we are'* (400). John talked of the organisation's commitment to the individual when he was asked, if he was told someone was high in GNS, what that would tell him about a person he said *'that possibly if we wanted to retain their services in the company we are going to have to invest in them'* (370).

3.6. Level of Activity

Participants described people high in GNS as being very active both in work and in the activities they pursued. All were asked what type of hobbies or outside interests a high GNS person would have and the majority described highly **energetic** sports such as snowboarding and bungee jumping. Brenda said a high GNS person would be *'a very active person... the kind of person who take up a new sport regularly'* (462) and Gemma too said such people would be fairly active. Outside interests were not however, described as exclusively high-energy sports. John felt that they would take up any pursuit be it academic or sporting, which occupied their minds (583). Nigel said that

people high in GNS would always be taking up new interest because if they did not *'they feel as if they are going brain dead or something'* (791).

Anne said that people high in GNS are *'quite competitive in nature and they might be doing sporting activities that involve competition'* (662). However, she said that this **competitiveness** was also exhibited in the work situation *'I think there are people who are probably really competitive in the workplace and really want to develop themselves but outside of work ... they're different'* (676).

3.7. The Role of Motivation

Participants were asked if they thought people acquired GNS or whether they brought it with them to the job. Sue said she thought there was an element of both but *'I think about 70% of it is probably whether you've got it or not'* (386). When asked to elaborate she continued *'You have got to **believe in yourself**, in your own personal attributes, in your technical ability whatever that may be, and you don't believe anybody should stand in your way'* (391).

Participants said that a person with high GNS would have a high degree of **self-motivation**. Anne said that when she thought about people who might be high in GNS she thought of people in marketing jobs and said *'they have their whinges and moans like everybody else, but overall they can get over that and they can still keep focussed on what they want to do'* (398).

Nigel described the type of person he would look for if a job offered the opportunity for personal growth and development and said *'I would probably look for someone that could get on with development themselves as well, because a lot of personal development... the onus is on you a lot of the time as the individual, to make sure you get the development'* (618). In a similar context Trisha described the type of person she would look for if the work was stimulating and challenging and said *'stimulating is something you recognise in an individual because you can see they are enthused or that they have moved something on, or that they have felt good about something, they are motivated'* (447). When Sue was asked to think of an alternative name for GNS she said *'for me it's about personal motivation and growth'* (280).

John said that the two people he talked about who he felt not only derived high satisfaction from their jobs but also were high in GNS. He implied that they seemed to get a lot of *stimulation from colleagues and the work* ‘*they both like ideas, they both like concepts and they both like being challenged*’ (615).

Brenda described people high in GNS and said ‘*they wanted a lot of career development, they needed constantly stimulating with new things for them to keep learning more and to move on to bigger and better things*’ (280).

3.8. Attitude Towards Life in General

Sue said that she saw people high in GNS as having ‘*quite a balanced view on life, but works hard and plays hard and has a family life*’ (505). Bob discussed the issue of balance at length and said ‘*there are probably people who are “growth need strength” in business terms but maybe if you look at the rest of their lives you will find a balance there*’ (359) and later ‘*This is a balanced individual who doesn’t channel all the growth need strength stuff into 16 hours a day*’ (551). Trisha described someone high in GNS as ‘*somebody who has strong standards for the life because I think it [home and family life] has to interact, because otherwise it wouldn’t work for them. I think you need to get it all balanced quite right but actually think that here is part of it, that both parts are just as important as each other*’ (653).

Brenda said she thought someone high in GNS would have a ‘*desire to see the nicer side of life*’ (465) and when asked to elaborate said ‘*nice home, nice cars, they have quite a high desire I think in a way*’ (471). She was then asked if a person’s need for personal growth included the things that money can buy and replied ‘*Yes, quite materialistic I suppose*’ (475) and later she summed up how she would recognise a person with high GNS ‘*if you were just talking to someone on a one-to-one basis I think you would quite easily be able to tell by the way in which they ... what they are doing with their life, perhaps looking at them through what I have said earlier about materialistic nature*’ (484)

Participants were asked what sort of characteristics high GNS people would look for in a job and Sue said ‘*The ability to be able to direct whatever part of their business the*

way that they want to do it. I think to people like that, a greater level of responsibility and freedom to do what they think is right in their area of the business, would be a major attraction. Plus probably a huge salary a big flashy car because I think that is important to certain people who have that drive' (483). Sue implied that this materialism was bound up with the **importance of personal image** to people high in GNS and Brenda said that such people 'quite a desire that people perceive them in the correct manner as well. What they see as correct – I'm not saying it is the correct thing in life' (470). Anne said that people high in GNS 'really enjoy what they are doing and they want to achieve and they have got a very **positive attitude**' (396).

John discussed the type of characteristics a person high in GNS might look for in a job and talked about how the technical and design staff in his organisation needed the creativity to design for the future. He said in his company a person may join as a software engineer and 'in 8 years time be designing IT systems, it's very creative' (558) and when prompted said that '**vision** is a good word for it as well. I think any truly creative individual has to have a vision of what they are trying to achieve' (563).

Sue likened GNS to being a 'high-achiever' (430) and when asked to define that term she said 'it's the ability to have a vision about the organisation, but also be very clear about what you want for yourself' (436).

Gemma described the type of person she would look for to fill a job which offered opportunities to learn new things from the work and said 'fairly adaptable, not somebody who's set in a particular mind set, they are going to have to be open to ideas, open to change' (440). Nigel echoed this **openness** when talking about the type of person he would look for if the job offered chances to exercise independent thought and action 'It's someone that has got to be open to ideas, but has got to be quite outspoken to a degree' (568). Finally, Anne described people with high GNS as really **enjoying** what they are doing (396) and Gemma said a high GNS person to her would be 'somebody who loves coming to work, loves to learn and likes to pass that knowledge on to other people possibly and generally enjoys what they are doing' (380).

Summary of Findings

The table on the following page shows a summary of the findings of Study 2. The themes which emerged from the interviews are shown on the left hand side of the table

and the three groups of people are shown at the top. An 'x' against each theme shows which were appropriate to each group of people.

Table 33

Theme	High Job Satisfaction People	People who Take up Training Opportunities More Readily	High GNS People
Desire for achievement	X	X	X
Determination to succeed	X	X	X
Drive	X		
Ambition	X		X
Clarity of sense of direction	X	X	X
Desire/need for personal change & development	X		X
Ability to cope with change			X
Making a difference to the business	X		X
Malleability			X
Desire/need for learning		X	X
Receptivity to learning		X	X
Curiosity		X	X
Seeking new experiences		X	X
Confidence	X	X	X
Competence	X		X
Experience & knowledge of the job	X		X
Interpersonal skills	X	X	X
Intelligence	X		
Creativity	X		
Rewards obtained from work	X		X
Personal commitment	X	X	X
Organisational commitment	X	X	X
Energetic	X		X
Competitiveness	X		X
Self-belief	X	X	X
Self-motivation	X		X
Stimulation from colleagues & the work	X	X	X
Balanced view of life			X
Materialism			X
Importance of personal image	X	X	X
Positive attitude	X		X
Forward thinking	X		
Vision	X		X
Openness	X	X	X
Enjoyment	X		X

Discussion

The aim of this study was find evidence from the world of work for the existence of GNS, it's nature and relationship to job satisfaction and its relationship to the taking up of training opportunities. In particular its purpose was find out if professionals in the field of human resources management had a notion of GNS when selecting people for employment and to see how well that concept matched Hackman and Oldham's (1975, p.163) idea of GNS of a malleable individual difference factor which affects the way people respond to jobs high in motivating potential.

The study used IPA to help interpret and describe these similarities and differences and the relationship between people high in GNS, those high in job satisfaction and those who take up training opportunities more readily and is highlighted by their shared/unique characteristics as perceived by the HR professionals. For example all three groups were described as having a desire to achieve, confident and competent in their abilities and a high level of both personal commitment and loyalty to the organisation. Each group is described as having a high level of self-belief and having an 'open' attitude towards life in general. However, unlike the high GNS group and the people who take up training opportunities, people who were perceived as deriving high satisfaction from their jobs were not perceived as having a particular desire or need to learn or seek new experiences from their work. Cain Smith (1992) suggests that there are many facets to job satisfaction such as pay, co-worker and supervisor satisfactions, so it is suggested here that although people who are perceived as deriving high satisfaction from their jobs may demonstrate a desire to achieve, this may not necessarily be through the process of learning. A further important point is that this study concerned participants ideas of what they thought *other people* might feel gives them job satisfaction and as such is highly subjective. Schneider et al (1992) suggest that when people think about job satisfaction they think of both past and present events (p.53) and that available opportunities are relevant to such satisfaction. It may be that although participants were asked to consider their past experiences of HR management their responses to the questions asked were more affected by their current experiences. They therefore may not have regarded the role of learning as important to job satisfaction as it may not have been a key factor in their current situation.

Participants reported that they perceived people high in GNS and those who took up training opportunities more readily as having a desire/need for learning, receptivity to learning, curiosity and seeking new experiences. This provides some support for a relationship between the two groups as suggested in the introduction to this thesis, and therefore confirmation that a test of whether or not GNS predicts people who take up training opportunities more readily than others, would be an appropriate method of concluding the current research.

Perceptions of Growth Need Strength

When discussing the meaning of Growth Need Strength all participants said that a particular 'type' of person came to mind. Although some of the characteristics described by the participants were also used to describe people perceived as high in job satisfaction and people who take up training opportunities more readily, certain characteristics applied only to high GNS people. What follows below are descriptors which could be applied to a person high in GNS, and is a composite of the participants' own words.

A person high in growth need strength is....

- Someone who has a desire for achievement, both in his/her work and in private life.
- Has a determination to succeed and has a clear sense of direction.
- Has a desire for personal development both in work and private life.
- Has a desire to learn, is receptive to learning opportunities and can be moulded by the organisation into what it wants him or her to be.
- Is curious about the things going on around her/him and actively seeks new experiences, is energetic and competitive both in work and private life.
- Is confident, relaxed and competent about his/her own abilities in work and has good interpersonal skills.
- Has a high level of personal and organisational commitment and values commitment offered by the organisation.
- Has a high level of self-belief and is self-motivated but also derives stimulation from colleagues and the work itself.
- Believes that personal image is important and has a sense of vision.

- Is open to ideas and those of other people and generally enjoys life both in and out of work.
- Is distinctive in that she or he has a balanced view of life, enjoys the material rewards that success can bring and has the ability to cope with change.

As suggested in the introduction to this thesis, the construct 'Growth Need Strength' has been poorly defined in spite of the fact that it has been tested in a number of studies. Studies 1 and 1a of the current research found support for Hackman and Oldham's (1975) prediction that GNS influences '*how positively an employee will respond to a job with objectively high motivating potential*' (p.163) and it is suggested that the results of Study 2 provide some support for GNS as a '*malleable individual difference factor*' (Hackman and Oldham, 1975, p. 163). Nigel (see page 112) said he thought the majority of people brought GNS with them to their jobs but there were others for whom it developed through their working lives. He himself used the word 'malleable' in discussing how he perceived people high in GNS as not necessarily having all of the right skills to do a particular job, but having the ability to be moulded by the organisation into what is required. All participants said that they thought that GNS was part of peoples' personality and was to a large extent determined by the influences of upbringing. However, they also perceived GNS as changing in level depending upon the environment in which people worked as demonstrated by Brenda's comment (see page 110) that '*the environment can either suppress it or expand it*'. Hackman and Oldham (1974b) suggest that it is possible that individuals' needs do indeed change or adjust to meet the demands of their situation (p.28), and that they become more 'growth oriented' when they are '*confronted with a complex job which seems to demand that the individual develop himself (sic) and exercise independent thought and action in his (sic) work*' (ibid., p.28).

The majority of participants used the term self-motivation when trying to define what they thought GNS might be and this is consistent with the conceptual basis of the Job Characteristics Model. The causal core of the Model are the three psychological states whereby individuals experience 'positive effect' (Hackman and Oldham, (see page 8) 1974b p.8) in that they learn that they personally have performed well on a task that they care about. Hackman and Oldham (1974a; 1974b; 1975; 1976) suggest that this

positive effect causes the individual to try and perform well in the future if he or she values the rewards which are gained as a result. Hackman and Oldham (1974b) describe this result as a *'self-perpetuating cycle of positive work motivation powered by self-generated rewards that is predicted to continue until one or more of the three psychological states is no longer present – or until the individual no longer values the internal rewards that derive from good performance'* (p. 8). Growth need strength, according to the model, is a useful way of conceptualising how individual differences among people moderate how they react to their work and is evidently perceived by Hackman and Oldham as the personal characteristic that is the root of the 'self-perpetuating cycle of positive work motivation' which they describe. The perceptions of the HR professionals are that GNS is an individual difference factor which is possibly part of peoples personality but which is affected by environmental and situational factors and this will be discussed in greater depth in Chapter 6.

The results of Study 2 suggest that people who are perceived by the participants as high in growth need strength share some personal characteristics with those who, in the experience of the participants, take up training opportunities more readily and this possible link forms the basis of the next study.

CHAPTER 5

Introduction

The work of Barrick and Mount (1991) suggests that the robustness of the five-factor model of personality provides a meaningful framework for formulating and testing hypotheses relating individual differences in personality - to a range of criteria including training and development (p. 23). The results of their study found that Openness to Experience, Conscientiousness and Extraversion were related to performance in training programmes and they suggested that future research should further investigate the relation of individual measures of personality to measures of training readiness and training success (p. 22). The current research investigated the possibility that the origins of the individual difference factor Growth Need Strength either lay in or were closely linked to Openness to Experience and because of this GNS too, may predict which individuals within the world of work are ready to take up training opportunities.

The results of Studies 1 and 1a showed that there was indeed a relationship between GNS (in 'job choice' format) and Openness to Experience and that Openness was a significant predictor of GNS (see page 72). The results of Study 2 provided some support for a relationship between people who were perceived by the participants as high in GNS and those who were perceived as taking up training opportunities more readily.

The results of the first three studies showed that there was a relationship between GNS, the values Achievement, Self-direction and Stimulation and the three work values. The three work values and the value Self-direction were found to be significant predictors of GNS and there was weak support for the value Achievement as a predictor. A relationship was also found between GNS and individuals who reported taking courses of study outside work, although the direction of the relationship would indicate that the higher the level of GNS the less likely this was. In spite of this, the fact that a relationship was found provides some support for investigating further the relationship between GNS and training and development activities.

It was suggested at the end of study 1a (see page 83) that GNS might be a personality trait and there was some evidence to support this from the participants' narratives described in Study 2. One of the requirements of a personality trait is that it may be used to explain why people are consistent in their behaviour across many different situations and over relatively long periods of time. The notion that individuals' personalities dispose them to act in certain, consistent ways appears to be fundamental to almost all theories of personality. As Thomas (1995) suggests, '*Without this underlying idea of consistency, the idea of personality seems to vanish, leaving peoples' behaviour unpredictable and perhaps no more than a reaction to the situations they find themselves in*' (p. 376). In a further step towards defining the nature of GNS, this study assesses the stability of the construct over time and whether it predicts a piece of observable workplace behaviour. To this end it was decided to measure GNS on two separate occasions to see whether peoples' scores for the measure of GNS were consistent and to test whether GNS predicted a readiness take up training opportunities.

Colquitt et al (2000) conducted a meta-analytic path analysis of 20 years of research into training and report that past research has focussed on the reaction, learning and behaviour change of trainees and the impact of these upon training effectiveness. They suggest that in line with more recent reviews of training research (e.g. Tannenbaum and Yukl, 1992), because the influence of variables such as setting and content of training varies among individuals, research should examine how personal characteristics relate to training effectiveness. One approach of such research has been to specify predictors of training motivation and their relationship to learning. Martocchio and Judge (1997) for example, examined the relationship between conscientiousness and learning in employee training. Mathieu et al (1992) examined the effects of career planning and job involvement on trainee motivation and Noe and Wilk (1993) examined the effects of self-efficacy and work environment on development activity. Tharenou (2001) discusses how research designed to explain participation in training, has not examined how motivation to train has actually predicted such participation. Tharenou proposed three explanations for how training motivation predicts participation in training and development and found that motivation through expectation and motivation to learn helped explain this behaviour. She suggests that future research should examine the antecedents of training motivation which as illustrated above, have been proposed as

both situational and individual. Tharenou further suggested that barriers to training such as high workload might moderate the relationship between motivation and participation but found no support for this.

The current study is designed to provide a model for how and under what conditions GNS predicts the taking up of training opportunities. An explanation is proposed for how GNS predicts those employees who will take up training opportunities more readily than others. Evidence from the literature regarding the relationship between personality and values indicates that an individual's personality determines the values she or he will hold and these values will determine how individuals will behave in a given situation. Roccas et al (2002) (see page 22) draw a distinction between personality and values which suggests that whereas personality traits are enduring dispositions, values refer to what people consider to be important and the goals they wish to pursue. One of the assumptions of Humanistic models of psychological understanding (e.g. Maslow, 1954; Kelly, 1955; Allport, 1961) is that people have the capacity to direct personal action and achieve personal growth. Maslow (1954) suggests that the need for self-fulfilment is a 'being' need which has no end-state, that it is fundamental to individuals to become everything they are capable of becoming. In this way the need for personal growth could be seen as part of peoples' personality which is manifested in different ways, at different times, for different people. The difference between such a need and a value is that a need leads a person to value certain things in his or her life and is necessary in order to realise those values, which are accomplished through goal-directed behaviour. It is proposed in the current study that GNS may be a personality trait that leads people to value certain things in their lives and as such they will wish to pursue certain goals. Thus, the behaviour that results from these feelings will reflect those values. Firstly however, it was necessary to ascertain whether GNS actually predicted general and work values for the data already collected and to do this the data collected for Study 1a was re-examined. A series of regression analyses was carried out with GNS as the independent variable and the values and work values purported earlier to be related to GNS, as dependent variables. The dependent variables tested were thus the general values (Schwartz, 1992) 'achievement', 'self-direction' and 'stimulation', and 'affective', 'cognitive' and 'instrumental' work values (Elizur, 1991) and the results are shown in Table 34.

Table 34

	Standardised Coefficients								
	Achievement			Self-direction			Stimulation		
	β	T	Sig.	β	t	Sig.	β	t	Sig.
Constant		18.41	.000		24.61	.000		17.02	.000
GNS 'job choice'	.22	6.03	.000	.24	6.55	.000	.16	4.28	.000

	Standardised Coefficients								
	Affective Work Values			Cognitive Work Values			Instrumental Work Values		
	β	T	Sig.	β	t	Sig.	β	t	Sig.
Constant		38.09	.000		26.04	.000		37.56	.000
GNS 'job choice'	.14	-3.76	.000	.26	7.17	.000	-.37	-10.42	.000

All six dependent variables are predicted by GNS and the results were highly significant for each one although the beta coefficients are not large. The coefficients for both affective and instrumental work values were in a negative direction, indicating that as scores for GNS were increasing, the scores for both of these work values were decreasing. Instrumental values include aspects of work such as pay and hours of work and affective work values are concerned with relationships with supervisors and co-workers and the recognition for doing a good job. As the discussion to Chapter 3 suggests, this is an understandable outcome as people who strongly value feelings of personal growth and development are less likely to value the aspects of work measured by affective and instrumental work values scales. Having found evidence from the data collected thus far for GNS as a predictor of values and work values, it was felt to be appropriate to continue the current research by devising a model that would test the predictive power of GNS.

Blinkhorn (1997) discusses test theory and its inception as a tool for either elaborating theories of individual differences or in the application of tests for practical purposes (p. 177). He points out that the uniqueness of test theory is that one of its main purposes is to '*provide indices of quality for commercial products*' (p. 175) and that in spite of the many technical advances in the measurement of validity, test theory has not made a great contribution to the understanding of individual differences. This, he says, is because the practical effectiveness of such tests has not improved over the years. Blinkhorn ends his paper with his ideas on prospects for the future of test theory and suggests that researchers should be finding ways to devise more effective tests with greater predictive validities. It is with these salient points in mind that the model for

Study 3 is proposed. In Figure 4, the conceptual model of training activity tested in this study is presented. The framework for the model is based on the work of Noe and Wilk (1993) who suggest that the influence of employees' perceptions of the work environment and self-efficacy, which includes employees' beliefs that they can cope with challenging situations (Bandura, 1977), is mediated by employees' perceptions of development needs, learning attitudes and perceived benefits from participation in development activities. The current model, however, suggests that employees' general and work values mediate the influence of their growth need strength on the taking up of training opportunities. The model further suggests that the mediating effect of the values causes employees to show an intention to take up training opportunities. The relationship between this intention and the actual taking up of such opportunities (training activity) is moderated by barriers to training. Although Tharenou (2001) did not find a moderating effect for barriers to training, conversations with the participating organisation for Study 3 revealed that the training manager had conducted a survey of non-attendance at voluntary training courses and found there to be a number of barriers reported by staff. It was therefore decided to include barriers to training in the model.

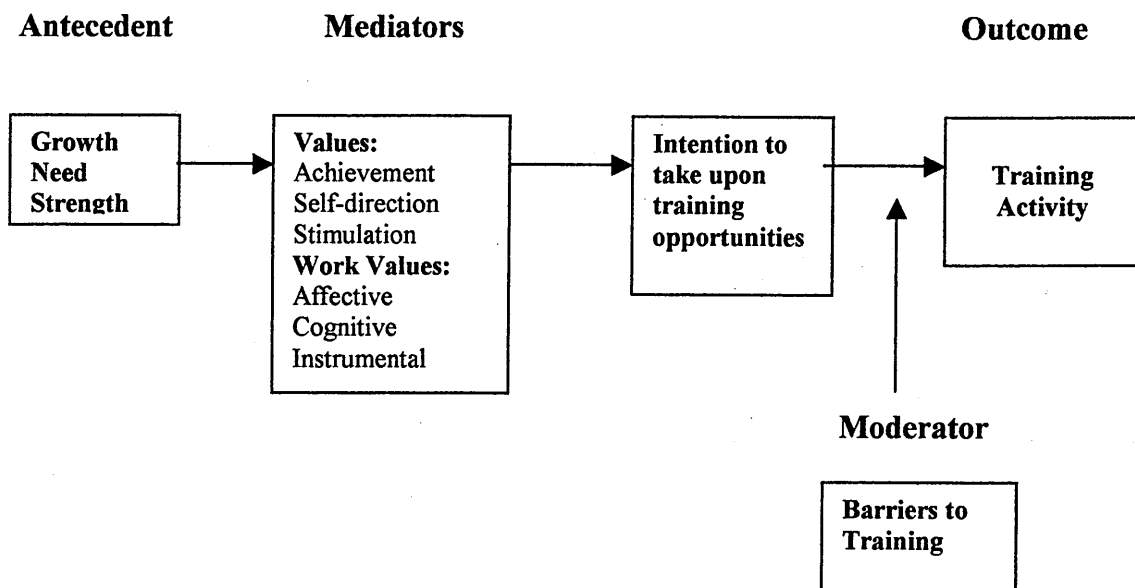


Figure 4. Conceptual Model of Training Activity

Antecedent

On the basis of the findings of Studies 1, 1a and 2, Growth Need Strength is shown as the antecedent to intention to train, as individuals who strongly desire personal growth and development are likely to seek such opportunities within the workplace, through the training and development courses offered by their employer. Training activity challenges employees to learn new skills and it is proposed here that GNS indirectly influences intention to training activity through its effects on general and work values, and the effect that these variables have on whether or not employees feel inclined to take up the opportunities offered by their organisation (intention to take up training opportunities). That is, the higher the individual's GNS the more he or she values certain goals, the more likely he or she is to display an intention to take up training opportunities. So GNS does not directly influence intentions to train but acts only via the mediation of values. Whether or not that intention to train results in actual training activity is moderated by barriers to training such as workload, location of training and attitudes of supervisors. In this way GNS and a lack of barriers to training will predict training activity (*Hyp.1*).

Mediating Variables

Baron and Kenny (1986) set out the mediating mechanism of variables and say that '*mediators explain how external physical events take on an internal psychological significance*' p.1176. They suggest that a variable can be said to function as a mediator '*to the extent that it accounts for the relation between the predictor and the criterion*' *ibid.* p. 1176. To illustrate the role of mediating variables, Baron and Kenny (1986) clarify the meaning of mediation by describing a model, which depicts a causal chain. This model is given below in Figure 5.

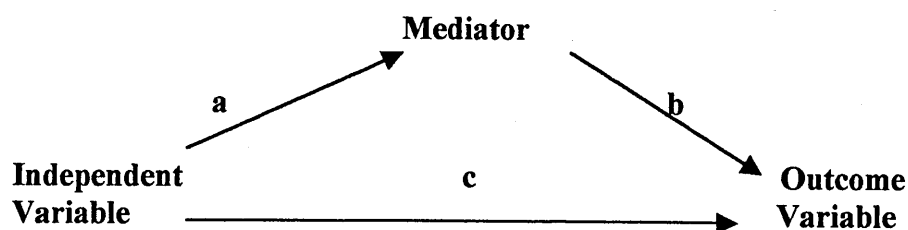


Figure 5. Mediational Model. Baron and Kenny, 1986, p.1176.

Baron and Kenny (1986) state that the model assumes a three-variable system where there are two causal paths to the outcome variable: Path c – the direct impact of the independent variable and Path b – the impact of the mediator. Path a is the path from the independent variable to the mediator. This model will be adopted in the current study. Noe and Wilk (1993) suggest that the decision to participate in development activities is '*likely influenced by employees' beliefs that development activity results in favourable outcomes*' (p. 292). The favourable outcomes they describe are increased chances for promotion, salary increases and recognition by managers. Here it is suggested that the decision to take part in training activities is influenced by the value employees attach to certain individual, social and workplace issues as guiding principles in their lives, which lead them to pursue certain goals. Therefore, following Baron and Kenny (1986) shown above, it is expected that there will be two causal paths to the dependent variable 'intention to train' – the direct impact of GNS, which would be expected to decrease if there is mediation, and the impact of the mediating values and work values. Further, there will be a causal path from GNS to the mediators. There are six potential mediators in the current model, the three general values Achievement, Self-direction and Stimulation and the three work values. The rationale for the mediating role of these is given below.

Achievement

Schwartz (1992) suggests that the defining goal of this value type is '*personal success through demonstrating competence according to social standards. Competent performance is a requirement if individuals are to obtain resources for survival*' (p.8). It is likely that individuals who are high in GNS will value the personal success described above, and will therefore be more ready to take up training opportunities offered by their organisation, as a means of gaining the knowledge and skills required to bring about that success.

Self-Direction

According to Schwartz (1992) the defining goal of this value type is '*independent thought and action – choosing, creating, exploring*' (p. 5). It is likely that people high in GNS will value such a goal and this will again lead to the predicted outcome as such

individuals will value making their own choices about the training and development they personally need to further their growth.

Stimulation

The defining goal of this value type is '*excitement, novelty and challenge in life*' (Schwartz, 1992 p. 8). It is likely that people high in GNS will value the opportunity to learn new and interesting things and will be stimulated to participate in training and development. Thus Achievement (*Hyp. 2*), Self-direction (*Hyp. 3*) and Stimulation (*Hyp. 4*) are expected to mediate the prediction by GNS of the intention to take up training opportunities.

Cognitive Work Values

Elizur et al (1991) describe cognitive work values as those aspects specific to the work environment which relate to outcomes such as interest, achievement and responsibility. It is likely that individuals high in GNS will value these outcomes and their training activity will increase as a result. Thus, Cognitive work values are expected to mediate the prediction by Growth Need Strength of the intention to take up training opportunities (*Hyp. 5*).

Affective Work Values

These values deal with issues concerning interpersonal relationships within the work environment and although it is likely that individuals high in GNS will value sharing the experiences of others and may indeed be stimulated by certain relationships, there may be relationships which they do not value as strongly. For this reason interpersonal relationship may not be valued by high GNS individuals as highly as cognitive work values. The regression analysis above shows that as scores for GNS increased, scores for affective work values decreased, so it might be expected that Affective work values mediate the prediction by GNS of the intention to take up training opportunities but as scores for GNS increase, scores for affective work values will decrease (*Hyp. 6*).

Instrumental Work Values

These values deal with work outcomes of a more material nature such as pay and benefits. It is likely that people high in GNS will not value such issues as highly as they

would the cognitive work values as they may not be as relevant to their goal of achieving personal growth. Because of this is expected that Instrumental work values mediate the prediction by Growth Need Strength of the intention to take up training opportunities, but as scores for GNS increase, those for instrumental work values will decrease (*Hyp. 7*).

Moderating Variables

Baron and Kenny (1986) suggest that whereas mediating variables explain *how or why* external physical events take on internal psychological significance, moderating variables specify *when* certain outcomes will occur (p. 1176). They suggest that a moderator is a quantitative or qualitative variable that affects the direction and/or strength of an independent and a dependent variable. Baron and Kenny (1986) summarise the essential properties of a moderator variable by using the model which is reproduced below.

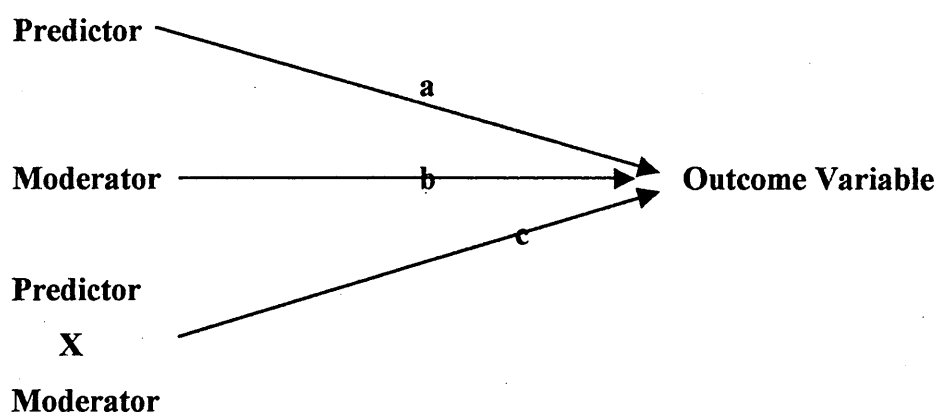


Figure 6. Moderator Model. Baron and Kenny, 1986 p. 1174.

It is hypothesised in the current model that GNS predicts Values and Work Values and these together induce in the individual's intent to undertake training and then to actual training activity. It is suggested that barriers to training moderate the relationship between intention to train and actual training activity. Therefore the smaller the number of barriers the higher the training activity, assuming there is an intention to train. Applying the model shown in Figure 3 to the current model, there are three causal paths that feed into the outcome variable training activity. Path 'a' is the impact of the

independent variable intention to train on the dependent variable training activity, Path 'b' is the impact of barriers to training on training activity and Path 'c' is the interaction between the two other causal paths. Support for the moderator hypothesis is found if the interaction is significant so it is expected that for employees who experience the least barriers to training the intention to train will be more related to training activity (*Hyp. 8*)

Method

Participants and Data Collection

The search for participants for Study 3 began among organisations which had structured employee training and development programmes. Such programmes provide staff with the opportunity to take up voluntary courses of study over and above any formal training, such as induction programmes, required for the job. This was an essential criterion for data collection as the main hypothesis was that GNS would predict those people who were more likely than others to take up training opportunities offered to them. Through personal contacts, an organisation was found which matched this criterion and agreed to take part. Mid-way through the planning stage of the study the organisation announced a large number of redundancies and all training programmes were curtailed. Thus the organisation had no choice but to withdraw from the research. There followed a period of some three months during which a search was made for another organisation and out of the 14 contacted 5 did not reply. Of those who replied, six had voluntary training schemes running concurrently with mandatory training. Two of these organisations expressed an interest in the research but withdrew as they felt that it was not appropriate to their needs. Finally, one organisation was found which not only fitted the criteria for the study but also was enthusiastic about taking part. The participating organisation was a large UK retail company which had an active programme of staff development. In April 2002 the company had opened a Training and Development Pool which 150 employees were invited to join on an entirely voluntary basis, and to select the courses they felt were most interesting or appropriate to their career development. It was agreed with the Training and Development Manager that the current study would be conducted in two parts with a 3-month gap between the two. The first part of the study would measure GNS, Personality and Intention to Take up Training Opportunities and the second part would measure GNS, Personality, General Values, Work Values and barriers to attending training courses and collect data about

how many training courses had actually been undertaken by participants during the intervening period. Additional data regarding the intention to train and actual training courses undertaken would also be provided by the organisation. The company agreed to partially fund the production of questionnaires for the study and it was agreed that a full report would be provided to them on the data specific to their organisation.

Questionnaire 1 (see Appendix Q)

A questionnaire was designed to measure GNS using the 'job choice' format plus three further questions designed as a result of Study 2. Also included in the questionnaire was the Big Five Index measure for personality (John et al, 1991) and a number of questions asking for demographic information and questions about individuals' experiences of and feelings about training undertaken in the past and to be undertaken in the future.

Growth Need Strength (Time 1)

Section 1 of the questionnaire contains the same measure for GNS 'job choice' as used in Studies 1 and 1a. Although the Studies carried out so far have shown that this measure has a complicated structure the results of Study 1a showed that it does moderate the relationship between job characteristics and job satisfaction as predicted by the Job Characteristics Model and is related to the personality domain Openness to Experience. Further it was shown that GNS predicted the taking of courses of study outside the work environment and thus there may be a link between it and intention to train and training activity in the work place. The results of Study 2 provided evidence from the participants' narratives, that GNS might be a personality trait and the consensus of opinion from the human resources professionals (HRPs) was that there were people who exhibited a stronger desire than others for personal growth and development. The HRPs also expressed the view that there was a 'type' of person who took up training opportunities in work more readily than others and analysis of the interview data from Study 2 showed that there were similarities between such people and those who were perceived as high in GNS.

Additional GNS Questions

The table on page 119 shows that three distinct characteristics emerged from the qualitative data analysis, which were peculiar to people high in GNS. These were the

ability to cope with change, a balanced view of life and materialism. Three new questions were designed in the same format as the original 'job choice' measure and following discussions with university colleagues, were included as questions 13, 14 and 15 in the questionnaire (see Appendix Q). Each new response choice was paired with one measuring one of the other work place issues used in the original format. The questions were added at the end of the job choice format in attempt to avoid introducing any new response artefacts. The rationale for adding the new questions was to investigate the possibility that there might be 'more to' GNS than was measured by Hackman and Oldham (1975). The scoring for the scale was as in Studies 1 and 1a, and it was intended that scores for question 13 would be reversed. The measure for GNS was the main measure in this questionnaire because it was felt important that if the individual difference factor it was designed to measure had predictive qualities then it would be necessary for it to have the ability to measure GNS independently from either the Job Diagnostic Survey or any other similar scale score.

Intention to take up training opportunities was measured in question 12 using a simple Yes/No format and from data provided from the organisation, which showed the numbers and names of courses employees on the training and development programme had signed up for. Demographic details were collected from responses to questions 1 to 4. The remaining questions were included after consultation with the Training and Development Manager of the participating organisation and were primarily for use in the report, which would be provided for the company.

Questionnaire 2 (see Appendix R)

The second questionnaire was designed to measure the following:

1. GNS (Time 2)
2. Personality (Time 2)
3. General Values
4. Work Values
5. Training undertaken since questionnaire 1
6. Intention to undertake further training in the future
7. Perceived barriers to Attending Training and Development Courses

The measures for GNS and personality were as for Time 1. The measures for General Values and Work Values were those used in Studies 1 and 1a details of which are given on page 40. Section 6 of the questionnaire asks respondents how many training courses they have attended since completing the first questionnaire and how many they intend to take in the future. Both responses are given on a 7 – point scale from ‘0’ courses to ‘6 or more’.

Barriers to Attending Training and Development Courses

These are measured using the 16 items in Section 7 of the questionnaire and are taken from the work of Tharenou (2001). According to Tharenou 8 of the items relate to barriers relating to employer support and 4 concern workload barriers. The remaining 4 items did not load onto either of these factors but it was intended that all 16 items would be analysed and the factors which emerged would then be used in the analysis to test their moderating effect on the relationship between intent to train and actual training activity.

It had been hoped that the model would be tested using data regarding training and development which had been undertaken during the three-month period between administration of the first and second questionnaires. However, this was not possible as the three months concerned covered the period of Christmas and New Year, when little or no training and development courses were run by the organisation due to its retail nature. Questionnaire 2 was therefore revised to ask participants the number of training courses they had actually attended since joining the training and Development Pool in April 2002. Data were also provided by the organisation from their records of how many courses participants had signed up for (intention to train) and how many they had actually attended. Initial analysis of the items from the measure for GNS showed that GNS was not only more complicated than for the first two studies but also had less reliability. In the JDS the ‘would like’ format for measuring GNS precedes the ‘job choice’ format in the questionnaire and this method was followed in Studies 1 and 1a. Because the measure for GNS was less reliable for Time 1 than for the earlier studies it was possible that the ‘would like’ format had had a priming effect on the ‘job choice’ items in both the JDS and the early studies in this research, inflating its reliability. The ‘would like’ format was, therefore added to Questionnaire 2 as Section 1 (prior to the

'job choice' format). There had been 8 cases for Studies 1 and 1a where all responses to the GNS 'would like' format' had been omitted by those respondents. It was decided that the probable reason for this was the situation of the measure inside the front cover of the questionnaire. For Study3, questionnaire 2, the measure was moved to the following page, leaving the inside front cover blank.

Results

Non-Responders to Questionnaire 2

The demographic details of 37 participants who did not respond to questionnaire 2 were compared with respondents and no differences between the two groups were found. All were comparable in their range of age, education and tenure and the male to female ratio was similar. The mean score for GNS12 for non-responders was 3.34 which was slightly lower than the mean of 3.52 for responders.

Screening of Data

Prior to analysis all data were examined through various SPSS programmes for accuracy of data entry, missing values and distributions. For questionnaire 1, 101 completed questionnaires were returned and all were included in the analyses. There were no further missing data. For questionnaire 2, 66 completed questionnaires were returned and of these, 2 could not be matched with the questionnaires from the first set of respondents as names had been omitted. These were excluded from the analyses. Of the remaining 64 questionnaires there were five where all responses to the GNS 'would like' measure were missing. These were replaced with the means for each item prior to analysis. To detect univariate outliers among continuous variables, z scores were calculated and there were no extreme standardised scores. Scores among the variables to be tested did not deviate substantially from normality

Examination of the data supplied by the organisation showed that in 13 cases information regarding intention to train was missing. In spite of the efforts of the staff within the organisation, these could not be found and so some hypotheses were tested using data from 51 employees and some from 64, on a variety of jobs ranging from supervisor to senior manager level in one organisation. The response rates were 67% for the first questionnaire and 63% for the second.

The following analyses were carried out:

- Validity and reliability analysis of all scales
- Test by bivariate correlation and simple regression of the relationship between GNS and training activity (Hypothesis 1)
- Test by series of regression models of the mediating function of the three general values and three work values on the relationship between GNS and training activity (Hypotheses 2 – 7)
- Test by moderated multiple regression using multiple interactive terms of the moderating effect of barriers to training and development on the intent to train-training activity relationship (Hypothesis 8)

The Measures of Growth Need Strength

The correlation between GNS 'job choice' Time 1 and Time 2 was $r = .64$ indicating a good test-retest reliability given the interval between the first and second questionnaire.

Exploratory factor analysis, through principle components analysis with direct oblimin rotation showed that of the new items only item 14, which was designed to measure 'a balanced way of life', was correlated with the original 12 items. To investigate whether item 14 had any effect on the correlations between GNS and the other variables tested, a separate GNS scale was created called here 'GNS13'. The items comprising the GNS 'would like' format all loaded onto one factor which was reliable at .78 and this measure was again psychometrically distinct from the GNS 'job choice' measure, indicating that the two are not measuring the same construct. The remaining two new items were excluded from the analyses.

Barriers to Training and Development

Principle Components Analysis with direct oblimin rotation revealed 3 factors. The first two factors 'Employer Support' and 'Workload Barriers' were similar in structure to those found by Tharenou (2001) and were clearly interpretable. However a third factor emerged from the analysis which is described as 'Inconvenience of Courses' here, and included items such as family commitments (item 5) and travelling distance/transport problems/location of training (item 6) (see Appendix R). Three items ('others wanting to go', 'courses fully booked' and 'lack of approval from supervisor') did not load

significantly onto any of the three factors above and examination of the responses to these items showed that over 90% of participants indicated that these issues did not represent a barrier to them attending courses. The three items were thus excluded from the analysis. The reliability of all scale scores is shown in Table 35.

Psychometric Properties of all Scales

Table 35

	Study 3 Time 1 (n=64)			Study 3 Time 2 (n=64)		
	Alpha	Mean	SD	Alpha	Mean	SD
GNS 12	.55	3.30	.39	.71	3.29	.45
GNS13	.60	3.21	.39	.73	3.19	.46
GNS 'would like'	-	-	-	.78	5.73	.72
Training intended	-	-	-	-	4.13	1.30
Training undertaken	-	-	-	-	3.76	1.79
Neuroticism	.82	2.47	.71	.75	2.47	.64
Extraversion	.83	3.90	.65	.85	3.82	.71
Openness	.83	3.50	.66	.85	3.45	.70
Agreeableness	.83	3.84	.61	.79	3.82	.61
Conscientiousness	.79	4.07	.54	.80	4.05	.55
Achievement	-	-	-	.78	4.73	.92
Benevolence	-	-	-	.76	4.36	.82
Conformity	-	-	-	.77	4.20	1.17
Hedonism	-	-	-	.49	5.10	.83
Power	-	-	-	.77	3.20	1.34
Security	-	-	-	.69	4.44	.98
Self-direction	-	-	-	.67	4.60	.79
Stimulation	-	-	-	.71	4.28	1.08
Tradition	-	-	-	.54	2.64	.86
Universalism	-	-	-	.80	3.97	.89
Affective Work Values	-	-	-	.59	3.88	.42
Cognitive Work Values	-	-	-	.73	3.77	.33
Instrumental Work Values	-	-	-	.59	3.42	.51
Barriers:						
Employer Support	-	-	-	.67	1.68	.54
Workload	-	-	-	.67	3.24	1.04
Inconvenience of courses	-	-	-	.61	1.36	.57

Relationship Among the Measures

Correlations among all questionnaire scale scores are presented in Appendix S. The correlation between GNS12 and Openness to Experience was higher than that between GNS12 and any of the other personality factors measured. The correlation between GNS12 and Openness was higher for Time 2, and significant at the .05 level (2-tailed) and the test-retest correlation for Openness to Experience was $r = 0.90$. The correlations between GNS12 and the values Achievement, Self-direction and Stimulation were higher than those for GNS12 and the remaining general values. The relationship was strongest between GNS12 and Self-direction ($r = 0.39$) and although modest, was in the

predicted direction and significant at the .01 level of significance. The correlations between GNS12 and both Cognitive ($r = 0.40$) and Instrumental ($r = -0.42$) work values were significant at the .01 level of significance but the relationship between GNS12 and Instrumental work values was in the negative direction. All of the above correlations were larger when the GNS13 scale was used and all are consistent with the findings of Studies 1 and 1a.

The Relationship Between GNS and Intention to Train

The first hypothesis predicted that there would be a relationship between GNS and intention to train, specifically, the higher the level of GNS the higher the level of intent to train. This was tested using both GNS12 and GNS13, although it should be noted that this relationship was tested using only 51 cases, as the information from the organisation was incomplete. Table 36 shows the results of the simple regressions which were performed to test this prediction between Intention to Train (organisation data) as the dependent variable and GNS as the independent variable. As can be seen, both GNS12 and GNS13 contributed significantly to the prediction of intention to train but the relationship was not in the expected direction, with high GNS predicting low intention to train. GNS 12 was a slightly better predictor of intention to train than GNS13. The amount of variance in training activity explained by both GNS12 and GNS13 was small, but marginally higher for GNS12.

Table 36

Standardised Coefficients							
Model	Beta	t	Sig.	Model	Beta	t	Sig.
Constant		5.692	.000	Constant		5.744	.000
GNS12	-.325	-2.426	.019	GNS13	-.320	-2.391	.021
F(1,50) = 5.886 p~0				F(1,50) = 5.716 p~0			
Adjusted R ² = .087				Adjusted R ² = .085			

Dependent Variable: Intention to Train

A possible reason for his unexpected result was that people with high GNS may indicate less intention to train than those with low GNS because they do not see the training courses as worthwhile, unless they perceive them as having the potential to satisfy their growth needs. To investigate this possibility the relationship between GNS and actual training activity was tested. It was found that although there was a significant relationship ($\beta = -.26$, $p < .05$), as scores for GNS increased the number of training courses undertaken decreased. If people who are high in GNS only attend courses which

they perceive as having the potential to satisfy their growth needs, the content of the courses they attended might be expected to reflect issues concerning personal growth. The data showed that four respondents had attended only 1 course and so their scores were examined further for level of GNS and demographic details to see if they were a) high in GNS and b) had undertaken courses which might be construed as having the potential to satisfy needs for personal growth. If this were the case then there would be some evidence for the explanation given earlier for the negative relationship between GNS and training activity. The results of this are shown in the table below:

Table 37

Respondent	GNS12 (Mean 3.29)	GNS13 (Mean 3.19)	Educational Attainment	Age	Tenure (years)	Gender
A	3.25	3.08	Degree	30-39	6-10	M
B	4.25	4.23	Degree	20-29	1-5	M
C	3.25	3.15	Degree	20-29	1-5	F
D	3.75	3.77	Degree	40-49	1-5	F

Means in brackets are for the total sample.

As can be seen, two respondents had GNS scores higher than the mean and two were slightly below. All were educated to degree level or higher. Respondents B, C and D had undertaken courses which were described by the Training Manager as 'in depth' courses relating to issues such as 'Influence with Impact' and 'Work/Life Balance'. These three respondents were all on senior grades earning salaries of over £50k per annum and were working in departments concerned with trading and marketing. There were 15 respondents who had taken 6 or more courses and of these, 10 had scores for GNS12 of below the mean (see Table 38).

Table 38

Respondent	GNS12 (Mean 3.29)	GNS13 (Mean 3.19)	Educational Attainment	Age	Tenure (years)	Gender
A	3.83	3.62	No degree	30-39	6-10	F
B	2.67	2.54	No degree	30-39	11-15	M
C	3.50	3.62	No degree	30-39	11-15	M
D	2.83	2.77	No degree	20-29	1-5	F
E	3.17	3.23	No degree	20-29	6-10	M
F	3.00	2.92	Degree	20-29	1-5	F
G	2.67	2.54	Degree	20-29	6-10	F
H	3.75	3.54	No degree	30-39	11-15	M
I	2.83	2.92	No degree	20-29	11-15	M
J	2.50	2.38	No degree	20-29	1-5	M
K	3.67	3.54	Degree	20-29	<1	M
L	3.00	2.92	Degree	30-39	1-5	F
M	3.58	3.54	No degree	40-49	16-20	M
N	2.92	2.77	Degree	30-39	16-20	M
O	2.58	2.54	Degree	30-39	6-10	M

Means in brackets are for the total sample.

All 15 respondents were on lower grades earning salaries of £30k or less and the courses they had undertaken were described by the Training Manager as 'skill oriented'. It should be noted here that data regarding courses undertaken had been supplied to the researcher prior to the analysis of data to maintain confidentiality. The implications of these results will be discussed at the end of this chapter.

The relationship between GNS, using both the 12 item and 13 item measures, and training activity was also tested and the results showed that both measures predicted low training activity.

The Mediating Effect of the Values and Work Values on the GNS Training Activity Relationship

Baron and Kenny (1986, p.1177) state that to test for mediation three regression equations should be estimated. First, regressing the mediator on the independent variable; second, regressing the dependent variable on the independent variable; and third, regressing the dependent variable on both the independent variable and the mediator. To establish mediation, the independent variable must affect the mediator in the first equation, the independent variable must be shown to affect the dependent variable in the second equation and finally the mediator must affect the dependent variable in the third equation. If all of these conditions are met in the predicted direction, then the effect of the independent variable on the dependent variable must be less in the third equation than in the second. Baron and Kenny go on to suggest that because the independent variable and the mediator must be correlated, this results in multicollinearity when the effects of the independent variable and the mediator on the dependent variable are estimated. Because of this, both the significance of the coefficient and their absolute size must be examined. The results of the first equation are shown in Table 39.

The Effect of GNS on Achievement, Self-direction, Stimulation; Affective, Cognitive and Instrumental Work Values

Examination of the absolute values in Table 39 shows that GNS12 predicted the values Self-direction and Stimulation and Cognitive Work Values and all were in the expected direction. For Instrumental Work Values, high GNS12 predicted low Instrumental Work

Values. The results for GNS13 show that it significantly predicted all three values in the expected direction and improved the coefficients for both Cognitive and Instrumental work values. For Affective values, although the coefficient was still small it changed to a negative direction. These results provide some support for the first equation.

Table 39

	Achievement	Self-direction	Stimulation	Affective Work Values	Cognitive Work Values	Instrumental Work values
GNS12	.22	.40	.31	.02	.40	-.42
GNS13	.27	.39	.33	-.04	.42	-.44

The results of the second equation are given in Table 36 (see page 140) and show that although there was a significant relationship between GNS and intention to train, this was not in the predicted direction. The results of the third equation are given below and show that none of the values or work values mediated the relationship between GNS and Intention to train and therefore no support is found for Hypotheses 2 – 7 that Achievement, Self-direction, Stimulation, and the three work values would mediate the above relationship..

Table 40

Dependant Variable: Intention to Train	Standardised Beta	R ²	Adj. R ²	Δ R ²	Δ F
Block					
1. GNS (IV)	-.30	.11	.09		5.89*
2. Achievement (mediator)	-.10	.11	.08	.01	.48
Block					
1. GNS (IV)	-.40	.11	.09		5.89*
2. Self-direction (mediator)	.20	.14	.11	.04	2.01
Block					
1. GNS (IV)	-.32	.11	.09		5.89*
2. Stimulation (mediator)	-.02	.11	.07	.00	.02
Block					
1. GNS (IV)	-.34	.11	.09		5.89*
2. Affective Work Values (mediator)	.21	.15	.12	.05	2.60
Block					
1. GNS (IV)	-.37	.11	.09		5.89*
2. Cognitive Work Values (mediator)	.10	.11	.08	.01	.51
Block					
1. GNS (IV)	-.31	.11	.09		5.89*
2. Instrumental Work Values (mediator)	.03	.11	.07	.00	.04

N.B. n = 51. *p < .05. Adj. R² = adjusted R²

Test of the Moderating Effect of Barriers to Training on the Intention to Train - Training Activity Relationship

The data used to measure intention to train again, came from the organisation's records and was incomplete, thus there were only 51 cases included in the regression analysis. The relationship between Intention to Train and Training Activity was significant ($\beta = .39$, $p < .01$ and was in the predicted direction.

The moderating effect of the barriers to training was tested using moderated multiple regression using multiplicative interactions terms, with training activity as the dependent variable and Intention to Train, Employer Support, Workload, Inconvenience of Courses and the interactions between Intention to Train and each of the barriers as independent variables. Each of the three proposed barriers was tested in turn and no support was found for the moderating effect of any of these on the intention to train-training activity relationship (see Appendix T).

Non-Responders to Questionnaire 2

There were 37 participants who responded to questionnaire 1 but not questionnaire 2. The GNS scores and demographic data for this group were examined and differences between responders and non-responders compared. The group was small but nonetheless there was no difference between the two groups for any the variables which could be compared with respondents' scores. The correlation between training intended and training undertaken using data supplied by the organisation was $r = .40$ which was similar to that for those who responded. As level of intention to train and training activity was the same for the two groups, the non-responses can only be attributed to lack of time or the fact that these 37 employees did not want to complete yet another questionnaire.

Discussion

This aim of Study 3 was to provide a model for how and under what conditions GNS predicts the intention to train and the taking up of training opportunities. Analysis of the GNS items showed item 14 did have some relationship with the items in the original scale. This item asked participants whether they would prefer a job which allow them to

put an equal amount of time into their work and their private life, over a job which gave them the chance for quick promotions, and was designed to measure the participants' 'balanced view of life' Adding this item to the scale did increase its reliability and also increased the size of its relationship with Openness to Experience, Achievement, Self-direction, Stimulation Cognitive and Instrumental Work Values.

Support was found for a relationship between GNS12 and the intention to train but the results showed that the higher the level of GNS12 the lower the number of courses signed up for (indicating intention to train). The addition of item 14 slightly increased the size of this relationship. The reason for the negative relationship between GNS and Intention to Train may be that individuals with high needs for personal growth do not want to take up as many training courses as those with a low need for growth. This could be because they do not see the courses as worthwhile to them, unless they are perceived as having the potential for satisfying those growth needs. Looking at the relationship the other way round it is possible that the individuals who are low in GNS are signing up for and attending a greater number of courses, not exclusively to satisfy needs for growth. They may be doing so because they either see them as a break from work, or they may feel that by being seen to sign up for and attend a high number of courses, this will impress their employer. Evidence from the human resources professionals who took part in Study 2 suggests that there are individuals who attend training courses with the aim of enhancing their career records. In fact, some of the HRP's referred to such individuals as 'CV enhancers'. High GNS individuals may have other motivations for attending courses and although the results show that they attended fewer courses, they may have been selective in their choice of courses, in only attending those which they perceived as having real benefit to them in terms of personal growth.

The post-hoc analyses carried out to assess the GNS scores for the individuals who took the fewest courses showed that two had scores in the high GNS range and two had scores in the medium range. Examination of the courses they had attended indicated that these were courses which the organisation's records show as being concerned with the development of more analytical skills, such as the influence of the organisation on its market and the work/life balance. All four individuals in this group were senior managers in highly paid positions within the trading and marketing areas of the

organisation, and all were educated to degree level. As such they may have reached a point in their careers where they are able to clearly identify their own growth needs and their training activity is reflecting this. An alternative theory is that they are less primed for training than their more junior colleagues, or indeed they may feel that signing up for training at their level of seniority might reveal an inadequacy on their part. Although the addition of item 14 was unlikely to have made any significant difference to the scale, it is interesting to note that of the four participants who had attended the least number of courses one who had a high score for GNS attended a course entitled 'Work/Life Balance'! This provides some evidence for the explanation of the negative relationship between GNS and training activity, that people with higher GNS undertake fewer courses than their low GNS counterparts because the quality of the courses offered is important to them. Nonetheless, it still remains that employees in a more senior position within the organisation may have less time to devote to their personal training and development.

The group which included those participants with the highest training activity was comprised of staff in the lower income bracket, in less senior positions within the organisation, who may have a lower workload and thus more time to attend training courses (Table 38). Although they were lower in GNS, they may be at a point in their careers where they are more training oriented as they are still gathering the skills which will enable them to seek promotion. It is possible that the level of GNS only increases at times when individuals' needs are not being met by the opportunities available within the organisation, so it would be interesting to assess the group with the highest training activity to see if and at what point their GNS increases. This may be the point at which employees are most ready to take up new opportunities and challenges within the workplace. Although this may indicate that GNS is a 'state' rather than a trait it does fit with Maslow's (1954) theory that self-fulfilment is a 'being' need with no end-state. If GNS is seen as a part of peoples' personality which is manifested in different ways at different times (as suggested in the introduction to this chapter), then it could be suggested that a 'personal growth' trait could be latent until it is not being met.

The way training activity is measured in this study does not reveal enough about training to be able to make an assessment as to whether high GNS individuals seek

quality rather than quantity in their training choices, because only the number of courses attended was measured. Future research should assess individuals' justification for attending courses or not and also ask individuals post-training, what they have gained most from each course.

The results of the tests of the mediating function of the values and work values showed that there was no support for this. The presence of a direct relationship between GNS and intention to train and training activity indicates that the values and work values people hold, although related to GNS have no relationship with training activity. There is a possibility that this mediating effect may be found with alternative workplace outcomes and this would again, provide interesting material for future research.

There was a strong relationship between Intention to Train and Training Activity, but this was not moderated by any of the three groups of barriers to attending training courses. Although there is limited support for the model proposed at the beginning of this chapter, there is evidence that GNS does predict workplace behaviour in terms of low training activity and this area at least is worthy of further research. The findings of this study will be discussed in relation to the others, in the final chapter of this thesis.

CHAPTER 6

What is GNS?

The meaning of human behaviour and the understanding of the similarities and differences among individuals, are at the heart of psychological theory. When researchers examine cause and effect they must look at if, how and when variables affect each other. Taking the relationship which forms the basis of the early studies in the research carried out for this thesis, why is it that under some circumstances the characteristics of jobs predicts job satisfaction? The JCM proposes that it is the level of peoples' need for personal growth, which makes the difference to that relationship – but what is GNS?

At the beginning of this thesis it was suggested that although a substantial amount of research had been carried out regarding the Job Characteristics Model, the nature of the individual difference factor Growth Need Strength had been under-defined both in the original research by Hackman and Oldham (1975; 1976) and subsequent researchers. Hackman and Oldham described GNS as a malleable individual difference factor which affected the way people responded to jobs high in motivating potential, but did not put forward a theory about the true nature of the factor. The review of the literature reported in Chapter 1 revealed that a number of studies (see Appendix A) replicated all or part of the research carried out by Hackman and Oldham (1976). Of these, some either did not use the 'job choice' format to measure GNS (as had Hackman and Oldham), or did not specify which format had been employed. The result of this ambiguity has been mixed support for GNS as a moderator of the job characteristics-job satisfaction relationship. To make matters more confusing, in most cases only the reliability of the measure was reported, with little or no mention of the structure and thus the validity of GNS as a measure. This made it difficult to make any real assumptions about what the 'job choice' items were actually measuring, as there were no historical data to refer to.

The aim of the research reported on here, was to isolate the specific individual difference factor, which is at the root of GNS. This would make it possible to identify those individuals who were likely to respond more positively to the factors in the JCM, concerning what it is about people which prompts certain work behaviour. If this were achieved then it would expand the model and increase its generalisability.

In Studies 1 and 1a, the main proposition of the JCM regarding GNS – that it moderates the relationship between job characteristics and job satisfaction, was tested. Support was found for both this hypothesis and the others proposed in the first studies, that there would be a relationship between GNS and Openness to Experience; GNS and the general values Achievement, Self-direction and Stimulation; and the three work values. None of the personality, values or work values moderated the same relationship as GNS ‘job choice’ indicating that it was measuring something that they were not.

In Study 2 evidence was sought, through methods of qualitative data collection and analysis, of the existence of GNS, its nature and relation to job satisfaction and its relation to the taking up of training opportunities offered within the workplace. The results of this analysis revealed some personal characteristics, which might be attributed to people who were perceived by the participants as high in GNS. Further, a link was established, again through the perceptions of the participants, between GNS and training activity.

As a result of the findings of Studies 1, 1a and 2, in Study 3 it was suggested that GNS might be part of peoples’ personalities and a model was presented for how and under what conditions GNS predicts the taking up of training opportunities. It was found that GNS did indeed predict intention to take up training opportunities but this was in a negative direction, with high GNS paradoxically predicting low intention to train.

Hackman and Oldham (1976) used the ‘job choice’ measure of GNS in their test of the JCM and showed that it moderated the job characteristics-job satisfaction relationship. However, some studies which followed this work (e.g. Abdel-Halim, 1979; de Jong et al, 2001), claiming to replicate it, used the ‘would like’ format and found support for it as a moderator of the relationship. The results of the studies reported in Chapters 2 and 3, found support for the moderating effect of the ‘job choice’ format only. The ‘would like’ format not only did not perform in the same way, it was also found to be psychometrically distinct from its companion. So it is suggested here that the two are not measuring the same construct but again, because of the paucity of historical data concerning the factor structure of GNS, it is impossible to say whether this is consistent with the findings of others. Richard Hackman has said in personal correspondence that the reason the ‘would like’ format was not used to test the moderating effect of GNS, was that it was so inviting of people to say they just wanted everything, when asked how important certain elements

were in their jobs. Here, it is accepted that, as Hackman and Oldham (1976) predict, GNS moderates the relationship between job characteristics and job satisfaction and this research has shown that none of the personality trait, values or work values moderated that relationship in the same way. The fact that support has been found for this prediction underpins the assumptions of this research.

The factor structure of the 'job choice' items for Studies 1 and 1a was found to be complicated, with no single clear factor emerging regardless of the methods of rotation employed. The two factors that emerged appeared to be uninterpretable until the non-response items were examined. This indicated that in answering questions about their preferences for jobs, employees' responses are affected by organisational issues such as co-worker relationships, pay and fringe benefits. If Maslow's (1943) theory, that human needs are organised in the form of a hierarchy is accepted, then these results are understandable. Individuals cannot consider their higher-order needs, such as needs for personal growth until other, more basic needs, are met. So it would be logical, if as Alderfer (1972) suggests, the desire for need fulfilment diminishes as each need is actually fulfilled, that the desire for the fulfilment of growth needs is only strengthened when the employee has as many friendly co-workers, and as much pay and remuneration as he or she needs at the time. As suggested on page 150, although this may point to GNS being a 'state' rather than a trait, if the need to become everything one is capable of becoming is a 'being' need with no end-state, GNS itself might be seen as a personality trait but its *level* rather than the trait itself is what has the potential for change. In this case there would be a level at which an individual's GNS is stable over time but there is a capacity for that level to change if the situations demands. An example of this from theories of personality could be shown in Eysenck's (1953) theory. An individual may demonstrate a high level of the personality trait 'sociability' consistently over time, but there will be occasions when that level of sociability changes depending on the individual's circumstances.

To answer the research question it was necessary to examine the two measures for GNS proposed by Hackman and Oldham, assess what they were actually measuring and how effective they were as predictors of workplace behaviour. As Blinkhorn (1997) suggested 'the classical issue of validity' (p. 179) needed to be addressed.

Growth Need Strength, using the 'job choice' format, was measured twice for Study 3, on the first occasion it was set in a questionnaire that also measured personality using the Big Five Index (John et al, 1991). On the second, it was measured alongside the BFI, general values and work values, so that comparisons could again be made between the personal characteristics measured by these scales and those for GNS. The three new items which had been devised as a result of Study 2 added nothing to the analysis on the first occasion. In the second questionnaire the 'would like' format for measuring GNS was added prior to the GNS 'job choice' scale and the factor structure of the original 12-item 'job choice' scale improved. All of the items loaded onto one factor and the scale was more reliable than for Studies 1 and 1a. There are two possible explanations for this increase in validity and reliability, apart from the fact that the sample was considerably smaller than for the previous studies. The first is that there was only a short period (3 months) between the administration of the first and second questionnaires, so respondents may have shown consistency effects. They had all seen the items before and could not only remember some of their first responses, but also may have developed some understanding of what was being measured by the questionnaire. The second explanation is that the addition of the 'would like' format prior to the 'job choice' items had a priming effect on the respondents. As mentioned in Chapter 1, Salancik and Pfeffer (1977) suggest that priming effects are an important issue and occur when various aspects of a situation are made more salient than they otherwise might be (p. 449). In the 'would like' format the respondents are asked how much of certain characteristics they would like to have in their jobs. Items concerned with GNS such as 'opportunities to learn new things from my work' are mixed with non-GNS items such as 'quick promotion'. The items for the 'job choice' format, although measured in a different way, are similarly worded. Here, it is suggested that the questioning process employed in the second questionnaire alerted respondents to those aspects of the job, which might be seen, not only by themselves as desirable in a job, but also what they thought the questioner wanted to see. In effect, people want others to think well of them and to be seen as having noble intentions! When this priming effect is coupled with the desire for consistency both in the two questionnaires and in both formats of GNS, the effects have the potential to be significant. The test-retest reliability of the scale with the original 12 items was good at .64 but the higher reliability for Time 2 is further evidence of possible priming and consistency effects, particularly as a result of the addition of the 'would like' format.

The only 'new' item for GNS that had a relationship with the original 12 was item 14 that was designed to measure the individual's desire for a 'balanced view of life'. This item increased the reliability of the scale from .71 to .73. The qualitative data study reported in Chapter 4 questioned whether the measure of GNS designed by Hackman and Oldham (1975), covered the whole domain of interest or whether there might be other personal characteristics that might be expected to be exhibited by people high in GNS. The results of the analysis of the 'job choice' items, with the inclusion of item 14 did not diminish the scale and it is suggested that this item may add to the definition of GNS. It may be that individuals see the attainment of a balance between an individual's work and private life as part of their personal growth. The results of Study 3 showed that of the people who attended the lowest number of training courses, one of the who was high in GNS attended a course entitled 'Work/Life Balance' which, although it is only one example, may be a demonstration of this premise.

If as suggested, GNS is part of peoples' personality then behaviour associated with it would not only be exhibited in the work environment. The fact that there is a relationship, however modest, between the original items measuring GNS and the need for a balanced way of life, suggests that individuals' feelings about their lives in general, affect the choices they make regarding employment issues. As this research is concerned with occupational psychology, the issue of peoples' lives outside work was only touched upon during Study 2. However, the participants in that study were united in their opinion that GNS was a personal characteristic which would be exhibited through peoples' choices of activities outside work. Indeed the Human Resources professionals said that asking interviewees for jobs, about their hobbies and interests gave them an insight into whether that person demonstrated, for example, a commitment to outside interests which might be transferred to their working life. The theme that emerged was that people high in GNS would seek new and interesting experiences, which would provide them with a sense of moving forward in their knowledge and understanding of life in general.

GNS as a Higher-Order Factor of Personality

Each time GNS 'job choice' was measured, its factor structure apart, it was positively related to the personality factor Openness to Experience. The correlation between GNS and Openness changed little on each occasion but slightly stronger between Openness and the GNS scale with item 14 added. GNS is not however, the same as Openness to

Experience although the two appear to be related. As suggested in Chapter 1, there has been considerable debate over both the number of factors that constitute personality and the nature of those factors. The factor which has caused the greatest concern to personality theorists (at least, those who accept that there are five factors) is the fifth factor which is accepted here as being called Openness to Experience.

Digman (1997) proposed that apart from the five basic factors, there were two higher-order factors that emerged from his examination of 14 studies supporting the Big Five model of personality and trait organisation. Digman described one of these two broad concepts as 'Personal growth versus personal constriction' (p. 1250) which is associated with the perspectives of personal growth theorists such as Rogers and Maslow. Maslow (1943) suggests that human beings have a basic desire to know and understand more about both themselves and the world around them. He says that this desire to know leads to self-actualisation and describes this as 'the desire to become more and more what one is, to become everything that one is capable of becoming' (p. 382). In their examination of need-satisfaction models, Salancik and Pfeffer (1977) suggest that regardless of whether needs are inherent or acquired there is an assumption that all people have them. Hackman and Oldham (1976) suggest that all of the respondents they tested had needs for personal growth but for some this need was stronger than for others. The majority of the research which followed their work drew the same conclusions. Here, those assumptions are accepted and are born out by the fact that the means and variances in scores for the GNS 'job choice' measure are similar in each study which has used that format, including the work contained in this thesis. Digman (1997) described personal growth in similar terms to self-actualisation, as an 'enlargement of self ... by being open to all experience, especially new experience and by unfettered use of one's intelligence' (p. 1250).

Costa and McCrae (1992a) describe the Openness factor that forms part of the five-factor model. They say that high scorers on Openness believe that an appreciation of art and beauty will enable them to develop a wider knowledge and appreciation of aesthetics; they are willing to attempt different activities; and prefer variety to familiarity and routine. They further suggest that such individuals will, over time, take up a number of different hobbies. Open people are also regarded as high in intellectual curiosity which results in an active pursuit of intellectual interests but Costa and McCrae (*ibid.*, p. 17) say

that an openness to ideas does not necessarily imply a high level of intelligence, although it can contribute to the development of intellectual potential.

In all of the above respects the descriptions given of people who are high in Openness to Experience are very similar to the descriptors used to measure GNS. People who are high in GNS are purported to be those who desire work where there is considerable opportunity to be creative and imaginative; a job, which has lots of variety; and a job that provides the opportunity to learn new and interesting things. However, it is suggested that GNS is more than Openness to Experience. Evidence from the participants in Study 2 (see page 121) shows that the description of a person high in GNS, although it includes some of the personal characteristics that relate to Openness to Experience, describes some personal characteristics which are distinctive. A high GNS person is someone who has a desire for achievement and a determination to succeed; is confident and competent and has a balanced view of life. The items used to measure GNS imply that those who score highly on this measure want the responsibility that results from doing high quality work; challenge in work and the chance to develop new skills and advance in the organisation. All of these aspects reflect a desire for achievement and a determination to succeed which is not reflected in the personality factor Openness to Experience.

Concerning the new item, which was added to the GNS scale to measure a balanced view of life, no evidence could be found in the literature on personality, specifically relating to this. The only link, albeit tenuous, is that people who are able to create a balance between their work and private life may be seeking stability or indeed may be more emotionally stable than others. If this is so then this proposed new aspect of GNS may be associated with another of the Big Five factors – Neuroticism. The intercorrelations among scale scores in Appendix S show mixed results for the relationship between GNS and Neuroticism. For Time 1, Neuroticism was significantly, negatively related to GNS both for the original GNS ‘job choice’ measure and for ‘GNS 13’. For Time 2, the relationship remained negative but it was not as strong. Nonetheless, the data indicate that, perhaps, people with high GNS are more stable than those with low GNS.

What does GNS Predict?

If the suggestion made here, that GNS is part of personality is true then it must be shown to do two things. Firstly, it should be possible to measure GNS independently from the

Job Diagnostic Survey and secondly, it should be shown to predict a piece of observable behaviour. The first of these conditions was partially met at the end of Study 1a when GNS was shown to moderate the relationship between job characteristics and job satisfaction using the same measure for GNS, but a different measure for job satisfaction. In Study 3 a model was proposed which set out how and under what conditions GNS predicts the taking up of training opportunities, in an attempt to meet the second condition. Although no support was found for the model as it stands, GNS was found to predict low intention to train and low training activity. It may be that people who are high in GNS do not necessarily seek the satisfaction of needs for growth through the training and development programmes offered by their employers. Alternatively, as suggested at the end of Chapter 5, high GNS people may be more selective about the training they sign up for, only choosing those courses which they feel are appropriate to their development needs at the time. When the scores for the respondents who had signed up for the lowest number of courses were examined, the results showed that the respondents were all senior managers within the organisation. Each of the respondents in that group had moderate to high levels of GNS but it may be that those needs were being met through the job itself, in that the respondents were being challenged and stimulated by their roles and that their desire to achieve and determination to succeed were being manifested in other ways. For people in higher grade jobs a lack of time for attending training and development courses may also have been a serious issue.

Examination of the group of respondents who signed up for the greatest number of courses also revealed some interesting details. All were at lower grades within the organisation and were in the low to moderate groups in terms of GNS scores. It may be that the respondents who showed the highest training activity had the lower levels of GNS because they were actually in the process of gathering knowledge and understanding through training, and so their needs for personal growth were being met. There may be a point at which such people feel their growth needs are not being met and if so their level of GNS may increase. If this is the case then repeated assessment of GNS may reveal the point at which the level changes, and this may be the time that employees are ready for new challenges in terms of personal growth. There is, however, the possibility that people high in GNS are generally more successful at work because they have taken up training opportunities early in their careers and maximised the usefulness of these. A comparison of the demographic details of high and low GNS individuals was made in the course of

Study 3, but there was no evidence to suggest a difference between the levels of education for the high and low groups. Before exploring this issue further it is important to return to the model proposed in Study 3.

Although there was no support for the mediating role of the values and work values in the GNS-training relationship, GNS did predict values and work values as the model had suggested (see Table 39, page 143). This provides some support for the hypothesis that GNS is an antecedent of the values Achievement, Self-direction and Stimulation and as such might be a personality trait that prompts people to value certain things both in and out of work. The relationship between personality and values was discussed in Chapter 1 and the data collected for this research is consistent with GNS as a part of personality, however in the absence of longitudinal data this assumption is speculative. Although there were correlations between GNS and the values and work values, these were not high so it is suggested that GNS is not a value, but people high in GNS might value certain things and that might result in goal-directed behaviour.

The lack of support for the mediating role of values and work values is explained in part by the negative relationship between GNS and intention to train. There was also no relationship between values and intention to train or training activity. Here, for values to act as mediators there needed to be a positive relationship between GNS and intention to train or training activity and it is suggested that the problem lies more with the dependent variable selected (the taking up of training opportunities) and the way in which it was measured, rather than its proposed predictors. The training data only concerned the number of courses respondents intended to take and actually undertook, and no information was available about the quality and nature of the courses undertaken. The reason for this was that measuring qualitative variables is far more problematic than measuring quantitative ones and the organisation was unlikely to have claimed to be running poor quality courses. For the test of mediation, independent and dependent variables must be measured on continuous scales which must accurately reflect what is being measured and for that reason the number of courses both intended and undertaken was chosen.

As a result of the findings of this research it is suggested that training was an inappropriate variable to use to test the predictive power of GNS. High GNS does not

appear to manifest itself through increased training activity; rather those respondents who showed the least intention to train were higher in GNS than others whose training activity was high. Thus, the ability of GNS to predict behaviour has not been shown through this research.

With regards to GNS it might be more pertinent to ask not just whether individuals have an inherent desire for personal growth but whether and under what circumstances those needs actually change. To do this it would be necessary to devise an accurate and reliable measure of GNS; to possibly find an alternative to training activity as the dependent variable; and to assess employees at different times and over a longer period than was possible for this research. It is suggested that the current method for measuring GNS, whilst it is devised in such a way that it reduces the possibility of respondents saying 'I want everything', may be improved by devising a format which does not include response alternatives. These appear to affect employees responses, as to answer the questions in the 'job choice' format they are giving a self-assessment of their level of GNS but to do so, they have to make a judgement about which of the alternatives is most inviting to them at the time. Now, it could be argued that this is a desirable outcome because of the earlier comments about the fulfilment of basic needs, prior to high-order needs, but if the premise is that GNS is a personality trait, then the response alternatives can only confuse the issue of what the scale is actually measuring. Here it is suggested that if GNS is part of personality it would be necessary to devise a means of measuring it, which follows the same format as other methods used in the measurement of personality.

The Big Five Index was found to be as reliable as the NEO-FFI and the correlations between Openness and GNS were similar for the two measures. The BFI is the measure of choice in respect of future research because the NEO-FFI is copyright protected and the license fees and cost per questionnaire used were high, even for research use. The NEO-FFI contains sixty items whereas the BFI contains only 44, which means the addition of extra items to measure GNS would only increase its size to that of the NEO-FFI. A possible scale is shown below which follows the format of the BFI and is made up of reworded items from the GNS 'job choice' scale plus the new item, which emerged from the qualitative data analysis for this research.

I see myself as someone who.....

1. wants the opportunity to be creative and imaginative in work (GNS)
2. sees work only as a way of making money
3. dislikes being required to make important decisions in my job (GNS-R)
4. tends to find fault with others
5. enjoys working hard for the responsibility it brings (GNS)
6. only goes to work because there are lots of nice people there
7. likes to decide on my own the best way to do my job (GNS)
8. prefers work that is routine (GNS-R)
9. is outgoing, sociable
10. has always worked hard to develop new skills (GNS)
11. doesn't feel the need to be challenged and tested in work activities (GNS-R)
12. is relaxed, handles stress well
13. wants to use my skills and abilities to the fullest extent (GNS)
14. needs to be able to put an equal amount of energy into my work and private lives (GNS)
15. is talkative
16. feels uncomfortable about learning new things (GNS-R)
17. wants more chance for promotion at work

Prior to including the items within a personality measure it would be necessary to test the scale alongside the old measure of GNS, to see whether it performs in the same way within a job characteristics-job satisfaction framework. If the results of this were encouraging then the next step would be to include the items measuring GNS in the BFI as above. There are, however, problems to be overcome when combining scales as the original validity and reliability of the personality scale may be reduced. Nonetheless, it is suggested that some exploratory work in this area may produce some interesting results.

Alternatives to Training as an Outcome Variable

The work for this thesis has shown that there is a predictive relationship between GNS and values and work values but that there was no relationship between these variables and either intention to train or training activity. As a result of this it is suggested that the area of training is too narrow a concept to be used as the dependent variable in the model proposed in Chapter 5, and that a variable be chosen which encompasses not just training

activity but more of the outcomes that people high in GNS might be seeking. Roccas et al (2002) suggest that values refer to what people consider to be important and the goals they wish to pursue. It is clear from Study3 that although to undertake more training in a particular area may be one specific goal for some people high in GNS, in itself training may be seen as no more than a means to an end. The defining goals of people of the three value types used in the model are set out on pages 130 and 131 and it is possible that such goals may be more useful outcomes with which to test the model. These could be taken from the defining goals of each value type so the three goals could be as follows

1. To become more competent in my work to provide myself with the opportunity for personal success (relating to Achievement) 'Personal Success'
2. To choose the areas I would like to learn more about and to gain ideas for my development into new and possibly different areas (relating to Self-direction) 'Development through Personal Choice'
3. To have the opportunity for greater novelty and challenge in my work (relating to Stimulation) 'Novelty and Challenge'

Work values are not included as separate goals as the only positive correlation with GNS was with Cognitive Work Values and it is felt that these would be encompassed by the three goals above. The revised model is shown in Figure 7.

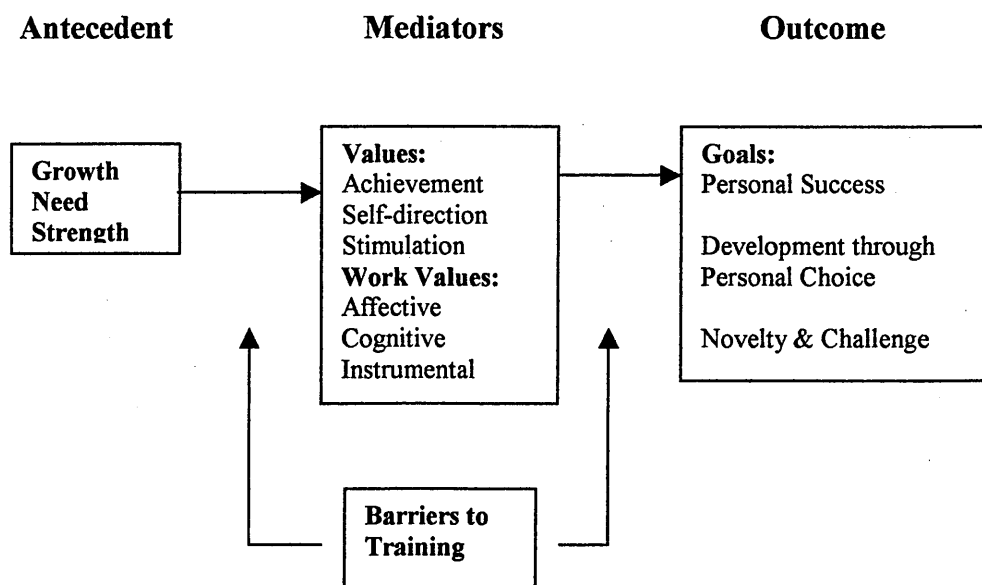


Figure 7

As can be seen from the diagram, barriers to attending training and development courses are included as potential moderators of the GNS-Goals relationship. Although no support was found for the moderating effect of barriers to training, as Tharenou (2001) suggests, employer support, both in terms of approval and funding and an organisational climate devoted to the personal development of employees is a key issue in training and development. So it is likely that such support is important in assisting employees in both the setting and achieving of goals, and the lack of it might be a potential barrier. Further, it was suggested earlier that a reason people high in GNS might take up fewer training opportunities, is that they may be in more senior positions within the organisation and as such may have less time due to higher workloads, so this may also be a potential barrier.

Finally, the combination of both quantitative and qualitative methods of data collection and analysis proved useful in this research. There are circumstances where human behaviour cannot be adequately explained by quantitative methods alone and the study reported in Chapter 4 added to the understanding of GNS as a construct. Human Beings use language to help to develop a 'common sense' of the meaning of their own and others' behaviour, so taking the 'common sense' view of participants added a dimension to the research that would not otherwise have been available. Here, the qualitative data collection and analysis was used to clarify and further examine some of the issues that arose from the earlier quantitative studies, which resulted in more quantitative work being carried out. Future researchers might wish to consider such a combination when examining individual difference factors whose definitions are proving elusive

Future Research

This research has provided some evidence that GNS might be a part of peoples' personality and as such has important implications for future research into GNS itself and the constituents of personality in general. This is particularly pertinent because of the ongoing debate about the nature of the fifth factor of personality that was accepted at the beginning of this thesis as Openness to Experience. At the start of this thesis it was suggested that an evaluation of the JCM and an in depth analysis of the nature of GNS may provide a clearer definition of the construct and thus make it more useful as a measurement tool in the field of occupational psychology. Future research should concentrate on the structure of GNS and its place in personality theory as a higher-order factor relating to personal growth. Such research should focus heavily on the validity of

and indeed other constructs that are used in test theory to help employers and employees make decisions about organisational issues. Such issues not only have serious financial implications but also have a potentially life-changing effect on individuals. Contribution to the personal development of employees can lead to great benefits in terms of the recruitment and retention of staff in that it is a demonstration of the value that organisations attach to their human resources. Littlewood calls for more rigorous research methods in the development of tests which are to be used for commercial purposes and future research should not only include in-depth assessments of validity but also report the results of those tests so that others may learn from them. Shying away from rigorous and diverse methods of construct analysis can only result in misleading and inappropriate tests being produced. Academics have a responsibility to employers and others to ensure that the theories they are proposing have sound bases, have a truly practical value and lead to genuine benefits to organisations and employees alike. For this reason, the assumptions made in this thesis need further investigation before a model can be devised which fulfils those requirements.

It is possible that Stress is a stable personal trait but that people's level of Stress may change according to that need, so future research could also include more longitudinal studies. Studies would provide the opportunity for GNS to be assessed at regular intervals and over a longer period of time. In this way might provide more support for the stability of the construct. Whilst it is accepted that some employees find the constant barrage of testing tiresome and often treat it with suspicion, if presented in such a way that their full support is obtained that testing could come to be seen by them as a valuable part of the development process. Because high GNS was found not to predict training activities, future research might concentrate on other work outcomes such as performance, or perhaps use outcomes decided upon by the employees themselves. In this way, research would be driven by the researcher, the employer and the employee alike and thus have greater practical value to the organisation.

Summary of findings

This research has provided evidence that there is an individual difference factor relating to personal goals, which moderates the relationship between job characteristics and job satisfaction. However, the lack of a clear definition from both academic and lay and subsequent researchers and the belief that sounds its measurement has made it

difficult to find indicators of the nature of GNS from past research. In an attempt to find a clearer definition of GNS, the studies reported here have shown that although GNS shares some similarities with other personality and values domains, it is measuring something that they do not. The 'would like' format was found to be psychometrically distinct from the 'job choice format', but as the wording of both sets of items is similar, the 'would like' format may have had a priming effect on the 'job choice' format both in the Job Diagnostic Survey and in the studies reported here. The addition of the new items to the 'job choice' scale resulted in a possible new dimension of GNS being found in that people who are high in GNS may be seeking a balance between their work and private lives. It may be that the stability that results from this balance gives high GNS people the base from which they can pursue personal growth and development.

GNS has been shown in this thesis to be different from the personality factor Openness to Experience. Evidence from the literature on the five-factor model indicates that there may be a further, higher-order, factor of personality which relates to personal growth. In order to show that GNS is a part of peoples' personalities the measure for it needs to be improved. Here, a method of doing this has been suggested which asks individuals their perceptions about themselves, rather than asking them for their preferences in terms of types of jobs. This measure could eventually be combined with the Big Five measure for personality and an examination of the properties of the GNS items could be made, alongside those for personality.

This research has shown that GNS as measured here, does not predict those people who take up training opportunities more readily, and so it was suggested that a more appropriate manifestation of the need for personal growth might be found in the goals individuals wish to pursue. Thus, a revised model was proposed which included these as behavioural outcomes. Finally, the results of this research suggest that GNS is an important individual difference factor which has the potential to provide a good basis for the prediction of workplace behaviour in the future.

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The Job Characteristics Model – Table of Selected Studies

	Authors	Organisation Type	Workers	Sample Size	Instruments	GNS format	Support for GNS as Moderator	Analysis
1.	Hackman & Oldham (1976)	Industrial & Service (7)	Blue collar White collar Professional	658	JDS JRF	Job choice	Yes	JCs & outcomes. JCs & CPSS. GNS as moderator. MPS & affective outcomes.
2.	* Dunham (1976)	Merchandising	Sales Assistants	3610	JDS	Not tested		JCs & work satisfaction. Factor analysis of JCs.
3.	* Oldham (1976)	Manufacturing	Clerical	64	JDS	Would like	Yes	Internal work motivation & job performance. Moderating effect of GNS & satisfactions on JCs & internal motivation.
4.	^ Oldham et al (1976)	Bank	Clerical	201	JDS JRF	Job choice	Yes	As study no. 1. Satisfaction with work context as moderator of JCs JS relationship.
5.	* Dunham et al (1977)	Service	Unskilled up to Director level	5945	Part JDS	Not tested		Factor analysis of job characteristics.
6.	* Hackman et al (1978)	Bank	Clerical	94	JDS (longitudinal)	Would like	Yes	Effects of changes in JCs on satisfactions, absenteeism & performance. Moderating effect of GNS.
7.	^ Wall et al (1978)	Sweet factory	Shop floor	47	JDS JRF	Not tested		Relationships betw. JCs, CPSS & outcomes using path analysis.
8.	# Abdel-Halim (1979)	Manufacturing	Managerial	87	JDS Job Involvement	Would like	Yes	Moderating effect of GNS & interpersonal satisfactions on relationship of JCs to job satisfaction & job involvement.
9.	# Evans et al (1979)	Car assembly	Supervisors & managers	343	JDS TAT measure of nAch	Job choice	Weak	JCs & CPSS & outcomes + moderating effect of GNS. Correlations with expectancy theory outcomes.

Table of Studies (continued)

	Authors	Organisation Type	Workers	Sample Size	Instruments	GNS format	Support for GNS as Moderator	Analysis
10.	^ Orpen (1979)	Quasi-federal agency	Clerical	72	JDS JDI (work satisfaction). Job involvement. (longitudinal)	Hackman & Lawler (1971) 12- item Would like scale	Yes	JCs & CPSs & Outcomes + moderating effect of GNS & context satisfactions.
11.	# Arnold & House (1980)	Manufacturing	Engineering	120	JDS Peer & supervisor ratings	Would like Job choice	No	JCs & CPSs & outcomes + moderating effect of GNS.
12.	# Champoux (1980)	R & D Federal agency Federal Agency	Mixed Mixed Mixed	1198 66 161	JDS	Tested but format not specified	Yes Yes Yes	Job scope & CPSs & outcomes + moderating effect of GNS.
13.	* Pokorney et al (1980)	Insurance	Supervisors & managers	173	JDS JRF	Tested but format not specified	Yes	Factor structure of JDS & JRF. Moderating effect of GNS.
14.	* Griffin (1982)	Manufacturing	Production	100	JDS JCI (Sims et al (1976) ERG (satisfaction scale Alderfer 1972)	Job Choice	Yes	JCs & Productivity & Satisfaction. Moderating effect of GNS.
15.	* Kemp & Cook (1983)	Manufacturing & Engineering	Blue collar	309 & 406	JDS. Job satisfaction, job complexity & Higher Order Need Strength (all Warr, Cook & Wall (1979)	Not tested (GNS measured as Higher Order Need Strength)	No	'GNS' & Job longevity as moderators of the job complexity - job satisfaction relationship.

Table of Studies (continued)

	Authors	Organisation Type	Workers	Sample Size	Instruments	GNS format	Support for GNS as Moderator	Analysis
16.	* Lee et al (1983)	Local Government	Mixed	1972	JDS, Triple Audit Opinion Survey (Dawis & Weitzel, 1970)	Not tested		JCs and satisfaction with work outcomes.
17.	* Harvey et al (1985)	National Guard	Guard members	2028	JDS	Not tested		Factor structure and dimensionality of job characteristics as measured using the JDS.
18.	^ Birnbaum et al (1986)	Banks, schools, phone company, labour department	Mixed	461	JDS, JRF, JDI	Not tested		Factor structure and dimensionality of job characteristics as measured using the JDS compared to JDI.
19.	* Graen et al (1986)	Government Service	Unknown	62	Part JDS	Job Choice	Yes	GNS as moderator compared with GN-GO model.
20.	* Fox & Feldman (1988)	Hospital, Government Agency	Middle management	92	JDS, Activity-related attention (ARA)	Not tested		Relations between attention state * JCs & outcomes + comparison with CPSs.
21.	# Jans & McMahon (1989)	Hospital, schools, armed forces	Mixed	90	JDS	Not tested		JCs & CPSs & outcomes + 6 th proposed dimension – self-expression.
22.	* Khandelwal & Aleem (1990)	Public sector	Mixed	160	JDS	Not tested		Factor structure of the JCs.
23.	# Medcof (1991)	Insurance	Clerical	82	JDS	Not tested		
24.	* Das (1991)	Students	Trained in assembly work	51	JDS	Tested but format not reported	Yes	Moderating effect of GNS. GN-GO model.
25.	^ Johns et al (1992)	Utility Company	Lower management	56	JDS	Tested but format not reported	Yes	Moderating effect of GNS.
25.				300	JDS	Job choice	Yes	Mediating effects of CPSs. Moderator effects of GNS, context satisfactions and performance.

Table of Studies (continued)

	Authors	Organisation Type	Workers	Sample Size	Instruments	GNS format	Support for GNS as Moderator	Analysis
26.	* Kahn & Robertson (1992)	Travel trade	Computer users	154	JDS	Not tested		JCs & outcomes. Training and experience as moderators.
27.	* Tiegs et al (1992)	Various	Various	6405 (database of Oldham et al, 1979)	JDS	Job choice	No	GNS & context satisfactions as moderators of the JCs & focal work outcomes.
28.	^ Barnabe & Burns (1994)	Schools	Teachers	247	JDS	Not tested		JCs & CPSS & outcomes
29.	^ Renn & Vandenberg (1995)	Not reported. Insurance.	Mixed Processing, customer services	188 90	JDS JDS	Not tested		Mediation of the CPSSs.
30.	^ Lee-Ross (1998)	Hotel	Seasonal workers	163	JDS	'would like' 'job choice'	Partial	As Study 1
31.	* de Jong et al (2001)	Various	Graduate students in part-time jobs	95	JDS Big Five	'would like'	Yes	

Key to symbols

NB Study 1 is the original test of the Job Characteristics Model

^ Close replication * Partial Replication # Extension

JDS – Job Diagnostic Survey

JRF – Job Rating Form

JDI – Job Diagnostic Index

This questionnaire was developed as part of a study of jobs and how people react to them.

On the following pages you will find several kinds of questions about your job and about yourself.

Specific instructions are given at the start of each section. Please read them carefully.

It should take no more than 40 minutes to complete the entire questionnaire. Please move through it quickly. You will find that some of the questions are very similar, this is intentional.

The questions are designed to obtain **your** perceptions of your job and **your** reactions to it, and **your** perceptions of yourself.

There are no trick questions. Your individual answers will be kept completely confidential.

Please answer each item as honestly and frankly as possible. **Please do not leave any questions unanswered.**

Thank you for your co-operation.

SECTION 1

Listed below are a number of characteristics which could be present in any job. People differ about how much they would like to have each one present in their own jobs. We are interested in learning *how much you personally would like* to have each one present in your job.

Using the scale below, please indicate the *degree* to which you *would like* to have each characteristic present in your job.

1	2	3	4	5	6	7
Would like having this only a moderate amount (or less)			Would like having this very much			Would like having this <i>extremely</i> much

- ☐ 1. High respect and fair treatment from my supervisor.
- ☐ 2. Stimulating and challenging work.
- ☐ 3. Chances to exercise independent thought and action in my job.
- ☐ 4. Great job security.
- ☐ 5. Very friendly co-workers.
- ☐ 6. Opportunities to learn new things from my work.
- ☐ 7. High salary and good fringe benefits.
- ☐ 8. Opportunities to be creative and imaginative in my work.
- ☐ 9. Quick promotions.
- ☐ 10. Opportunities for personal growth and development in my job.
- ☐ 11. A sense of worthwhile accomplishment in my work.

SECTION 2

People differ in the kinds of jobs they would most like to hold. The questions in this section give you a chance to say just what it is about a job that is most important to *you*.

For each question, two different kinds of jobs are briefly described. You are to indicate which of the jobs you personally would prefer - if you had to make a choice between them.

In answering each question, assume that everything else about the job is the same. Pay attention only to the two characteristics actually listed.

Two examples are given below.

JOB A			JOB B	
A job requiring work with mechanical equipment most of the day			A job requiring work with other people most of the day	
1-----	2-----	3-----	4-----	5
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you like working with people and working with equipment equally well, you would circle number 3, as has been done in the example.

* * * * *

Here is another example. This one asks for a harder choice - between two jobs which both have some undesirable features.

JOB A			JOB B	
A job requiring you to expose yourself to considerable physical danger			A job located 200 miles from your home and family	
1-----	2-----	3-----	4-----	5
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you would slightly prefer risking physical danger to working far from your home, you would circle number 2, as has been done in the example.

JOB A

1. A job where the pay is very good.

JOB B

A job where there is considerable opportunity to be creative & innovative.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

2. A job where you are often required to make important decisions.

JOB B

A job with many pleasant people to work with.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

3. A job in which greater responsibility is given to those who do the best work.

JOB B

A job in which greater responsibility is given to loyal employees who have the most seniority.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

4. A job in an organisation which is in financial trouble - and might have to close down within the year.

JOB B

A job in which you are not allowed to have any say whatever in how your work is scheduled, or in the procedures to be used in carrying it out.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

5. A very routine job.

A job where co-workers are not very friendly.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

6. A job with a supervisor who is often very critical of you and your work in front of other people.

A job which prevents you from using a number of skills that you worked hard to develop.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

7. A job with a supervisor who respects you and treats you fairly.

A job which provides constant opportunities for you to learn new and interesting things.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

8. A job where there is a real chance you could be laid off.

A job with very little chance to do challenging work.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

9. A job in which there is a real chance for you to develop new skills and advance in the organisation.

A job which provides lots of vacation time and an excellent fringe benefit package.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

10. A job with little freedom and independence to do your work in the way you think best.

A job where the working conditions are poor.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

11. A job with very satisfying teamwork.

A job which allows you to use your skills and abilities to the fullest extent.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

12. A job which offers little or no challenge.

A job which requires you to be completely isolated from co-workers.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

SECTION 3

In this section you are asked to rate how important each aspect of work is for you, using the scale provided.

The rating scale we want you to use runs:

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Use '1' if the aspect of work is **not at all important to you**

Use '4' if the aspect of work is **moderately important to you**

Use '7' if the aspect of work is **extremely important to you**

List of Work Aspects

Circle a number under each work aspect, according to the scale above

1. **Worthwhile accomplishment**

1.....2.....3.....4.....5.....6.....7

2. **Doing high quality work**

1.....2.....3.....4.....5.....6.....7

3. **A feeling of self-fulfilment**

1.....2.....3.....4.....5.....6.....7

4. **Knowing I have done a job well**

1.....2.....3.....4.....5.....6.....7

5. **Knowing that my job requires me to use my abilities**

1.....2.....3.....4.....5.....6.....7

6. **Personal growth and development**

1.....2.....3.....4.....5.....6.....7

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SECTION 4

This section of the questionnaire contains 60 statements. Please read each item carefully and mark in the box next to it the number which best corresponds to your agreement or disagreement.

1.....	2.....	3.....	4.....	5.....
Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

There are no right or wrong answers, and you need not be an "expert" to complete this part of the questionnaire. Describe yourself honestly and state your opinions as accurately as possible.

- | | | |
|-----|---|-------|
| 1. | I am not a worrier. | [] |
| 2. | I like to have a lot of people around me. | [] |
| 3. | I don't like to waste my time day dreaming. | [] |
| 4. | I try to be courteous to everyone I meet. | [] |
| 5. | I keep my belongings clean and neat. | [] |
| 6. | I often feel inferior to others. | [] |
| 7. | I laugh easily. | [] |
| 8. | Once I find the right way to do something, I stick to it. | [] |
| 9. | I often get into arguments with my family and co-workers. | [] |
| 10. | I'm pretty good about pacing myself so as to get things done on time. | [] |
| 11. | When I'm under a great deal of stress, sometimes I feel like I'm going to pieces. | [] |
| 12. | I don't consider myself especially "light-hearted". | [] |
| 13. | I am intrigued by the patterns I find in art and nature. | [] |
| 14. | Some people think I'm selfish and egotistical. | [] |
| 15. | I am not a very methodical person. | [] |
| 16. | I rarely feel lonely or blue. | [] |

	1.....	2.....	3.....	4.....	5.....	
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	
17.	I really enjoy talking to people.					[]
18.	I believe letting students hear controversial speakers can only confuse and mislead them.					[]
19.	I would rather cooperate with others than compete with them.					[]
20.	I try to perform all the tasks assigned to me conscientiously.					[]
21.	I often feel tense and jittery.					[]
22.	I like to be where the action is.					[]
23.	Poetry has little or no effect on me					[]
24.	I tend to be cynical and sceptical of other's intentions.					[]
25.	I have a clear set of goals and work toward them in an orderly fashion					[]
26.	Sometimes I feel completely worthless.					[]
27.	I usually prefer to do things alone.					[]
28.	I often try new and foreign foods.					[]
29.	I believe that most people will take advantage of you if you let them.					[]
30.	I waste a lot of time before settling down to work.					[]
31.	I rarely feel fearful or anxious.					[]
32.	I often feel as if I'm bursting with energy.					[]
33.	I seldom notice the moods or feelings that different environments produce.					[]
34.	Most people I know like me.					[]
35.	I work hard to accomplish my goals.					[]
36.	I often get angry at the way people treat me.					[]
37.	I am a cheerful, high-spirited person.					[]
38.	I believe we should look to our religious authorities for decisions on moral issues.					[]
39.	Some people think of me as cold and calculating.					[]
40.	When I make a commitment, I can always be counted on to follow through					[]
41.	Too often, when things go wrong, I get discouraged and feel like giving up.					[]

	1.....	2.....	3.....	4.....	5.....
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
42.	I am not a cheerful optimist.				[]
43.	Sometimes when I am reading poetry or looking at a work of art, I feel a chill or wave of excitement.				[]
44.	I'm hard-headed and tough-minded in my attitudes.				[]
45.	Sometimes I'm not as dependable or reliable as I should be.				[]
46.	I am seldom sad or depressed.				[]
47.	My life is fast paced.				[]
48.	I have little interest in speculating on the nature of the universe or the human condition.				[]
49.	I generally try to be thoughtful and considerate.				[]
50.	I am a productive person who always gets the job done.				[]
51.	I often feel helpless and want someone else to solve my problems.				[]
52.	I am a very active person.				[]
53.	I have a lot of intellectual curiosity.				[]
54.	If I don't like people I let them know it.				[]
55.	I never seem to be able to get organised.				[]
56.	At times I have been so ashamed I just wanted to hide.				[]
57.	I would rather go my own way than be a leader of others.				[]
58.	I often enjoy playing with theories or abstract ideas.				[]
59.	If necessary, I am willing to manipulate people to get what I want.				[]
60.	I strive for excellence in everything I do.				[]

SECTION 5

This section asks you to describe your job as accurately as you can. It does so by asking you two things

- a. How much of a certain characteristic is present in your job and**
- b. How important this job characteristic is to you**

Please use the scales below and circle the number you have chosen.

- 1a. **How much variety is there in your job?**

1-----2-----3-----4-----5-----6-----7

Very little

Moderate
Variety

Very Much

- 1b. **How important is this to you?**

1-----2-----3-----4-----5-----6-----7

Not at all
Important

Somewhat
Important

Extremely
Important

- 2a. **To what extent does your job give you the chance to do entire piece of work from beginning to end?**

1-----2-----3-----4-----5-----6-----7

Very little

A moderate
chance

Very much

- 2b. **How important is this to you?**

1-----2-----3-----4-----5-----6-----7

Not at all
Important

Somewhat
Important

Extremely
Important

- 3a. **How significant or important is your job? That is are the results of your work likely to significantly affect the lives or well-being of others?**

1-----2-----3-----4-----5-----6-----7

Not very
Significant

Moderately
Significant

Highly
Significant

- 3b. **How important is this to you?**

1-----2-----3-----4-----5-----6-----7

Not at all
Important

Somewhat
Important

Extremely
Important

- 4a. **How much autonomy is there in your job? That is, to what extent does your job permit you to decide on your own how to go about doing the work?**

1-----2-----3-----4-----5-----6-----7

Very little

Moderate
Autonomy

Very much

- 4b. **How important is this to you?**

1-----2-----3-----4-----5-----6-----7

Not at all
Important

Somewhat
Important

Extremely
Important

- 5a. **To what extent does doing the job itself provide you with information about your work performance? (That is apart from any feedback you may get from other people.)**

1-----2-----3-----4-----5-----6-----7

Very little

Moderate
Feedback

Very much

- 5b. **How important is this to you?**

1-----2-----3-----4-----5-----6-----7

Not at all
Important

Somewhat
Important

Extremely
Important

SECTION 6

In this section you are asked to rate how important each value is for you as a guiding principle in YOUR life, using the scale provided. Try to distinguish as much as possible between the values by using all of the numbers. Since only a limited number of the values can be "of supreme importance" do not over-use this highest category. The rating scale we want you to use runs:

-1	0	1	2	3	4	5	6	7
----	---	---	---	---	---	---	---	---

Use '-1' if any values are opposed to the principles that guide you
 Use '7' for any value of supreme importance as a guiding principle in your life

Most values will fall in the range '0' to '6'
 where 0 means the value is **not at all important to you**
 and 6 means it is **very important to you**

Value List

Circle a number under each value, according to the scale above.

1. EQUALITY (equal opportunity for all)

-1—0—1—2—3—4—5—6—7

2. INNER HARMONY (at peace with myself)

-1—0—1—2—3—4—5—6—7

3. SOCIAL POWER (control over others, dominance)

-1—0—1—2—3—4—5—6—7

4. PLEASURE (gratification of desires)

-1—0—1—2—3—4—5—6—7

5. FREEDOM (freedom of action and thought)

-1—0—1—2—3—4—5—6—7

6. A SPIRITUAL LIFE (emphasis on spiritual not material matters)

-1—0—1—2—3—4—5—6—7

7. SOCIAL ORDER (stability of society)

-1—0—1—2—3—4—5—6—7

8. AN EXCITING LIFE (stimulating experiences)

-1—0—1—2—3—4—5—6—7

9. MEANING IN LIFE (a purpose in life)

-1—0—1—2—3—4—5—6—7

10. POLITENESS (courtesy, good manners)

-1—0—1—2—3—4—5—6—7

11. WEALTH (material possessions, money)

-1—0—1—2—3—4—5—6—7

12. NATIONAL SECURITY (protection of my nation from enemies)

-1—0—1—2—3—4—5—6—7

13. RECIPROCATION OF FAVOURS (avoidance of indebtedness)

-1—0—1—2—3—4—5—6—7

14. CREATIVITY (uniqueness, imagination)

-1—0—1—2—3—4—5—6—7

15. A WORLD AT PEACE (free of war and conflict)

-1—0—1—2—3—4—5—6—7

16. RESPECT FOR TRADITION (preservation of time-honoured customs)

-1—0—1—2—3—4—5—6—7

Use '-1' if any values are opposed to the principles that guide you
 Use '7' for any value of supreme importance as a guiding principle in your life

Most values will fall in the range '0' to '6'
 where 0 means the value is not at all important to you
 and 6 means it is very important to you

17. **SELF-DISCIPLINE** (self-restraint, resistance to temptation)

-1—0—1—2—3—4—5—6—7

18. **DETACHMENT** (from worldly concerns)

-1—0—1—2—3—4—5—6—7

19. **FAMILY SECURITY** (safety for loved ones)

-1—0—1—2—3—4—5—6—7

20. **UNITY WITH NATURE** (fitting into nature)

-1—0—1—2—3—4—5—6—7

21. **A VARIED LIFE** (filled with challenge, novelty and change)

-1—0—1—2—3—4—5—6—7

22. **WISDOM** (a mature understanding of life)

-1—0—1—2—3—4—5—6—7

23. **AUTHORITY** (the right to lead or command)

-1—0—1—2—3—4—5—6—7

24. **A WORLD OF BEAUTY** (beauty of nature and the arts)

-1—0—1—2—3—4—5—6—7

25. **SOCIAL JUSTICE** (correcting injustice, care for the weak)

-1—0—1—2—3—4—5—6—7

26. **INDEPENDENT** (self-reliant, self-sufficient)

-1—0—1—2—3—4—5—6—7

27. **MODERATE** (avoiding extremes of feeling and action)

-1—0—1—2—3—4—5—6—7

28. **LOYAL** (faithful to my friends, group)

-1—0—1—2—3—4—5—6—7

29. **AMBITIOUS** (hardworking, aspiring)

-1—0—1—2—3—4—5—6—7

30. **BROAD MINDED** (tolerant of different ideas and beliefs)

-1—0—1—2—3—4—5—6—7

31. **HUMBLE** (modest, self-effacing)

-1—0—1—2—3—4—5—6—7

32. **DARING** (seeking adventure, risk)

-1—0—1—2—3—4—5—6—7

33. **PROTECTING THE ENVIRONMENT** (preserving nature)

-1—0—1—2—3—4—5—6—7

34. **INFLUENTIAL** (having an impact on people and events)

-1—0—1—2—3—4—5—6—7

35. **HONOURING OF PARENTS AND ELDERS** (showing respect)

-1—0—1—2—3—4—5—6—7

36. **CHOOSING OWN GOALS** (selecting own purposes)

-1—0—1—2—3—4—5—6—7

37. **CAPABLE** (competent, effective, efficient)

-1—0—1—2—3—4—5—6—7

38. **ACCEPTING MY PORTION IN LIFE** (submitting to life's circumstances)

-1—0—1—2—3—4—5—6—7

Use '-1' if any values are opposed to the principles that guide you
 Use '7' for any value of supreme importance as a guiding principle in your life

Most values will fall in the range '0' to '6'
 where 0 means the value is not at all important to you
 and 6 means it is very important to you

39. **HONEST** (genuine, sincere)

-1—0—1—2—3—4—5—6—7

40. **OBEDIENT** (dutiful, meeting obligations)

-1—0—1—2—3—4—5—6—7

41. **HELPFUL** (working for the welfare of others)

-1—0—1—2—3—4—5—6—7

42. **ENJOYING LIFE** (enjoying food, sex, leisure, etc.)

-1—0—1—2—3—4—5—6—7

43. **DEVOUT** (holding to religious faith and belief)

-1—0—1—2—3—4—5—6—7

44. **RESPONSIBLE** (dependable, reliable)

-1—0—1—2—3—4—5—6—7

45. **CURIOUS** (interested in everything, exploring)

-1—0—1—2—3—4—5—6—7

46. **FORGIVING** (willing to pardon others)

-1—0—1—2—3—4—5—6—7

47. **SUCCESSFUL** (achieving goals)

-1—0—1—2—3—4—5—6—7

48. **CLEAN** (neat, tidy)

-1—0—1—2—3—4—5—6—7

SECTION 7

In this section you are asked to rate how important each aspect of work is for you, using the scale provided.

The rating scale we want you to use runs:

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Use '1' if the aspect of work is not at all important to you

Use '4' if the aspect of work is moderately important to you

Use '7' if the aspect of work is extremely important to you

Work Aspects list

Circle a number under each work aspect, according to the scale above

1. **Using your skills to the maximum**

1.....2.....3.....4.....5.....6.....7

2. **Achieving something that you personally value**

1.....2.....3.....4.....5.....6.....7

Use '1' if the aspect of work is **not at all** important to you
Use '4' if the aspect of work is **moderately** important to you
Use '7' if the aspect of work is **extremely** important to you

3.

The opportunity to make your own decisions

1.....2.....3.....4.....5.....6.....7
4.

The opportunity to learn new things

1.....2.....3.....4.....5.....6.....7
5.

Challenging work

1.....2.....3.....4.....5.....6.....7
6.

Extending your range of abilities

1.....2.....3.....4.....5.....6.....7

SECTION 8

In this section you are asked to rate how important each **work value** is to you, using the scale provided.
The rating scale we want you to use runs:

1	2	3	4	5
---	---	---	---	---

Use '1' if the work value is **not at all** important to you
Use '3' if the work value is **moderately** important to you
Use '5' if the work value is **extremely** important to you

Work Values list

Circle a number under each work aspect, according to the scale above

1.

Achievement in work

1-----2-----3-----4-----5
2.

Advancement, chances for promotion

1-----2-----3-----4-----5
3.

Benefits, holiday, sick leave, pension, insurance etc.

1-----2-----3-----4-----5
4.

Company, to be employed by a company for which you are proud to work

1-----2-----3-----4-----5

Use '1' if the work value is **not at all important to you**
 Use '3' if the work value is **moderately important to you**
 Use '5' if the work value is **extremely important to you**

- | | |
|--|--|
| <p>5. Contributions to society</p> <p>1-----2-----3-----4-----5</p> | <p>15. Job status</p> <p>1-----2-----3-----4-----5</p> |
| <p>6. Convenient hours of work</p> <p>1-----2-----3-----4-----5</p> | <p>16. Meaningful work</p> <p>1-----2-----3-----4-----5</p> |
| <p>7. Co-workers, fellow workers who are pleasant and agreeable</p> <p>1-----2-----3-----4-----5</p> | <p>17. Opportunity for personal growth</p> <p>1-----2-----3-----4-----5</p> |
| <p>8. Esteem, that you are valued as a person</p> <p>1-----2-----3-----4-----5</p> | <p>18. Opportunity to meet people and interact with them</p> <p>1-----2-----3-----4-----5</p> |
| <p>9. Feedback concerning the results of your work</p> <p>1-----2-----3-----4-----5</p> | <p>19. Pay, the amount of money you receive</p> <p>1-----2-----3-----4-----5</p> |
| <p>10. Independence in work</p> <p>1-----2-----3-----4-----5</p> | <p>20. Recognition for doing a good job</p> <p>1-----2-----3-----4-----5</p> |
| <p>11. Influence in the organisation</p> <p>1-----2-----3-----4-----5</p> | <p>21. Responsibility</p> <p>1-----2-----3-----4-----5</p> |
| <p>12. Influence in work</p> <p>1-----2-----3-----4-----5</p> | <p>22. Supervisor, a fair and considerate boss</p> <p>1-----2-----3-----4-----5</p> |
| <p>13. Job interest, to do work which is interesting to you</p> <p>1-----2-----3-----4-----5</p> | <p>23. Use of ability and knowledge in your work</p> <p>1-----2-----3-----4-----5</p> |
| <p>14. Job security, permanent job</p> <p>1-----2-----3-----4-----5</p> | <p>24. Work conditions, comfortable and clean</p> <p>1-----2-----3-----4-----5</p> |

SECTION 9

In this section you are asked how much you **agree** or **disagree** with each of the following statements about your job. **Tick one for each statement.**

	4-----3-----2-----1-----0	Strongly Agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree
1. I find real enjoyment in my job		()	()	()	()	()
2. I consider my job rather unpleasant		()	()	()	()	()
3. I am often bored with my job		()	()	()	()	()
4. I am fairly well satisfied with my job		()	()	()	()	()
5. I definitely dislike my job		()	()	()	()	()
6. Each day on my job seems like it will never end		()	()	()	()	()
7. Most days I am enthusiastic about my job		()	()	()	()	()

SECTION 10

In this final section please give the following details about yourself.

1. Sex: Male _____ Female: _____

2. Age (tick one):

_____ under 20

_____ 40-49

_____ 20-29

_____ 50-59

_____ 30-39

_____ 60 or over

3. Education (tick one):

_____ Secondary School to GCSE level

_____ Secondary School to A level

_____ Some Business College or Technical School Experience

_____ Some College Experience (other than business or technical school)

_____ Business College or Technical School Degree

_____ University Degree

_____ Master's or Higher Degree

_____ Distance Learning Course of Study

4. What is your brief job title?

Please turn the page

5. Have you ever undertaken a course of study outside working hours, e.g. night school? YES [] NO []

6. If YES please describe the course in the space below

7. Did you undertake this course of study mainly to further your career? YES [] NO []

OR

8. Was the course of study undertaken mainly for your own interest or enjoyment YES [] NO []

PLEASE MAKE SURE THAT YOU HAVE ANSWERED EVERY QUESTION

Thank you very much for your time

If you would like to take part in the draw for the cash prize, please give your name and department below. These details will not be seen by anyone but the researcher.

NAME: _____ DEPARTMENT: _____

Ph.D Research

Dear Staff Member,

I am research student at the University of Surrey working for a Ph.D in Occupational Psychology. Your Chief Executive has kindly agreed to allow me to ask you if you would take part in my work.

All you are asked to do is complete the questionnaire which accompanies this letter and return it directly to me the reply paid envelope.

None of the completed questionnaires will **ever** be seen by anyone but me and my research colleagues. Although the completion of the questionnaire is for academic research purposes, the results of my work may help employers to better understand the training and career needs of their staff.

The deadline for returning your completed questionnaire is Friday 17th September and after this I will be holding a draw for a **cash prize of £100**. If you would like to take part in the draw you **must** fill out the box on the back page of your completed questionnaire, giving your name and department so that I can contact the winner. The slip containing your name will be removed from the questionnaire and destroyed once the draw has taken place to ensure your anonymity thereafter.

Please, please return your completed questionnaire and I shall be eternally grateful!!

Thank you.

Yours very sincerely,

Jane Sutton

Study 1 - Correlations Among Questionnaire Scales Scores

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.	19.	20.	21.	22.	23.
Skill Variety	-																						
Task Identity	.33	-																					
Task Significance	.18	.10	-																				
Autonomy	.35	.27	.13	-																			
Feedback (job)	.38	.27	.26	.22	-																		
MPS	.54	.48	.34	.66	.78	-																	
MPSA	.69	.65	.52	.61	.67	.89	-																
MPSM	.60	.58	.38	.55	.64	.94	.88	-															
Job Satisfaction	.51	.36	.25	.30	.37	.46	.57	.47	-														
GNS 'would like'	.23	.11	.23	.22	.20	.25	.31	.25	.14	-													
GNS 'job choice'	.17	.08	.04	.23	.07	.17	.19	.18	.13	.42	-												
HONS	.19	.09	.27	.21	.21	.27	.30	.27	.19	.54	.40	-											
SANS	.17	.11	.28	.19	.21	.26	.30	.25	.13	.53	.23	.50	-										
Importance of ICs	.34	.27	.44	.39	.35	.52	.56	.54	.23	.42	.31	.52	.49	-									
Neuroticism	-.17	-.17	-.08	-.12	-.17	-.22	-.22	-.24	-.36	-.07	-.15	-.14	-.08	-.14	-								
Extraversion	.18	.11	.20	.08	.18	.21	.24	.24	.24	.25	.12	.35	.23	.27	-.36	-							
Openness	.06	.03	.11	.07	.09	.10	.11	.09	.00	.25	.29	.22	.16	.20	.04	.11	-						
Agreeableness	.04	.07	.03	-.03	.10	.04	.07	.03	.17	.06	-.18	-.04	.07	-.03	-.13	.16	-.05	-					
Conscientiousness	.01	.04	.05	.01	.10	.10	.11	.12	.17	.13	.04	.24	.29	.22	-.30	.31	-.04	.19	-				
Achievement	.06	.05	.23	.08	.15	.18	.18	.17	.13	.37	.27	.63	.41	.37	-.21	.35	.09	-.10	.40	-			
Benevolence	.05	.08	.14	.02	.13	.15	.14	.17	.13	.15	.03	.40	.37	.30	-.04	.22	.12	.20	.30	.48	-		
Conformity	-.06	.00	.04	-.04	.08	.03	.01	.01	.05	.05	-.12	.29	.28	.14	-.02	.14	-.15	.17	.35	.47	.61	-	
Hedonism	.04	.05	.11	.03	.11	.10	.11	.09	-.02	.22	-.06	.28	.20	.22	-.06	.36	.05	-.02	.13	.36	.19	.22	-
Power	-.03	.04	.10	.02	.10	.10	.08	.11	-.05	.17	.12	.26	.08	.16	-.09	.22	.01	-.28	.15	.50	.10	.25	.29
Security	-.01	.11	.08	-.06	.12	.08	.08	.08	.06	.08	-.07	.27	.30	.22	-.06	.08	-.03	.03	.36	.44	.51	.60	.27
Self-direction	.07	.10	.15	.08	.13	.16	.17	.17	.00	.35	.29	.56	.38	.38	-.06	.23	.37	-.15	.24	.64	.51	.37	.39
Stimulation	.11	.04	.12	.05	.14	.16	.14	.17	-.02	.35	.23	.49	.28	.31	-.09	.41	.19	-.16	.10	.54	.38	.26	.49
Tradition	-.07	-.03	.03	-.13	.07	-.01	-.04	-.01	.03	.00	-.17	.21	.17	.09	.07	.06	-.09	.08	.18	.41	.59	.67	.09
Universalism	.01	.07	.05	-.02	.09	.07	.06	.07	-.02	.21	.05	.34	.34	.27	-.03	.09	.36	.05	.18	.38	.63	.43	.26
Affective WV	.04	-.02	.20	.01	.14	.13	.11	.12	.00	.19	-.13	.35	.39	.30	.09	.27	-.04	.11	.21	.37	.42	.36	.31
Cognitive WV	.08	.10	.33	.19	.25	.33	.33	.33	.19	.43	.31	.74	.54	.56	-.14	.32	.19	-.03	.29	.65	.46	.33	.28
Instrumental WV	-.08	.01	.12	-.09	.03	-.01	.00	.00	-.06	.02	-.34	.19	.23	.13	-.02	.10	-.14	.03	.23	.33	.31	.49	.31

n = 486

Correlations of 0.09 and above are significant at the 0.05 level (2-tailed).

Correlations of 0.12 and above are significant at the 0.01 level (2-tailed).

Study 1 - Correlations Among Questionnaire Scales Scores (continued)

	24.	25.	26.	27.	28.	29.	30.	31.	32.
Skill Variety									
Task Identity									
Task Significance									
Autonomy									
Feedback (job)									
MPS									
MPSA									
MPSM									
Job Satisfaction									
GNS 'would like'									
GNS 'job choice'									
HONS									
SANS									
Importance of JCs									
Neuroticism									
Extraversion									
Openness									
Agreeableness									
Conscientiousness									
Achievement									
Benevolence									
Conformity									
Hedonism									
Power	-								
Security	.33	-							
Self-direction	.34	.51	-						
Stimulation	.32	.27	.64	-					
Tradition	.26	.53	.35	.28	-				
Universalism	.10	.51	.62	.44	.49	-			
Affective WV	.13	.29	.33	.30	.29	.33	-		
Cognitive WV	.34	.37	.54	.46	.27	.42	.55	-	
Instrumental WV	.28	.50	.29	.22	.46	.34	.50	.41	-

n = 486

Correlations of 0.09 and above are significant at the 0.05 level (2-tailed).

Correlations of 0.12 and above are significant at the 0.01 level (2-tailed)

The questions below form part of a research project by a PhD student at Surrey University into the study of jobs and how people react to them.

You will find 5 questions asking you to describe your job as accurately as you can. Each question asks you how much of a certain characteristic is present in your job.

You will not be asked to give your name and all answers will be kept completely confidential.

Please use the scales below and circle the number you have chosen.

1. How much variety is there in your job?

1.....2.....3.....4.....5.....6.....7
Very little Moderate Very much
Variety

2. To what extent does your job give you the chance to do an entire piece of work from beginning to end?

1.....2.....3.....4.....5.....6.....7
Very little A Moderate Chance Very much

3. How significant or important is your job? That is, are the results of your work likely to significantly affect the lives or well-being of others?

1.....2.....3.....4.....5.....6.....7
Very little Moderately Significant Very much

4. How much autonomy is there in your job? That is, to what extent does your job permit you to decide on your own how to go about doing the work?

1.....2.....3.....4.....5.....6.....7
Very little Moderate Very much
Autonomy

5. To what extent does the job itself provide you with information about your work performance? (That is, apart from any feedback you may get from other people.)

1.....2.....3.....4.....5.....6.....7
Very little Moderate Very much
Feedback

Study 1a – Correlations Among Questionnaire Scale Scores

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.	19.	20.	21.	22.	23.
Skill Variety	-																						
Task Identity	.38	-																					
Task Significance	.40	.23	-																				
Autonomy	.50	.35	.38	-																			
Feedback (job)	.22	.19	.16	.17	-																		
MPS	.62	.51	.47	.73	.64	-																	
MPSA	.76	.66	.65	.73	.51	.89	-																
MPSM	.66	.58	.48	.61	.50	.93	.85	-															
Job Satisfaction	.60	.38	.46	.49	.26	.56	.67	.54	-														
GNS 'would like'	.13	.07	.18	.11	.22	.20	.21	.19	.08	-													
GNS 'job choice'	.14	.02	.07	.12	.02	.11	.11	.13	.06	.37	-												
HONS	.13	.08	.25	.17	.19	.24	.25	.24	.16	.51	.35	-											
SANS	.20	.12	.31	.18	.14	.25	.29	.26	.20	.44	.24	.54	-										
Importance of JCs	.42	.30	.57	.44	.27	.55	.61	.56	.39	.33	.28	.48	.48	-									
Neuroticism	-.06	-.10	-.07	-.07	-.13	-.15	-.13	-.16	-.25	-.09	-.13	-.19	-.11	-.12	-								
Extraversion	.10	.12	.16	.08	.14	.18	.18	.20	.20	.22	.07	.33	.23	.26	-.34	-							
Openness	.07	.00	.12	.06	.05	.08	.09	.07	.01	.21	.30	.13	.09	.22	.03	.07	-						
Agreeableness	.11	.11	.14	.04	.07	.11	.15	.10	.24	.03	-.13	-.06	.09	.08	-.14	.15	.02	-					
Conscientiousness	.07	.15	.17	.06	.10	.15	.17	.15	.24	.16	.08	.30	.36	.30	-.32	.30	-.01	.22	-				
Achievement	.03	.06	.20	.05	.15	.14	.14	.14	.10	.35	.22	.65	.42	.35	-.22	.34	.01	-.12	.40	-			
Benevolence	.06	.08	.16	.03	.11	.13	.13	.15	.14	.19	.00	.38	.32	.28	-.05	.20	.10	.20	.30	.48	-		
Conformity	-.04	.02	.07	-.04	.07	.03	.03	.02	.07	.12	-.11	.34	.32	.16	-.05	.15	-.17	.16	.35	.50	.60	-	
Hedonism	-.08	-.02	.03	-.04	.09	.03	.00	.02	-.08	.24	-.07	.29	.21	.12	-.07	.35	.02	-.07	.09	.41	.23	.27	-
Power	-.09	.00	.04	-.05	.10	.03	.00	.04	-.09	.18	.10	.30	.13	.10	-.10	.22	-.07	-.30	.15	.54	.13	.28	.35
Security	-.01	.10	.07	-.06	.11	.06	.06	.06	.02	.14	-.07	.30	.31	.17	-.07	.10	-.05	.03	.33	.44	.48	.60	.30
Self-direction	.00	.05	.07	-.02	.11	.08	.06	.09	-.05	.36	.24	.54	.35	.29	-.11	.24	.30	-.12	.24	.66	.50	.42	.45
Stimulation	-.54	-.02	.02	-.08	.13	.04	-.01	.04	-.12	.35	.16	.49	.26	.17	-.13	.39	.09	-.18	.10	.58	.37	.31	.56
Tradition	-.03	.02	.04	-.07	.06	.01	.01	.01	.03	.07	-.17	.24	.17	.06	.06	.09	-.13	.08	.19	.42	.56	.64	.14
Universalism	-.01	.08	.07	-.03	.07	.06	.06	.07	-.02	.24	.04	.34	.32	.24	-.04	.09	.33	.12	.20	.38	.59	.45	.27
Affective WV	.05	.04	.18	.03	.15	.13	.13	.13	.06	.22	-.14	.41	.39	.31	.03	.25	-.05	.13	.26	.41	.43	.42	.32
Cognitive WV	.15	.11	.31	.16	.21	.29	.29	.31	.18	.43	.26	.74	.56	.51	-.17	.31	.11	-.03	.34	.68	.43	.39	.32
Instrumental WV	-.12	.00	.05	-.09	.06	-.02	-.03	-.02	-.06	.08	-.37	.22	.20	.06	-.07	.15	-.19	.02	.20	.34	.31	.49	.35

n = 692

Correlations of 0.08 and above are significant at the 0.05 level (2-tailed).

Correlations of 0.11 and above are significant at the 0.01 level (2-tailed).

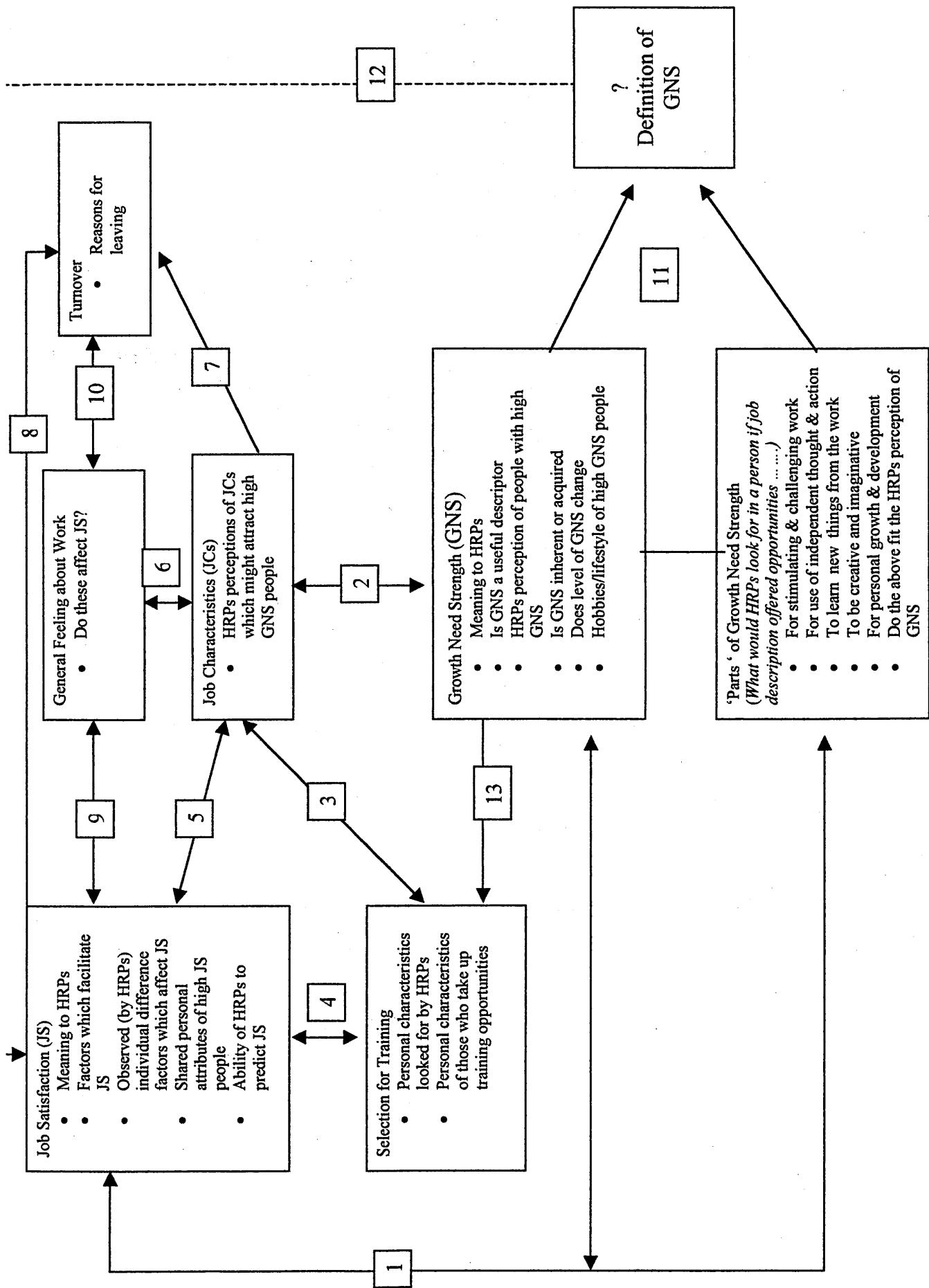
Study 1a – Correlations Among Questionnaire Scale Scores

	24.	25.	26.	27.	28.	29.	30.	31.	32.
Skill Variety									
Task Identity									
Task Significance									
Autonomy									
Feedback (job)									
MPS									
MPSA									
MPSM									
Job Satisfaction									
GNS 'would like'									
GNS 'job choice'									
HONS									
SANS									
Importance of JCs									
Neuroticism									
Extraversion									
Openness									
Agreeableness									
Conscientiousness									
Achievement									
Benevolence									
Conformity									
Hedonism									
Power	-								
Security	.35	-							
Self-direction	.38	.50	-						
Stimulation	.40	.31	.64	-					
Tradition	.31	.51	.34	.32	-				
Universalism	.12	.52	.60	.42	.49	-			
Affective WV	.16	.37	.35	.30	.30	.39	-		
Cognitive WV	.37	.38	.54	.47	.31	.43	.59	-	
Instrumental WV	.30	.50	.30	.28	.43	.33	.55	.43	-

n = 692

Correlations of 0.08 and above are significant at the 0.05 level (2-tailed).

Correlations of 0.11 and above are significant at the 0.01 level (2-tailed).



Conceptual Framework for Qualitative Data Analysis
N.B. HRP's = Human Resources Professionals

Semi-Structured Interview - Questions

1. What gives you personally, satisfaction with your job?
2. How does that make you feel?
3. Can you think of two people you know in similar jobs - one who seems to get a lot of job satisfaction and one who does not? Can you compare them and tell me what's different about them?
4. Do you think job satisfaction means the same to everyone?
5. Can you tell about any factors you think help people get satisfaction from their jobs?
6. Is the way people feel about work in general important to the amount of satisfaction they get?
7. Thinking of people you know who derive high satisfaction from their jobs are there any personal attributes which they share?
8. When you interview people are you able to predict which ones will get high satisfaction from their jobs?
9. What is it about them that makes you think that?
10. If you conduct exit interviews what are the main reasons people give for leaving?
11. What type of training opportunities do you offer your staff?
12. When you interview people for jobs which offer opportunities for training, what would you look for in people?
13. In your experience, are people who take up training opportunities more readily than others?
14. If so can you describe anything about such people which sets them apart from others?
15. What would you want to know about an individual if you were selecting people for training?
16. I'm interested in an individual difference factor which affects the way people respond to their job and how much satisfaction they derive from it. This factor is called 'growth need strength'. What do you think 'growth need strength' is?

(Give loose definition)
17. Is this a useful way of describing people?
18. Would you have a different name for this factor 'growth need strength'?
19. If so, what would you call it?

20. If I said a person was 'high' in 'growth need strength', what would it tell you about that person?
21. If, in a job description, the work was described as 'stimulating and challenging' what would you look for in a person in order to fill that role?
22. If, in a job description, the work offered 'chances to exercise independent thought and action', what would you look for in a person in order to fill that role?
23. If, in a job description, the work offered 'opportunities to learn new things from the work', what would you look for in a person in order to fill that role?
24. If, in a job description, the work offered 'opportunities to be creative and imaginative', what would you look for in a person in order to fill that role?
25. If, in a job description, the work offered opportunities for personal growth and development', what would you look for in a person in order to fill that role?
26. Thinking back to the questions I've just asked about job descriptions, do you think that people with the attributes you've described could be said to be high in growth need strength?
27. Do you think people acquire 'growth need strength' or do you think the job holders bring it with them?
28. Is there a difference between 'growth need strength' and 'autonomy'?
29. If so what is that difference?
30. Do you think that 'growth need strength' is part of peoples' personality?
31. Do you think that 'growth need strength' comes from the environment we live and work in?
32. Do you think peoples' 'level' of growth need strength changes depending on what they are doing?
33. What do you think affects that level?
34. What sort of job characteristics do you think a person high in growth need strength would be attracted to?
35. How do you think a person high in 'growth need strength' might behave in their life outside work ie. what sort of hobbies might they have?
36. How would you know that someone had high Growth Need Strength?
37. How long have you worked in HR?
38. Do you feel you have advanced as much as you had hoped at the start of your career?
39. Do you feel you are rewarded enough for your work?

Explanation for Participants -Pilot Study

I am designing a semi-structured questionnaire which I will be able to use in my Occupational Psychology Research. During this interview I will be reading out the explanation which I hope to give people when they take part in the main part of the research. Before I conduct the main research I need to make sure that people understand my explanation for the interview and the questions I am going to be asking.

Your part in this is to tell me whether you understand the explanation and then whether you understand the actual questions. Are you happy for me to tape our conversation?

Please tell me what you think of the questions as we go along and any reservations you may have about the research in general, drawing on your experience as an HR professional. Anything you have to say is important so don't hold back!

Explanation for Participants in Interview Study

Thank you for agreeing to take part in this research . I would like to record our conversation - do you have any objections to my doing so?

The purpose of this interview is to find out your opinions about attributes which you would look for when interviewing people for jobs. To do this, I will be asking you a number of questions about what you would look for in people, and some general questions about what you think helps individuals to derive job satisfaction from their work and about things which might affect turnover.

Do you have any questions at this point?

I will begin with some general questions. Please feel free to speak openly and give me your opinions and thoughts. The tape will only be heard by Surrey University Research staff and will be held in confidence.

Demographic Details of Participants for Study 2

Sue was a human resources manager for a large retail organisation and had 16 years experience in the area. She had begun her career in teaching but found little satisfaction in that role. Her job in HR was concerned mostly with recruitment to all areas of the organisation, so her experience of interviewing people for jobs was particularly useful to the study. Sue said she was very happy in her job and did not intend to pursue any new career moves.

John was head of human resources for a software design and development company. He had only been in that role for less than a year and had worked in HR for only 3 years. His early career had been in the insurance industry but following redundancy and having an interest in HR, he used his redundancy money to fund a degree in human resources management. He said that he loved his job but felt somewhat constrained as he had lots of new ideas which he was unable to put into action at that time. John said he intended to stay in the job because he believed his skills and understanding of HR could be put to good use.

Nigel was a human resources manager for a nationwide retail organisation who was full of enthusiasm for his job and was determined to seek career progression within the organisation. He had begun his career working in one of the company's stores and had worked his way up, taking extra training in HR to help him in his role. Nigel had worked in HR for 5 years and his job included recruitment and training. He said did not intend to move to another company because he was committed to the job he was in and had prospects of promotion in the future.

Bob had spent the first twenty years of his career working as an HR manager for what he described as a 'high-powered' organisation. He had left that job to start his own HR consultancy company which he had been involved in for the past 10 years and said it was the best thing he had ever done. Bob said he had never enjoyed working for other people and loved the autonomy that running his own company brought him. He had a significant and wide-ranging understanding of HR and was currently involved with executive recruitment, which he said he would be happy to do until he retired.

Trisha was a human resources manager for a retail organisation and had work in HR for 15 years. She was not particularly happy in her job as she said she felt that it did not give her the opportunity to develop her career. Trisha said that having put aside her career to raise a family, she would now like to concentrate on herself and her job. She had extensive experience of recruitment and the development of training programmes with the organisation.

Gemma had worked in HR for 8 years. She worked for a financial services organisation and had held several positions in the company before joining the HR team. She had originally been involved with recruitment but had recently moved to a more administrative role. She said that she was satisfied with her job and had no intentions of moving on.

Ann was a consultant in HR management who had been running her own business for 6 years. Her early career in HR had been with an insurance company and she had moved into consultancy work in order to spend more time with her family. Ann had extensive experience of designing and running training and development schemes and was currently involved with an ongoing training project for a large financial services organisation. She said she was very happy in her role but sometimes missed working as part of a team.

Brenda was a human resources assistant for a financial services organisation and had worked in HR for 3 years in two different companies. She said she was very interested in all areas of HR and would like to undergo further training in the area, but could see no prospect of that in her current role. Ann's experience had been mainly in recruitment and at the time of the interview she was looking for another job, as she wanted greater opportunity to extend her career.

Initial Coding

1. Job satisfaction
2. Personal satisfaction (of HRPs)
3. High job satisfaction people
4. Low job satisfaction people
5. Predicting job satisfaction
6. Feelings about work on general (effect on job satisfaction)
7. People who take up training opportunities
8. Turnover
9. Job characteristics sought by high GNS people
10. Usefulness of GNS
11. Alternatives to GNS
12. High GNS people
13. Low GNS people
14. Personality and GNS
15. Environment and GNS
16. Autonomy and GNS
17. Hobbies expected to be taken up by high GNS people
18. What is GNS?
19. Relationships - Job satisfaction and GNS
20. Job characteristics and GNS
21. High and low GNS people
22. Job satisfaction and taking up training opportunities
23. Job satisfaction and turnover
24. Job characteristics and turnover
25. Job satisfaction and feelings about work in general

Theming the Analysis for the Interview Data

Ability to succeed
Achievement: capacity for
Achievement: desire for
Achievement: desire for learning
Achievement: desire for personal
Achievement: determination to achieve
Achievement: need for
Achievement: of results
Achievement: personal
Achievement: sense of purpose
Activeness
Adaptability
Altruism
Ambitious
Assertiveness: lack of
Attention to detail
Authority
Autonomy
Balanced view of life
Capacity for learning
Challenging: the status quo
Challenge: desire for
Challenge: lack of
Change: ability to cope with
Change: career
Change: inability to cope with changing nature of workplace
Change: making a difference to the business
Change: personal
Common sense
Communication: lack of
Competitiveness
Creativity
Curiosity
Decision-making
Depressing
Desire to succeed
Determination to succeed
Development: career
Development: clarity of sense of direction
Development: confidence in personal direction
Development: continuing professional
Development: desire for
Development: personal

Energetic
Enjoyment
Enthusiasm
Expectations of the organisation

Extraversion
Feeling valued
Flexibility
Forgetful
Forward-thinking
Honesty: about motivation for learning
Insularity
Intelligence
Interest in people
Interpersonal relationships: problems with manager/other colleagues
Interpersonal relationships: teamwork
Interpersonal relationships: wanting more interaction with people
Introversion
Job characteristics: autonomy
Job characteristics: variety
Job satisfaction: low
Knowledge of opportunities
Malleability
Materialistic
Motivation: from others or the work
Obstructing organisational development
Openness: in communication
Openness: to challenge
Openness: to change
Openness: to ideas
Openness: to new experiences
Openness: to others
Optimistic
Organisational commitment and loyalty
Organisational politics: avoidance of
Organisational support
Personal Commitment
Personal Image: importance of
Persuasive
Questioning
Receptivity to learning
Relaxed
Responsibility: willingness to accept
Restlessness
Rewards: fairness of
Rewards: financial
Rewards: other benefits (e.g. pension)
Rewards: respect from others
Seeking new experiences
Seen by organisation as worth investing in
Self-belief
Self-conviction
Self-direction
Self-motivation
Self-satisfaction
Setting high standards for self and others
Sharing knowledge

Skills: aptitude for learning
Skills: confidence
Skills: competence
Skills: experience of the job
Skills: feelings of inadequacy in knowledge
Skills: interpersonal
Skills: lack of confidence in personal knowledge
Skills: lack of confidence in personal relationships
Skills: lack of correct skills for the job
Skills: lack of knowledge about the business
Skills: lack of understanding of the job
Skills: full utilisation of
Stimulating
Stimulation: from colleagues
Stimulation: mental
Support from managers
Training: lack of
Trusting
Vision
Willingness to learn
Working conditions: physical
Working conditions: work processes

Theming the Analysis for the Interview Data

1. Ability to succeed
2. Achievement: capacity for
3. Achievement: desire for
4. Achievement: desire for learning
5. Achievement: desire for personal
6. Achievement: determination to achieve
7. Achievement: need for
8. Achievement: of results
9. Achievement: personal
10. Achievement: sense of purpose
11. Activeness
12. Adaptability
13. Aggressive determination
14. Altruism
15. Ambition
16. Assertiveness: lack of
17. Attention to detail
18. Authority
19. Autonomy
20. Autonomy: restricted opportunity for use of experience
21. Balanced view of life
22. Challenging the status quo
23. Challenge: desire for
24. Challenge: lack of
25. Change: ability to cope with
26. Change: career
27. Change: desire for
28. Change: inability to cope with changing nature of the workplace
29. Change: making a difference to the business
30. Change: personal
31. Common sense
32. Communication: lack of
33. Competitiveness
34. Creativity
35. Curiosity
36. Decision-making
37. Decision-making: responsibility for
38. Depressing
39. Desire to succeed
40. Determination to succeed
41. Development: career
42. Development: clarity of sense of direction
43. Development: confidence in personal direction
44. Development: continuing professional
45. Development: desire for
46. Development: personal
47. Energetic

48. Enjoyment
49. Enthusiasm
50. Expectations of the organisation
51. Extraversion
52. Feeling valued
53. Flexibility
54. Forgetful
55. Forward thinking
56. Honesty: about motivation for learning
57. Insularity
58. Intelligence
59. Interest in people
60. Interpersonal relationships: problems with manager/other colleagues
61. Interpersonal relationships: teamwork
62. Interpersonal relationships: wanting more interaction with people
63. Introversion
64. Job characteristics: autonomy
65. Job characteristics: variety
66. Job satisfaction: low
67. Learning: capacity for
68. Learning: desire for
69. Learning: pleasure in
70. Learning: receptivity to
71. Learning: willingness to learn
72. Knowledge of opportunities
73. Malleability
74. Materialism
75. Motivation: from others or the work
76. Obstructing organisational development
77. Openness: in communication
78. Openness to challenge
79. Openness: to change
80. Openness to ideas
81. Openness: to new experiences
82. Openness: to others
83. Optimistic
84. Organisational commitment and loyalty
85. Organisational politics: dislike of
86. Organisational support
87. Personal commitment
88. Personal image: importance of
89. Persuasive
90. Positive attitude
91. Questioning
92. Relaxed
93. Responsibility: willingness to accept
94. Restlessness
95. Rewards: fairness of
96. Rewards: financial
97. Rewards: other benefits (e.g. pension)
98. Rewards: respect from others
99. Rewards: status

100. Seeking new experiences
101. Seen by organisation as worth investing in
102. Self-belief
103. Self-conviction
104. Self-direction
105. Self-motivation
106. Self-satisfaction
107. Setting high standards for self and others
108. Sharing knowledge
109. Skills: aptitude for learning
110. Skills: confidence
111. Skills: competence
112. Skills: experience of the job
113. Skills: feelings of inadequacy in knowledge
114. Skills: full utilisation of
115. Skills: interpersonal
116. Skills: lack of confidence in personal knowledge
117. Skills: lack of confidence in interpersonal relationships
118. Skills: lack of correct skills for the job
119. Skills: lack of knowledge about the business
120. Skills: lack of understanding of the job
121. Skills: opportunity to develop
122. Stimulating
123. Stimulation from colleagues
124. Stimulation: mental
125. Success: history of
126. Support from managers
127. Training: lack of
128. Trusting
129. Unquestioning trust
130. Vision
131. Working condition: physical
132. Working condition: work processes

Final Themes

1. Ability to cope with change
2. Ambition
3. Balanced view of life
4. Clarity of sense of direction (includes sense of purpose)
5. Competence
6. Competitiveness
7. Confidence
8. Creativity
9. Curiosity (includes questioning)
10. Desire for achievement
11. Desire/need for learning
12. Desire/need for personal development
13. Determination to succeed (includes success in work)
14. Drive
15. Energetic
16. Enjoyment
17. Experience and knowledge of the job
18. Forward thinking
19. Intelligence
20. Interpersonal skills (includes teamwork and wanting to interact with others)
21. Making a difference to the business
22. Malleability
23. Materialistic
24. Openness
25. Organisational commitment
26. Personal commitment
27. Personal image: importance of
28. Positive attitude
29. Receptivity to learning
30. Rewards obtained from work (includes internal and external rewards)
31. Seeking new experiences
32. Self-belief
33. Self-motivation
34. Stimulation from colleagues and the work
35. Vision

Themes – High GNS People

Achievement	Change & Development	Learning	Personal Knowledge & Skills	Commitment & Loyalty	Activity
Desire for	Clarity of sense of direction	Desire/need for	Confidence	Personal commitment	Level of Energy
Determination to succeed	Desire for personal	Receptivity to	Competence	Organisational commitment	Competitiveness
Ambition	Ability to cope with change	Curiosity	Experience and knowledge of the job		
	Making a difference to the business	Seeking new experiences	Interpersonal		
	Malleability		Intelligence		
Motivation	Attitude Towards Life				
Self-belief	Balanced view of life				
Self-motivation	Materialistic				
Stimulation from colleagues and the work	Importance of personal image				
	Positive attitude				
	Forward thinking				
	Vision				
	Openness				
	Enjoyment				

This questionnaire was developed as part of a study of jobs and how people react to them.

On the following pages you will find several kinds of questions about your job and about yourself.

Specific instructions are given at the start of each section. Please read them carefully.

It should take you no more than 20 minutes to complete the questionnaire. You will find that some of the questions are similar, this is intentional.

The questions are designed to obtain **your** perceptions of jobs you would prefer and **your** perceptions of yourself.

There are no trick questions. Your individual answers will be kept completely confidential.

Please answer each item as honestly and frankly as possible. **Please do not leave any questions unanswered.**

Thank you for your co-operation.

Important

Before you complete this questionnaire it is important that you read and sign the following statement.

I agree to take part in this study and understand that the details contained in this questionnaire will remain confidential to the researchers. I understand that the results of this study will be used for **research purposes only** and will not adversely affect my employment in any way. I understand that I may be asked for supplementary information at a later date and to do so the researchers will need to pass my name to HR so that I may be contacted – no other information will be divulged to my employer. I give my permission for the researcher to contact me directly, should this be necessary.

Signed: _____ Date: _____

Please print your name here: _____ Department _____

SECTION 1

People differ in the kinds of jobs they would most like to hold. The questions in this section give you a chance to say what it is about your job that is most important to *you*. For each section, two different kinds of jobs are briefly described. You are to indicate which of the jobs you personally would prefer – if you had to make a choice between them.

In answering each question, assume that everything else about the job is the same. Pay attention only to the two characteristics actually listed.

Two examples are given below.

Job A A job requiring work with mechanical equipment most of the day	Job B A job requiring work with other people most of the day			
1-----2-----3-----4-----5				
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you like working with people and working with equipment equally well, you would circle number 3, as has been done in the example.

* * * * *

Here is another example. This one asks for a harder choice – between two jobs which both have some undesirable features.

Job A A job requiring you to expose yourself to considerable physical danger	Job B A job located 200 miles from your home and family			
1-----2-----3-----4-----5				
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you would slightly prefer risking physical danger to working far from home, you would circle number 2, as has been done in the example.

JOB A**JOB B**

1. A job where the pay is very good.

A job where there is considerable opportunity to be creative & imaginative.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

2. A job where you are often required to make important decisions.

A job with many pleasant people to work with.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

3. A job in which greater responsibility is given to those who do the best work.

A job in which greater responsibility is given to loyal employees who have the most seniority.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

4. A job in an organisation which is in financial trouble - and might have to close down within the year.

A job in which you are not allowed to have any say whatever in how your work is scheduled, or in the procedures to be used in carrying it out.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

5. A very routine job.

A job where co-workers are not very friendly.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

6. A job with a supervisor who is often very critical of you and your work in front of other people.

A job which prevents you from using a number of skills that you have worked hard to develop.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

7. A job with a supervisor who respects you and treats you fairly.

A job which provides constant opportunities for you to learn new and interesting things.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

8. A job where there is a real chance you could be laid off.

A job with very little chance to do challenging work.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

9. A job in which there is a real chance for you to develop new skills and advance in the organisation.

A job which provides lots of vacation time and an excellent fringe benefit package.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

10. A job with little freedom and independence to do your work in the way you think best.

A job where the working conditions are poor.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

11. A job with very satisfying teamwork.

A job which allows you to use your skills and abilities to the fullest extent.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

12. A job which offers little or no challenge.

A job which requires you to be completely isolated from co-workers.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

13. A job where the physical working conditions are poor.

A job which requires you to make regular changes to the way you work.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

14. A job where you can put an equal amount of energy into your work and your private life.

A job where you have the chance of quick promotion.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

15. A job where your manager appreciates your work.

A job which provides you with the resources to have lots of personal possessions.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

SECTION 2

Here are a number of characteristics that may or may not apply to you. For example, do you agree that you are someone who likes to spend time with others? Please write a number next to each statement to indicate the extent to which you agree or disagree with that statement.

Disagree Strongly	Disagree a little	Neither agree nor disagree	Agree a little	Agree strongly
1	2	3	4	5

I see Myself as Someone Who.....

- | | |
|--|--|
| ___ 1. Is talkative | ___ 23. Tends to be lazy |
| ___ 2. Tends to find fault with other | ___ 24. Is emotionally stable, not easily upset |
| ___ 3. Does a thorough job | ___ 25. Is inventive |
| ___ 4. Is depressed, blue | ___ 26. Has an assertive personality |
| ___ 5. Is original, comes up with new ideas | ___ 27. Can be cold and aloof |
| ___ 6. Is reserved | ___ 28. Perseveres until the task is finished |
| ___ 7. Is helpful and unselfish with others | ___ 29. Can be moody |
| ___ 8. Can be somewhat careless | ___ 30. Values artistic, aesthetic experiences |
| ___ 9. Is relaxed, handles stress well | ___ 31. Is sometimes shy, inhibited |
| ___ 10. Is curious about many different things | ___ 32. Is considerate and kind to almost everyone |
| ___ 11. Is full of energy | ___ 33. Does things efficiently |
| ___ 12. Starts quarrels with others | ___ 34. Remains calm in tense situations |
| ___ 13. Is a reliable worker | ___ 35. Prefers work that is routine |
| ___ 14. Can be tense | ___ 36. Is outgoing, sociable |
| ___ 15. Is ingenious, a deep thinker | ___ 37. Is sometimes rude to others |
| ___ 16. Generates a lot of enthusiasm | ___ 38. Makes plans and follows through with them |
| ___ 17. Has a forgiving nature | ___ 39. Gets nervous easily |
| ___ 18. Tends to be disorganised | ___ 40. Likes to reflect, play with ideas |
| ___ 19. Worries a lot | ___ 41. Has few artistic interests |
| ___ 20. Has an active imagination | ___ 42. Likes to co-operate with others |
| ___ 21. Tends to be quiet | ___ 43. Is easily distracted |
| ___ 22. Is generally trusting | ___ 44. Is sophisticated in art, music or literature |

Please check: Did you write a number in front of each statement?

SECTION 3

1. What is your age group? (tick one)

☐ under 20 ☐ 40 - 49
☐ 20 - 29 ☐ 50 - 59
☐ 30 - 39 ☐ 60 or over

2. How long have you worked for your current employer? (tick one)

☐ up to 1 year ☐ 11-15 years
☐ 1 - 5 years ☐ 16 - 20 years
☐ 6 - 10 years ☐ over 20 years

3. What is your educational background? (tick one)

☐ GCSE level
☐ A or A/S level
☐ Business/Technical College
☐ College/University degree
☐ Masters or higher degree
☐ Distance learning course (e.g. Open University)

4. Are you ☐ Male ☐ Female (please tick)

5. Did you enjoy your experience of education? **Yes No (circle)**

6. Do you have a clear idea of the direction you would like your career to go in? **Yes No (circle)**

7. Have you taken training courses concerned with your work, in the past? **Yes No (circle)**

Please describe this training in the space below:

8. Do you have hobbies or interests outside work? **Yes No (circle)**
If so, what are they?
9. Do you enjoy team sports? **Yes No (circle)**
10. If you have worked for other companies did they have personal development programmes? **Yes No (circle)**
If so, what did they involve?
11. Are you currently undergoing any training concerned with your work? **Yes No (circle)**
Please describe this training in the space below:
12. Do you intend to undertake training in the future? **Yes No (circle)**
13. If you were offered the opportunity for further training in your work what would you like to do? (please describe in the space below)

14. Do you enjoy competitive sports and activities? **Yes No (circle)**

15. Do you think your current employer offers enough opportunities for promotion? **Yes No (circle)**

16. Where do you see yourself in 3 years time, within your current organisation?
(please describe in the space below)

17. Have you ever taken a course of study outside working hours? **Yes No (circle)**

18. Was this mainly

a. to further your career **Yes No (circle)**

b. for your own interest or enjoyment **Yes No (circle)**

Now please make sure that you have answered every question and thank you very much for your time.

This questionnaire was developed as part of a study of jobs and how people react to them.

On the following pages you will find several kinds of questions about your job and about yourself, some of which you may recognise from the first questionnaire you completed. We would like you to answer them again, please.

Specific instructions are given at the start of each section. Please read them carefully.

It should take no more than 30 minutes to complete the entire questionnaire. Please move through it quickly. You will find that some of the questions are very similar, this is intentional.

The questions are designed to obtain **your** perceptions of your job, **your** reactions to it and **your** perceptions of yourself.

There are no trick questions. Your individual answers will be kept completely confidential.

Please answer each item as honestly and frankly as possible. **Please do not leave any questions unanswered.**

Thank you again for your co-operation.

Important

Before you complete this questionnaire it is important that you read and sign the following statement.

I agree to take part in this study and understand that the details contained in this questionnaire will remain confidential to the researchers. I understand that the results of this study will be used for **research purpose only** and will not adversely affect my employment in any way. I give my permission for the researcher to contact me directly, should this be necessary.

Signed:_____ Date:_____

Please print your name here:_____ Department:_____

SECTION 1

Listed below are a number of characteristics which could be present in any job. People differ about how much they would like to have each one present in their own job. We are interested in learning *how much you personally would like* to have each one present in your job.

Using the scale below, please indicate the *degree* to which you *would like* to have each characteristic present in your job.

1	2	3	4	5	6	7
Would like having this only a moderate amount (or less)			Would like having this very much			Would like having this a very great deal

- ☐ 1. High respect and fair treatment from my supervisor.
- ☐ 2. Stimulating and challenging work.
- ☐ 3. Chances to exercise independent thought and action in my job.
- ☐ 4. Great job security.
- ☐ 5. Very friendly co-workers.
- ☐ 6. Opportunities to learn new things from my work.
- ☐ 7. High salary and good fringe benefits.
- ☐ 8. Opportunities to be creative and imaginative in my work.
- ☐ 9. Quick promotions.
- ☐ 10. Opportunities for personal growth and development in my job.
- ☐ 11. A sense of worthwhile accomplishment in my work.

SECTION 2

People differ in the kinds of jobs they would most like to hold. The questions in this section give you a chance to say just what it is about a job that is most important to *you*.

For each question, two different kinds of jobs are briefly described. You are to indicate which of the jobs you personally would prefer - if you had to make a choice between them.

In answering each question, assume that everything else about the job is the same. Pay attention only to the two characteristics actually listed.

Two examples are given below.

JOB A A job requiring work with mechanical equipment most of the day			JOB B A job requiring work with other people most of the day	
1-----2-----3-----4-----5				
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you like working with people and working with equipment equally well, you would circle number 3, as has been done in the example.

* * * * *

Here is another example. This one asks for a harder choice - between two jobs which both have some undesirable features.

JOB A A job requiring you to expose yourself to considerable physical danger			JOB B A job located 200 miles from your home and family	
1-----2-----3-----4-----5				
Strongly Prefer A	Slightly Prefer A	Neutral	Slightly Prefer B	Strongly Prefer B

If you would slightly prefer risking physical danger to working far from your home, you would circle number 2, as has been done in the example.

JOB A**JOB B**

1. A job where the pay is very good

A job where there is Considerable opportunity

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

2. A job where you are often pleasant required to make important with. decisions.

A job with many people to work

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

3. A job in which greater responsibility is given to those who do the best work.

A job in which greater responsibility is given to loyal employees who have the most seniority.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

4. A job in an organisation which is in financial trouble - and might have to close down within the year.

A job in which you are not allowed to have any say whatever in how your work is scheduled, or in the procedures to be used in carrying it out.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

JOB B

5. A very routine job.

A job where co-workers are not very friendly.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

JOB B

6. A job with a supervisor who is often very critical of you and your work in front of other people.

A job which prevents you from using a number of skills that you worked hard to develop.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

JOB B

7. A job with a supervisor who respects you and treats you fairly.

A job which provides constant opportunities for you to learn new and interesting things.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A

JOB B

8. A job where there is a real chance you could be laid off.

A job with very little chance to do challenging work.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

9. A job in which there is a real chance for you to develop new skills and advance in the organisation.

A job which provides lots of vacation time and an excellent fringe benefit package.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

10. A job with little freedom and independence to do your work in the way you think best.

A job where the working conditions are poor.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

11. A job with very satisfying teamwork.

A job which allows you to use your skills and abilities to the fullest extent.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

JOB A**JOB B**

12. A job which offers little or no challenge.

A job which requires you to be completely isolated from co-workers.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

13. A job where the physical
working conditions are poor.

A job which requires you
To make regular changes to the
way you work.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

14. A job where you can put an
equal amount of energy into
your work and your private life

A job where you have the
chance of quick promotion.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

15. A job where your manager
appreciates your work.

A job which provides you
With the resources to have lots
of personal possessions.

1-----2-----3-----4-----5

Strongly
Prefer A

Slightly
Prefer A

Neutral

Slightly
Prefer B

Strongly
Prefer B

SECTION 3

Here are a number of characteristics that may or may not apply to you. For example, do you agree that you are someone who likes to spend time with others? Please write a number next to each statement to indicate the extent to which you agree or disagree with that statement.

Disagree Strongly	Disagree A little	Neither agree Nor disagree	Agree A little	Agree Strongly
1	2	3	4	5

I see Myself as Someone Who

- | | |
|--|--|
| ___ 1. Is talkative | ___ 23. Tends to be lazy |
| ___ 2. Tends to find fault with others | ___ 24. Is emotionally stable, not easily upset |
| ___ 3. Does a thorough job | ___ 25. Is inventive |
| ___ 4. Is depressed, blue | ___ 26. Has an assertive personality |
| ___ 5. Is original, comes up with new ideas | ___ 27. Can be cold and aloof |
| ___ 6. Is reserved | ___ 28. Perseveres until the task is finished |
| ___ 7. Is helpful and unselfish with others | ___ 29. Can be moody |
| ___ 8. Can be somewhat careless | ___ 30. Values artistic, aesthetic experiences |
| ___ 9. Is relaxed, handles stress well | ___ 31. Is sometimes shy, inhibited |
| ___ 10. Is curious about many different things | ___ 32. Is considerate and kind to almost everyone |
| ___ 11. Is full of energy | ___ 33. Does things efficiently |
| ___ 12. Starts quarrels with others | ___ 34. Remains calm in tense situations |
| ___ 13. Is a reliable workers | ___ 35. Prefers work that is routine |
| ___ 14. Can be tense | ___ 36. Is outgoing, sociable |
| ___ 15. Is ingenious, a deep thinker | ___ 37. Is sometimes rude to others |
| ___ 16. Generates a lot of enthusiasm | ___ 38. Makes plans and follows through with them |
| ___ 17. Has a forgiving nature | ___ 39. Get nervous easily |
| ___ 18. Tends to be disorganised | ___ 40. Likes to reflect, play with ideas |
| ___ 19. Worries a lot | ___ 41. Has few artistic interests |
| ___ 20. Has an active imagination | ___ 42. Likes to co-operate with others |
| ___ 21. Tends to be quiet | ___ 43. Is easily distracted |
| ___ 22. Is generally trusting | ___ 44. Is sophisticated in art, music or literature |

Please check: Did you write a number in front of each statement?

SECTION 4

In this section you are asked to rate how important each value is for you as a guiding principle in YOUR life, using the scale provided.
The rating scale we want you to use runs:

-1	0	1	2	3	4	5	6	7
----	---	---	---	---	---	---	---	---

Use '-1' if any values are opposed to the principles that guide you
Use '7' for any value of supreme importance as a guiding principle in your life

Most values will fall in the range '0' to '6'
where 0 means the value is **not at all important** to you
and 6 means it is **very important** to you

Value List

Circle a number under each value, according to the scale above.

1. **EQUALITY** (equal opportunity for all)

-1—0—1—2—3—4—5—6—7

2. **INNER HARMONY** (at peace with myself)

-1—0—1—2—3—4—5—6—7

3. **SOCIAL POWER** (control over others, dominance)

-1—0—1—2—3—4—5—6—7

4. **PLEASURE** (gratification of desires)

-1—0—1—2—3—4—5—6—7

5. **FREEDOM** (freedom of action and thought)

-1—0—1—2—3—4—5—6—7

6. **A SPIRITUAL LIFE** (emphasis on spiritual not material matters)

-1—0—1—2—3—4—5—6—7

7. **SOCIAL ORDER** (stability of society)

-1—0—1—2—3—4—5—6—7

8. **AN EXCITING LIFE** (stimulating experiences)

-1—0—1—2—3—4—5—6—7

9. **MEANING IN LIFE** (a purpose in life)

-1—0—1—2—3—4—5—6—7

10. **POLITENESS** (courtesy, good manners)

-1—0—1—2—3—4—5—6—7

11. **WEALTH** (material possessions, money)

-1—0—1—2—3—4—5—6—7

12. **NATIONAL SECURITY** (protection of my nation from enemies)

-1—0—1—2—3—4—5—6—7

13. **RECIPROCATION OF FAVOURS** (avoidance of indebtedness)

-1—0—1—2—3—4—5—6—7

14. **CREATIVITY** (uniqueness, imagination)

-1—0—1—2—3—4—5—6—7

15. **A WORLD AT PEACE** (free of war and conflict)

-1—0—1—2—3—4—5—6—7

16. **RESPECT FOR TRADITION** (preservation of time-honoured customs)

-1—0—1—2—3—4—5—6—7

17. **SELF-DISCIPLINE** (self-restraint, resistance to temptation)

-1—0—1—2—3—4—5—6—7

18. **DETACHMENT** (from worldly concerns)

-1—0—1—2—3—4—5—6—7

19. **FAMILY SECURITY** (safety for loved ones)

-1—0—1—2—3—4—5—6—7

20. **UNITY WITH NATURE** (fitting into nature)

-1—0—1—2—3—4—5—6—7

21. **A VARIED LIFE** (filled with challenge, novelty and change)

-1—0—1—2—3—4—5—6—7

22. **WISDOM** (a mature understanding of life)

-1—0—1—2—3—4—5—6—7

23. **AUTHORITY** (the right to lead or command)

-1—0—1—2—3—4—5—6—7

24. **A WORLD OF BEAUTY** (beauty of nature and the arts)

-1—0—1—2—3—4—5—6—7

25. **SOCIAL JUSTICE** (correcting injustice, care for the weak)

-1—0—1—2—3—4—5—6—7

26. **INDEPENDENT** (self-reliant, self-sufficient)

-1—0—1—2—3—4—5—6—7

27. **MODERATE** (avoiding extremes of feeling and action)

-1—0—1—2—3—4—5—6—7

28. **LOYAL** (faithful to my friends, group)

-1—0—1—2—3—4—5—6—7

29. **AMBITIOUS** (hardworking, aspiring)

-1—0—1—2—3—4—5—6—7

30. **BROAD MINDED** (tolerant of different ideas and beliefs)

-1—0—1—2—3—4—5—6—7

31. **HUMBLE** (modest, self-effacing)

-1—0—1—2—3—4—5—6—7

32. **DARING** (seeking adventure, risk)

-1—0—1—2—3—4—5—6—7

33. **PROTECTING THE ENVIRONMENT** (preserving nature)

-1—0—1—2—3—4—5—6—7

34. **INFLUENTIAL** (having an impact on people and events)

-1—0—1—2—3—4—5—6—7

35. **HONOURING OF PARENTS AND ELDERS** (showing respect)

-1—0—1—2—3—4—5—6—7

36. **CHOOSING OWN GOALS** (selecting own purposes)

-1—0—1—2—3—4—5—6—7

37. **CAPABLE** (competent, effective, efficient)

-1—0—1—2—3—4—5—6—7

38. **ACCEPTING MY PORTION IN LIFE** (submitting to life's circumstances)

-1—0—1—2—3—4—5—6—7

39. **HONEST** (genuine, sincere)

-1—0—1—2—3—4—5—6—7

40. **OBEDIENT** (dutiful, meeting obligations)

-1—0—1—2—3—4—5—6—7

41. **HELPFUL** (working for the welfare of others)

-1—0—1—2—3—4—5—6—7

42. **ENJOYING LIFE** (enjoying food, sex, leisure, etc.)

-1—0—1—2—3—4—5—6—7

43. **DEVOUT** (holding to religious faith and belief)

-1—0—1—2—3—4—5—6—7

44. **RESPONSIBLE** (dependable, reliable)

-1—0—1—2—3—4—5—6—7

45. **CURIOUS** (interested in everything, exploring)

-1—0—1—2—3—4—5—6—7

46. **FORGIVING** (willing to pardon others)

-1—0—1—2—3—4—5—6—7

47. **SUCCESSFUL** (achieving goals)

-1—0—1—2—3—4—5—6—7

48. **CLEAN** (neat, tidy)

-1—0—1—2—3—4—5—6—7

SECTION 5

In this section you are asked to rate how important each **work value** is to you, using the scale provided. As in the previous section try to distinguish as much as possible between the values by using all of the numbers. Since only a limited number of the values can be “extremely important to you” do not over-use this highest category.

The rating scale we want you to use runs:

1	2	3	4	5
---	---	---	---	---

Use ‘1’ if the work value is not at all important to you

Use ‘3’ if the work value is moderately important to you

Use ‘5’ if the work value is extremely important to you

Work Values list

Circle a number under each work aspect, according to the scale above

- | | |
|--|--|
| <p>1. Achievement in work</p> <p>1-----2-----3-----4-----5</p> | <p>9. Feedback concerning the results of your work</p> <p>1-----2-----3-----4-----5</p> |
| <p>2. Advancement, chances for promotion</p> <p>1-----2-----3-----4-----5</p> | <p>10. Independence in work</p> <p>1-----2-----3-----4-----5</p> |
| <p>3. Benefits, holiday, sick leave, pension, insurance etc.</p> <p>1-----2-----3-----4-----5</p> | <p>11. Influence in the organisation</p> <p>1-----2-----3-----4-----5</p> |
| <p>4. Company, to be employed by a company for which you are proud to work</p> <p>1-----2-----3-----4-----5</p> | <p>12. Influence in work</p> <p>1-----2-----3-----4-----5</p> |
| <p>5. Contributions to society</p> <p>1-----2-----3-----4-----5</p> | <p>13. Job interest, to do work which is interesting to you</p> <p>1-----2-----3-----4-----5</p> |
| <p>6. Convenient hours of work</p> <p>1-----2-----3-----4-----5</p> | <p>14. Job security, permanent job</p> <p>1-----2-----3-----4-----5</p> |
| <p>7. Co-workers, fellow workers who are pleasant and agreeable</p> <p>1-----2-----3-----4-----5</p> | <p>15. Job status</p> <p>1-----2-----3-----4-----5</p> |
| <p>8. Esteem, that you are valued as a person</p> <p>1-----2-----3-----4-----5</p> | <p>16. Meaningful work</p> <p>1-----2-----3-----4-----5</p> |
| | <p>17. Opportunity for personal growth</p> <p>1-----2-----3-----4-----5</p> |

- | | |
|--|--|
| <p>18. Opportunity to meet people and interact with them</p> <p>1-----2-----3-----4-----5</p> | <p>21. Responsibility</p> <p>1-----2-----3-----4-----5</p> |
| <p>19. Pay, the amount of money you receive</p> <p>1-----2-----3-----4-----5</p> | <p>22. Supervisor, a fair and considerate boss</p> <p>1-----2-----3-----4-----5</p> |
| <p>20. Recognition for doing a good job</p> <p>1-----2-----3-----4-----5</p> | <p>23. Use of ability and knowledge in your work</p> <p>1-----2-----3-----4-----5</p> |
| | <p>24. Work conditions, comfortable and clean</p> <p>1-----2-----3-----4-----5</p> |

SECTION 6

The questions in this section ask you about the training you have undertaken since April 2002 when you entered the Training and Development Programme.

1. How many training courses have you attended since April 2002?

Please circle the number appropriate to you

0 1 2 3 4 5 6 or more

2. How many training courses have you signed up for which you intend to undertake in the future?

Please circle the number appropriate to you

0 1 2 3 4 5 6 or more

Now please answer the questions overleaf

SECTION 7

There may be some factors which have prevented you from attending the training courses you would have liked, so the following questions ask you what factors have prevented you from attending training and development courses.

Using the scale below, please indicate the *degree* to which you believe each of the following individual factors prevented you from attending the courses.

	1	2	3	4	5
	Not at all		Moderately		Very Significantly
☐	1. No courses offered/available.				
☐	2. Courses fully booked.				
☐	3. No time available/time restrictions.				
☐	4. Lack of funds.				
☐	5. Family commitments.				
☐	6. Travelling distance/transport problems/location of training.				
☐	7. Lack of approval from supervisor.				
☐	8. Others wanting to go.				
☐	9. Work pressures/workloads/deadlines.				
☐	10. Inadequacy of courses offered.				
☐	11. Cost of courses/too expensive.				
☐	12. Scheduling of work and training.				
☐	13. Lack of employer/my organisation's support.				
☐	14. Lack of information on available courses.				
☐	15. Not interested or motivated.				
☐	16. Little difference to work prospects.				

Finally, please go back and make sure you have answered every question. Thank you very much for taking part in this research, your help has been invaluable.

Kind Regards,
Jane.

Study 3 - Correlations Among Questionnaire Scales Scores

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
GNS 'job choice' 12 Time 1	-																
GNS 'job choice' 12 Time 2	.64	-															
GNS 'job choice' 13 Time 1	.97	.62	-														
GNS 'job choice' 13 Time 2	.65	.98	.66	-													
GNS 'would like'	.28	.46	.29	-.44	-												
Training Intended (org. data)*	-.04	-.33	-.05	-.32	.00	-											
Training Taken (org. data)*	-.31	-.20	-.29	-.22	-.23	.21	-										
Training Intended (resp. data)	-.26	-.26	-.24	-.26	-.12	.39	.41	-									
Training Taken (resp. data)	.03	-.23	.05	-.23	.01	.47	.26	.26	-								
Neuroticism Time 1	-.32	-.12	-.30	-.16	.22	.19	.22	-.02	-.13	-							
Neuroticism Time 2	-.26	-.07	-.28	-.11	.12	.17	.20	.12	-.16	.82	-						
Extraversion Time 1	-.17	-.22	-.19	-.24	.01	.15	.03	.12	.15	.21	.13	-					
Extraversion Time 2	-.13	-.04	-.14	-.06	.14	-.01	.02	.08	.12	.21	.11	.84	-				
Openness Time 1	.23	.23	.23	.24	.47	.09	.06	-.07	.16	.04	-.03	.04	.03	-			
Openness Time 2	.24	.28	.24	.29	.47	.11	.03	-.08	.19	.14	.11	.06	.08	.90	-		
Agreeableness Time 1	-.23	-.10	-.26	-.12	.18	-.02	-.14	.06	.02	-.23	-.17	.05	-.07	-.05	-.15	-	
Agreeableness Time 2	-.13	-.06	-.17	-.07	.16	-.07	-.03	.14	.11	-.25	-.28	.07	.08	-.03	-.16	.80	-
Conscientiousness Time 1	-.01	.05	-.01	.06	.10	-.34	-.37	-.20	-.17	-.23	-.21	-.14	-.12	-.16	-.12	.33	.30
Conscientiousness Time 1	.02	.15	.06	.16	.28	-.40	-.30	-.19	-.09	-.13	-.22	-.34	-.19	-.09	-.04	.26	.26
Achievement	.17	.22	.22	.27	.28	-.16	-.49	-.24	-.06	-.17	-.15	-.10	.01	-.07	-.03	.10	.10
Benevolence	.14	.21	.11	.21	.40	.07	-.12	-.07	.11	-.10	-.09	-.09	-.04	.22	-.15	.38	.43
Conformity	.04	.11	.02	.13	.23	-.06	-.34	.04	-.10	-.14	-.16	.01	.04	-.10	-.19	.44	.46
Hedonism	-.03	.03	.02	.03	.18	-.13	.23	-.05	.23	.10	.05	.03	.22	.07	.10	-.04	.03
Power	.12	.07	.21	.12	.16	-.08	-.19	-.24	-.09	.14	-.08	.10	.15	-.01	.04	-.08	-.09
Security	-.01	.06	-.01	.06	.24	-.13	-.37	-.12	-.07	-.06	-.18	-.16	-.11	-.10	-.11	.23	.22
Self-direction	.29	.39	.27	.39	.48	.05	-.10	-.13	.25	-.06	-.05	.01	.16	.52	.51	.03	.13
Stimulation	.12	.31	.13	.33	.30	-.12	-.05	-.19	.02	-.02	-.08	.06	.16	.32	.34	-.05	.12
Tradition	.01	.01	.01	-.01	.00	-.24	-.30	.08	-.10	-.13	-.13	-.09	-.14	-.14	-.20	.28	.26
Universalism	.08	.19	.01	.16	.45	.03	-.02	-.04	.15	.01	.03	-.04	.05	.45	.47	.16	.17
Affective WV	-.18	.02	-.22	-.04	.21	.20	.02	.06	.04	.06	.15	.16	.15	.16	.12	.41	.27
Cognitive WV	.18	.40	.22	.42	.54	-.04	-.24	-.15	.08	-.01	-.07	-.13	.08	.26	.31	.11	.07
Instrumental WV	-.38	-.42	-.33	-.44	-.06	.15	.11	.14	.21	-.02	-.17	.23	.08	-.05	-.05	.26	.18
Employer Support	.07	-.02	.10	.00	-.04	-.29	-.11	-.36	-.30	.03	-.04	-.04	.00	-.06	.01	-.01	.02
Workload	.03	.16	-.06	.17	.03	-.17	-.28	-.23	-.27	.05	.20	-.05	.04	-.07	.01	-.12	.15
Inconvenience	-.09	-.01	-.11	-.05	.13	.22	-.12	.19	.07	.12	.06	.18	.20	.07	.04	.06	.04

For variables marked * n = 51. For all remaining variables n = 64
Correlations of .25 and above are significant at the 0.05 level (2-tailed)
Correlations of .34 and above are significant at the 0.05 level (2-tailed)

Study 3 - Correlations Among Questionnaire Scales Scores

	18.	19.	20.	21.	22.	23.	24.	25.	26.	27.	28.	29.	30.	31.	32.	33.	34.
GNS 'job choice' 12 Time 1																	
GNS 'job choice' 12 Time 2																	
GNS 'job choice' 13 Time 1																	
GNS 'job choice' 13 Time 2																	
GNS 'would like'																	
Training Intended (org. data)*																	
Training Taken (org. data)*																	
Training Intended (resp. data)																	
Training Taken (resp. data)																	
Neuroticism Time 1																	
Neuroticism Time 2																	
Extraversion Time 1																	
Extraversion Time 2																	
Openness Time 1																	
Openness Time 2																	
Agreeableness Time 1																	
Agreeableness Time 2																	
Conscientiousness Time 1																	
Conscientiousness Time 2	.80	-															
Achievement	.26	.38	-														
Benevolence	.17	.18	.39	-													
Conformity	.43	.40	.53	.60	-												
Hedonism	-.05	.07	.20	.19	.05	-											
Power	.02	.14	.55	.30	.39	.31	-										
Security	.43	.51	.59	.60	.74	.12	.49	-									
Self-direction	-.10	-.03	.43	.66	.34	.40	.34	.37	-								
Stimulation	.01	-.01	.29	.39	.19	.49	.33	.27	.53	-							
Tradition	.19	.16	.32	.38	.50	-.06	.29	.45	.16	.03	-						
Universalism	-.03	.03	.07	.65	.20	.18	.06	.34	.68	.30	.18	-					
Affective WV	-.01	-.07	-.03	.27	.20	-.10	-.09	.02	.16	-.05	-.02	.28	-				
Cognitive WV	.08	.33	.54	.46	.40	.19	.42	.45	.53	.38	.11	.39	.39	-			
Instrumental WV	.10	.02	.06	.17	.21	.10	.28	.29	-.02	.11	.31	.07	.21	.08	-		
Employer Support	.15	.09	.23	-.01	.08	.03	.31	.07	.04	.13	.06	-.07	-.10	-.01	.13	-	
Workload	-.22	-.16	.13	-.17	-.08	.01	-.09	-.12	.09	-.03	-.07	.08	-.04	.01	-.24	.02	-
Inconvenience	-.08	-.13	-.09	.25	.05	.05	.00	.17	.18	.01	.19	.24	.05	-.09	.19	-.35	-.11

For variables marked * n = 51. For all remaining variables n = 64
Correlations of .25 and above are significant at the 0.05 level (2-tailed)
Correlations of .34 and above are significant at the 0.05 level (2-tailed)

Results of the Test of the Moderating Effect of Barriers to Attending Training on the Intention to Train – Training Activity Relationship

Standardised Coefficients			
	Beta	t	Sig.
Constant		14.600	.000
Training Intended	.300	1.930	.060
Employer Support	-.290	-1.565	.124
Interaction	-.001	-.007	.994
F (3, 48) = 4.784		Adjusted R ² = 0.18	

Dependent Variable: Training Undertaken (Respondents' Data)

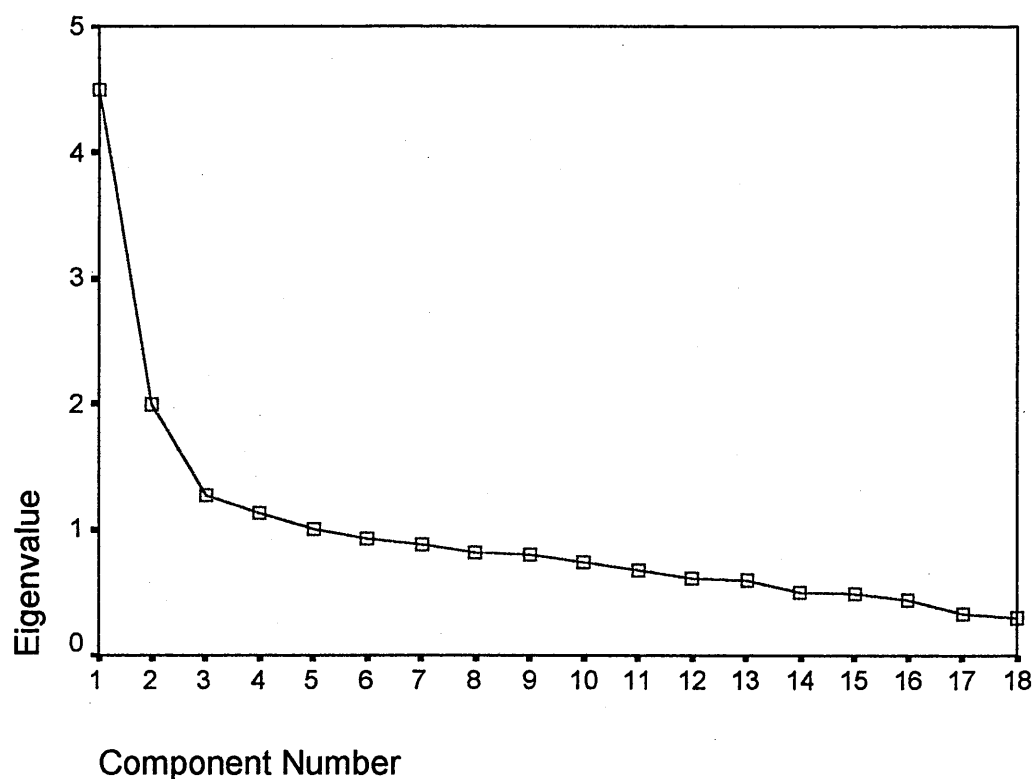
Standardised Coefficients			
	Beta	t	Sig.
Constant		15.890	.000
Training Intended	.373	2.606	.012
Workload	-.246	-1.886	.065
Interaction	-.047	-.332	.742
F (3, 48) = 4.356		Adjusted R ² = 0.17	

Dependent Variable: Training Undertaken (Respondents' Data)

Standardised Coefficients			
	Beta	t	Sig.
Constant		15.322	.000
Training Intended	.389	2.875	.006
Inconvenience	.118	.794	.431
Interaction	-.122	-.828	.412
F (3, 48) = 3.312		Adjusted R ² = 0.12	

Dependent Variable: Training Undertaken (Respondents' Data)

Scree Plot for Data from Study 1



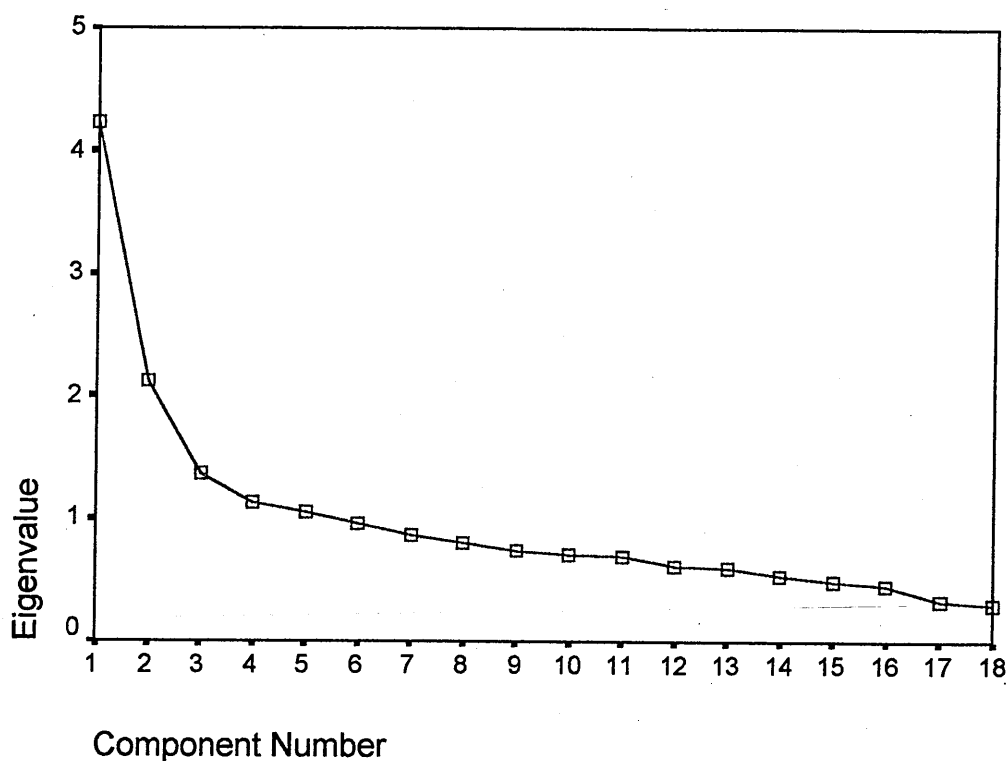
Variance Accounted for Before Oblique Rotation

	Component		
	1	2	3
Variance Accounted for (%)	24.94	11.09	7.10

Pattern Matrix for Data from Study 1

	Component		
	1	2	3
GNS 10	.85	.03	.06
GNS 6	.80	.01	.11
GNS 11	.75	-.20	-.07
GNS 2	.73	.06	-.11
GNS 3	.71	.05	-.11
GNS 8	.69	.08	.06
GNS 14	.22	.11	-.18
GNS 17	-.08	.61	.10
GNS 16	-.12	.56	-.28
GNS 23	-.09	.55	-.12
GNS 22	.05	.53	.04
GNS 13	.12	.52	-.09
GNS 18	.24	.48	.18
GNS 20	.12	.39	-.17
GNS 15	.08	-.18	-.74
GNS 19	-.04	.13	-.70
GNS 21	-.06	.09	-.61
GNS 12	.17	.04	-.48

Scree Plot for Data from Study 1a



Variance Accounted for Before Oblique Rotation

	Component		
	1	2	3
Variance Accounted for (%)	23.53	11.76	7.58

Pattern Matrix for Data from Study 1a

	Component		
	1	2	3
GNS 10	.82	.02	-.09
GNS 6	.79	.02	-.12
GNS 11	.74	-.23	.09
GNS 3	.72	.09	.09
GNS 8	.71	.11	-.11
GNS 2	.69	.09	.16
GNS 14	.25	.04	.20
GNS 22	.03	.66	-.18
GNS 23	-.08	.61	.04
GNS 18	.14	.56	-.16
GNS 16	-.08	.55	.27
GNS 13	.09	.53	.11
GNS 17	-.03	.39	.07
GNS 20	.12	.34	.16
GNS 19	-.06	.12	.73
GNS 15	.08	-.20	.71
GNS 21	-.11	.10	.61
GNS 12	.11	.15	.45